

ISC Class XII Physics

Part 2 Ray Optics to Semiconductor Electronics

243 Numericals Answer Key

Step-by-Step ISC Board Solutions

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Master Index — All Numericals

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Chapter 9: Ray Optics and Optical Instruments

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ISC Class XII Physics

Chapter 9

Ray Optics and Optical Instruments

111 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

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- II. Convex Mirror
- III. Snell's Law and Refractive Index
- IV. Apparent Depth of Object
- V. Total Internal Reflection
- VI. Refraction at Spherical Surface
- VII. Lens Maker's Formula
- VIII. Effect of Immersing Lens in a Liquid
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- XII. Power of the Lens
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- XVII. Dispersion due to Prism
- XVIII. Simple Microscope
- XIX. Compound Microscope
- XX. Telescope
- XXI. Miscellaneous

Formula Sheet

I–II. Spherical Mirrors

Mirror Formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

u = object distance, v = image distance, f = focal length ($f = R/2$), all measured from pole.

Magnification (Mirror)

$$m = -\frac{v}{u} = \frac{f}{f - u}$$

Negative m = real, inverted; positive m = virtual, erect.

III. Snell's Law and Refractive Index

Snell's Law

$$n_1 \sin i = n_2 \sin r$$

i = angle of incidence, r = angle of refraction, n_1, n_2 = refractive indices of the two media.

Refractive Index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

c = speed of light in vacuum, v = speed in medium.

IV. Apparent Depth

Apparent Depth

$$n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

Valid for near-normal viewing from a rarer medium.

Normal Shift

$$\Delta t = t \left(1 - \frac{1}{n} \right)$$

t = thickness of slab/liquid; Δt = apparent upward shift of the object.

V. Total Internal Reflection

Critical Angle

$$\sin C = \frac{1}{n}$$

n = refractive index of denser medium relative to the rarer medium; C = critical angle.

VI. Refraction at a Spherical Surface

Spherical Refraction

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

n_1 = incident-side index, n_2 = refracted-side index, R = radius of curvature of the surface.

VII–VIII. Lens Maker's Formula

Lens Maker's Formula

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

n = refractive index of lens material relative to surrounding medium; R_1, R_2 = radii of the two surfaces.

Lens in a Liquid

$$\frac{1}{f_L} = \left(\frac{n_{\text{lens}}}{n_{\text{liquid}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

f_L = focal length of the lens when immersed in a liquid of index n_{liquid} .

IX–XI. Lens Formula, Magnification, Combination

Thin Lens Formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

All distances measured from the optical centre.

Magnification (Lens)

$$m = \frac{v}{u} = \frac{f}{f + u}$$

Positive m = erect (virtual for convex, always virtual for concave); negative m = real, inverted.

Lenses in Contact

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

F = equivalent focal length of two thin lenses held in contact.

XII. Power of a Lens**Power**

$$P = \frac{1}{f \text{ (in metres)}}$$

Unit: dioptre (D). $P_{\text{total}} = P_1 + P_2$ for lenses in contact.

XIII. Combination of Lenses and Mirrors**Sequential Refraction/Reflection**

v_{final} found by applying lens/mirror formula stage by stage

Image of one element becomes the object for the next; effective mirror power $P_{\text{eff}} = P_{\text{lens}} + P_{\text{mirror}} + P_{\text{lens}}$ for lens-mirror silvered systems.

XIV–XVII. Prism**Prism Angle Relation**

$$A = r_1 + r_2$$

A = angle of prism, r_1, r_2 = angles of refraction at the two faces.

Deviation

$$\delta = i + e - A$$

i = angle of incidence, e = angle of emergence.

Refractive Index (Min. Deviation)

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

δ_m = angle of minimum deviation (occurs when $i = e$, $r_1 = r_2$).

Thin Prism Deviation

$$\delta = (n - 1)A$$

Valid for small prism angle A (thin prism approximation).

Angular Dispersion

$$\theta = (n_v - n_r)A$$

n_v, n_r = refractive indices for violet and red light.

XVIII–XIX. Microscopes**Simple Microscope**

$$M = 1 + \frac{D}{f} \text{ (image at } D\text{)}; \quad M = \frac{D}{f} \text{ (image at } \infty\text{)}$$

D = least distance of distinct vision (25 cm), f = focal length of lens.

Compound Microscope

$$M = m_o \times m_e = \frac{v_o}{u_o} \times \left(1 + \frac{D}{f_e}\right)$$

m_o = linear magnification of objective, m_e = angular magnification of eyepiece.

XX. Telescope**Astronomical Telescope**

$$M = -\frac{f_o}{f_e} \text{ (relaxed eye)}; \quad L = f_o + f_e$$

f_o, f_e = focal lengths of objective and eyepiece; L = tube length for image at infinity.

Telescope (Image at D)

$$M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$$

Used when the final image is formed at the least distance of distinct vision.

I. Concave Mirror

Question 1

An object is placed in front of a concave mirror of radius of curvature 40 cm at a distance of 10 cm. Find the position, nature and magnification of the image.

Given: $R = 40 \text{ cm} \Rightarrow f = -20 \text{ cm}$ (concave); $u = -10 \text{ cm}$

Formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, \quad m = -\frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-20} - \frac{1}{-10} = -\frac{1}{20} + \frac{1}{10} = \frac{1}{20}$$

$\Rightarrow v = +20 \text{ cm}$ (positive $\Rightarrow$ behind the mirror, virtual)

$$m = -\frac{v}{u} = -\frac{20}{-10} = +2$$

Final Answer

Image is 20 cm behind the mirror, virtual, erect, magnification = +2.

Common Mistake: Do not forget the sign of f : for a concave mirror f is always negative in the Cartesian convention.

Question 2

A 1 cm long flame is placed at a distance of 1.5 m from a wall. How far from the wall should a concave mirror be placed in order to obtain a 2 cm long flame on the same wall? What is the focal length of the mirror?

Given: Flame-wall distance = 1.5 m; image (on wall) is real, $|m| = 2$

Formula: $|u| + |v| = 1.5 \text{ m}, \quad |m| = \frac{|v|}{|u|} = 2$

Solution:

Let mirror be at distance d from the wall. Then $|v| = d$ and $|u| = 1.5 - d$.

Since $|v| = 2|u|$: $d = 2(1.5 - d) \Rightarrow 3d = 3 \Rightarrow d = 1 \text{ m}$

So $|u| = 0.5 \text{ m}$, $|v| = 1 \text{ m}$ (both real, so $u = -0.5 \text{ m}$, $v = -1 \text{ m}$)

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{-1} + \frac{1}{-0.5} = -1 - 2 = -3 \Rightarrow f = -\frac{1}{3} \text{ m}$$

Final Answer

Mirror should be placed 1 m from the wall; $f = \frac{1}{3} \text{ m} \approx 33.3 \text{ cm}$ (both real/negative).

Question 3

An object is placed at a distance of 30 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 10 cm away from the mirror, find the displacement of the image.

★ Likely Exam Question

Given: $f = -15 \text{ cm}$; $u_1 = -30 \text{ cm}$; $u_2 = -40 \text{ cm}$

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

Solution:

Position 1: $\frac{1}{v_1} = \frac{1}{-15} - \frac{1}{-30} = -\frac{1}{15} + \frac{1}{30} = -\frac{1}{30} \Rightarrow v_1 = -30 \text{ cm}$

Position 2: $\frac{1}{v_2} = \frac{1}{-15} - \frac{1}{-40} = -\frac{1}{15} + \frac{1}{40} = -\frac{1}{24} \Rightarrow v_2 = -24 \text{ cm}$

Displacement of image = $|v_1 - v_2| = |-30 - (-24)| = 6 \text{ cm}$; since $|v|$ decreased, the image has moved towards the mirror.

Final Answer

Image displacement = 6 cm towards the mirror.

Common Mistake: Students often assume the image moves the same direction and distance as the object — this is only true far from f and $2f$; always recompute v separately.

Question 4

A concave mirror has a radius of curvature of 30 cm. How far must it be placed from a small object in order to give a five times magnified virtual image?

Given: $R = 30 \text{ cm} \Rightarrow f = -15 \text{ cm}$; $m = +5$ (virtual, erect)

Formula: $m = \frac{f}{f - u}$

Solution:

$$5 = \frac{-15}{-15 - u} \Rightarrow -15 - u = \frac{-15}{5} = -3 \Rightarrow -u = -3 + 15 = 12 \Rightarrow u = -12 \text{ cm}$$

Final Answer

Object should be placed 12 cm in front of the mirror.

Question 5

A concave mirror produces a real image of half the size of an object placed at 60 cm in front of it. Where should the object be placed to obtain a virtual image of double the size of the object?

Given: $u_1 = -60 \text{ cm}$, $m_1 = -\frac{1}{2}$ (real); required $m_2 = +2$ (virtual)

Formula: $m = -\frac{v}{u}$, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Solution:

Step 1 — find f : $-\frac{1}{2} = -\frac{v_1}{-60} \Rightarrow v_1 = -30 \text{ cm}$

$$\frac{1}{f} = \frac{1}{-30} + \frac{1}{-60} = -\frac{1}{20} \Rightarrow f = -20 \text{ cm}$$

Step 2 — for virtual image, $m_2 = -\frac{v_2}{u_2} = 2 \Rightarrow v_2 = -2u_2$. Substituting in mirror formula:

$$\frac{1}{-2u_2} + \frac{1}{u_2} = \frac{1}{f} \Rightarrow \frac{1}{2u_2} = \frac{1}{f} \Rightarrow u_2 = \frac{f}{2} = \frac{-20}{2} = -10 \text{ cm}$$

Final Answer

Object should be placed 10 cm in front of the mirror.

Common Mistake: A virtual, magnified image in a concave mirror is only possible when the object lies between the pole and the focus — always check $|u| < |f|$.

Question 6

An object is placed first at a distance of 25 cm and then at a distance of 40 cm from a concave mirror. The magnification in the first case is four times than that in the second case, the image being real in both cases. What is the focal length of the mirror?

★ Likely Exam Question

Given: $u_1 = -25 \text{ cm}$, $u_2 = -40 \text{ cm}$, $m_1 = 4m_2$ (both real)

Formula: $m = \frac{f}{f - u}$

Solution:

$$\frac{f}{f + 25} = 4 \times \frac{f}{f + 40}$$

$$\frac{1}{f + 25} = \frac{4}{f + 40}$$

$$f + 40 = 4(f + 25) = 4f + 100$$

$$-60 = 3f \Rightarrow f = -20 \text{ cm}$$

Final Answer

Focal length of the mirror = -20 cm .

Question 7

A concave mirror for face viewing has focal length 36.0 cm . Find the distance at which you should hold the mirror from your face in order to see your image upright with a magnification of 4.

Given: $f = -36 \text{ cm}$; $m = +4$ (virtual, erect)

Formula: $m = \frac{f}{f - u}$

Solution:

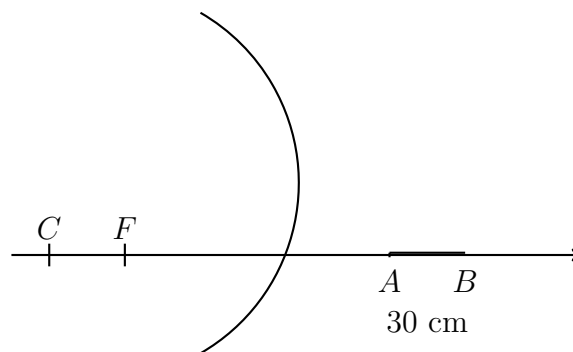
$$4 = \frac{-36}{-36 - u} \Rightarrow -36 - u = \frac{-36}{4} = -9 \Rightarrow u = -36 + 9 = -27 \text{ cm}$$

Final Answer

Mirror should be held 27.0 cm from the face.

Question 8

An object AB is placed along the axis of a concave mirror of focal length 10 cm . End A (nearer end) of the object is at 30 cm from the mirror. Find the length of the image, if: (i) length of object is 5 cm , (ii) length of object is 1 mm , and (iii) show the image's position in case (i).



Object AB (5 cm) placed along the principal axis, near end A at 30 cm from a concave mirror of $f = 10$ cm

Given: $f = -10$ cm; $u_A = -30$ cm; object length (i) 5 cm, (ii) 1 mm

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$, longitudinal magnification $m_L = \left(\frac{v}{u}\right)^2$

Solution:

(i) End A : $u = -30$: $\frac{1}{v_A} = \frac{1}{-10} - \frac{1}{-30} = -\frac{1}{15} \Rightarrow v_A = -15$ cm

End B (far end, $u = -35$): $\frac{1}{v_B} = \frac{1}{-10} - \frac{1}{-35} = -\frac{1}{14} \Rightarrow v_B = -14$ cm

Image length $= |v_A - v_B| = |-15 - (-14)| = 1$ cm

(ii) Near A : $m_L = \left(\frac{v_A}{u_A}\right)^2 = \left(\frac{-15}{-30}\right)^2 = 0.25$

$dv = m_L \times du = 0.25 \times 1 \text{ mm} = 0.25 \text{ mm}$

Final Answer

(i) Image length = 1 cm (along the principal axis); (ii) Image length = 0.25 mm; (iii) Image $A'B'$ is real, inverted, and lies between 14 cm and 15 cm from the mirror.

Common Mistake: Longitudinal magnification is $(v/u)^2$, NOT the same as lateral magnification $-v/u$ — a very common exam slip.

II. Convex Mirror

Question 9

An object is placed at 18 cm in front of a spherical mirror. The image is formed at 3.0 cm behind the mirror. Find the focal length and the nature of the mirror.

Given: $u = -18$ cm; $v = +3$ cm (behind mirror $\Rightarrow$ virtual)

Formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Solution:

$$\frac{1}{f} = \frac{1}{3} + \frac{1}{-18} = \frac{6}{18} - \frac{1}{18} = \frac{5}{18}$$

$$f = \frac{18}{5} = 3.6 \text{ cm (positive } \Rightarrow \text{ convex mirror)}$$

Final Answer

Focal length = +3.6 cm; mirror is convex.

Question 10

An object is placed 15 cm away from a convex mirror of radius of curvature 90 cm. Find the image's position and magnification.

Given: $R = 90$ cm $\Rightarrow f = +45$ cm; $u = -15$ cm

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}, \quad m = -\frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{45} - \frac{1}{-15} = \frac{1}{45} + \frac{3}{45} = \frac{4}{45}$$

$$v = \frac{45}{4} = 11.25 \text{ cm (behind the mirror, virtual)}$$

$$m = -\frac{11.25}{-15} = +0.75$$

Final Answer

Image is 11.25 cm behind the mirror, virtual; magnification = +0.75.

Question 11

A 2 cm long object is placed at a distance of 100 cm from a convex mirror of radius of curvature 50 cm. Find the position, nature and length of the image.

Given: $R = 50 \text{ cm} \Rightarrow f = +25 \text{ cm}$; $u = -100 \text{ cm}$; object height = 2 cm

Formula: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}, \quad m = -\frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{25} - \frac{1}{-100} = \frac{4}{100} + \frac{1}{100} = \frac{5}{100}$$

$v = 20 \text{ cm}$ (behind mirror, virtual)

$$m = -\frac{20}{-100} = +0.2$$

Image length = $2 \times 0.2 = 0.4 \text{ cm}$

Final Answer

Image is 20 cm behind the mirror, virtual, erect, length = 0.4 cm.

III. Snell's Law and Refractive Index

Question 12

The refractive index of diamond relative to air is 2.42 and that of glass is 1.51. How much faster does light travel in glass than in diamond?

Given: $n_d = 2.42$; $n_g = 1.51$

Formula: $n = \frac{c}{v} \Rightarrow v = \frac{c}{n}$

Solution:

$$\frac{v_g}{v_d} = \frac{c/n_g}{c/n_d} = \frac{n_d}{n_g} = \frac{2.42}{1.51} = 1.60$$

Final Answer

Light travels 1.60 times faster in glass than in diamond.

Question 13

A ray of light in air is incident on a glass plate at 45° . The refractive index of glass is 1.5. What is the angle of refraction?

Given: $i = 45^\circ$; $n = 1.5$

Formula: $n = \frac{\sin i}{\sin r}$

Solution:

$$\sin r = \frac{\sin i}{n} = \frac{\sin 45^\circ}{1.5} = \frac{0.7071}{1.5} = 0.4714$$

$$r = \sin^{-1}(0.4714) = 28^\circ$$

Final Answer

Angle of refraction $\approx 28^\circ$.

Question 14

A ray of light which is incident in air on the surface of a glass block is deviated through 15° . The angle of incidence in air is 60° . What is the angle of refraction in glass? Also calculate the critical angle for the glass-air interface.

★ Likely Exam Question

Given: $i = 60^\circ$; deviation $= 15^\circ$

Formula: Deviation $= i - r$; $n = \frac{\sin i}{\sin r}$; $\sin C = \frac{1}{n}$

Solution:

$$r = i - 15^\circ = 60^\circ - 15^\circ = 45^\circ$$

$$n = \frac{\sin 60^\circ}{\sin 45^\circ} = \frac{0.8660}{0.7071} = 1.225$$

$$\sin C = \frac{1}{1.225} = 0.8165 \Rightarrow C = 54.7^\circ$$

Final Answer

Angle of refraction $= 45^\circ$; critical angle $= 54.7^\circ$.

IV. Apparent Depth of Object

Question 15

How deep will a 4 m deep tank appear when seen in air due to optical refraction? The refractive index of water is $4/3$.

Given: Real depth = 4 m; $n = 4/3$

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$

Solution:

$$\text{Apparent depth} = \frac{\text{Real Depth}}{n} = \frac{4}{4/3} = 4 \times \frac{3}{4} = 3 \text{ m}$$

Final Answer

Apparent depth = 3 m.

Question 16

A beaker is filled with water to a height of 12.5 cm. The apparent depth of a needle at the bottom of the beaker is 9.4 cm. Find refractive index of water.

Given: Real depth = 12.5 cm; apparent depth = 9.4 cm

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$

Solution:

$$n = \frac{12.5}{9.4} = 1.330$$

Final Answer

Refractive index of water = 1.33.

Question 17

The apparent depth of an object at the bottom of a tank filled with a liquid of refractive index 1.3 is 7.7 cm. What is the actual depth?

Given: $n = 1.3$; apparent depth = 7.7 cm

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}$

Solution:

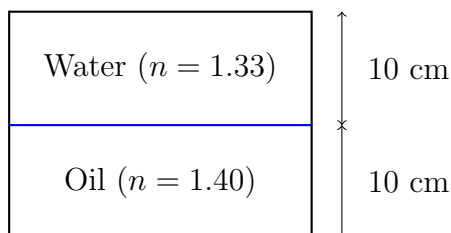
$$\text{Real depth} = n \times \text{apparent depth} = 1.3 \times 7.7 = 10.01 \text{ cm} \approx 10 \text{ cm}$$

Final Answer

Actual depth $\approx 10 \text{ cm}$.

Question 18

A beaker 20 cm high is half filled with an oil of refractive index 1.40 and the upper half filled with water of refractive index 1.33. What is the apparent depth of the beaker as observed normally?



Two 10 cm layers – water (top) over oil (bottom) – viewed normally from above

Given: Oil layer = 10 cm, $n_{\text{oil}} = 1.40$; water layer = 10 cm, $n_{\text{water}} = 1.33$

Formula: Apparent depth (multi-layer) = $\sum \frac{t_i}{n_i}$

Solution:

$$\text{Apparent depth} = \frac{10}{1.40} + \frac{10}{1.33} = 7.143 + 7.519 = 14.66 \text{ cm}$$

Final Answer

Apparent depth of the beaker = 14.66 cm.

Common Mistake: For layered media, apparent depth is the SUM of each layer's own t/n — never divide the total depth by an average n .

Question 19

A needle lying on a horizontal surface is viewed vertically from a distance of 100 cm. A glass slab of 12 cm thickness is placed on the needle. Find the distance of the vertical image of the needle from the eye, taking $n_g = 1.5$.

Given: Eye-to-needle distance = 100 cm; slab thickness $t = 12 \text{ cm}$; $n = 1.5$

Formula: $\Delta t = t \left(1 - \frac{1}{n} \right)$

Solution:

$$\Delta t = 12 \left(1 - \frac{1}{1.5} \right) = 12 \times \frac{1}{3} = 4 \text{ cm}$$

The needle appears shifted up (closer) by 4 cm, so apparent distance = $100 - 4 = 96 \text{ cm}$

Final Answer

Distance of the vertical image from the eye = 96 cm.

Question 20

A microscope focussed on a pin lying at the bottom of a beaker reads 3.965 cm. When a liquid is poured up to a height of 2.537 cm into the beaker, the microscope focussed again on the pin reads 3.348 cm. Find the refractive index of the liquid.

Given: Microscope shift = $3.965 - 3.348 = 0.617 \text{ cm}$; liquid depth $t = 2.537 \text{ cm}$

Formula: $\Delta t = t \left(1 - \frac{1}{n} \right)$

Solution:

$$0.617 = 2.537 \left(1 - \frac{1}{n} \right)$$

$$1 - \frac{1}{n} = \frac{0.617}{2.537} = 0.2432$$

$$\frac{1}{n} = 0.7568 \Rightarrow n = 1.321$$

Final Answer

Refractive index of the liquid = 1.321.

Question 21

A beaker is filled with a liquid to a height of 14 cm. The apparent depth of a needle fixed at the bottom of the beaker is measured to be 10 cm by a microscope. What is the refractive index of the liquid? The height of the liquid in the beaker is now raised to 21 cm. By what distance would the microscope have to be moved to focus on the needle again?

Given: Real depth = 14 cm, apparent depth = 10 cm; new liquid height = 21 cm

Formula: $n = \frac{\text{Real Depth}}{\text{Apparent Depth}}, \quad \Delta t = t \left(1 - \frac{1}{n} \right)$

Solution:

$$n = \frac{14}{10} = 1.4$$

Additional liquid column added = $21 - 14 = 7$ cm

$$\text{Extra apparent shift due to this added liquid} = 7 \left(1 - \frac{1}{1.4} \right) = 7 \times 0.2857 = 2.0 \text{ cm}$$

Final Answer

Refractive index = 1.4; microscope must be raised by 2.0 cm to refocus.

Common Mistake: The microscope must be moved by the CHANGE in apparent shift caused by the added liquid column, not by the apparent depth of the new total column.

Question 22

A beaker is filled with water ($n = 1.5$) to a height h . Height of the beaker is 15 cm. When the bottom of the beaker is viewed vertically by an observer, the beaker appears to be half filled. Find the value of h .

Given: Beaker height $H = 15$ cm; $n = 1.5$

Formula: Apparent depth of water column (from surface) = $\frac{h}{n}$

Solution:

"Appears half filled" means the apparent depth of the bottom (measured up from the water surface) equals the empty (air) column above the water:

$$\frac{h}{n} = H - h$$

$$\frac{h}{1.5} = 15 - h$$

$$h = 1.5(15 - h) = 22.5 - 1.5h$$

$$2.5h = 22.5 \Rightarrow h = 9.0 \text{ cm}$$

Final Answer

$h = 9.0$ cm.

V. Total Internal Reflection

Question 23

Critical angle for total internal reflection from a medium to vacuum is 60° . Find the velocity of light in the given medium.

Given: $C = 60^\circ$; $c = 3 \times 10^8$ m/s

Formula: $\sin C = \frac{1}{n}$, $n = \frac{c}{v}$

Solution:

$$n = \frac{1}{\sin 60^\circ} = \frac{1}{0.8660} = 1.1547$$

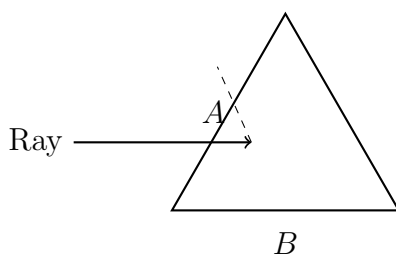
$$v = \frac{c}{n} = c \sin C = 3 \times 10^8 \times 0.8660 = 2.6 \times 10^8 \text{ m/s}$$

Final Answer

Velocity of light in the medium = $2.6 \times 10^8 \text{ ms}^{-1}$.

Question 24

A ray of light falls normally on the face of a glass prism ($n = 1.5$). Find the angle of the prism, if the ray just fails to emerge from the second face.



Ray enters normally on face 1 (no bending) and strikes face 2 at the critical angle

Given: $n = 1.5$; ray enters normally $\Rightarrow$ internal angle at 2nd face = A (prism angle)

Formula: $\sin C = \frac{1}{n}$; for normal incidence, angle at 2nd face = A

Solution:

Since the ray enters the first face normally, it does not bend, and by geometry the angle

of incidence at the second face equals the prism angle A . For the ray to "just fail" to emerge, $A = C$:

$$\sin C = \frac{1}{1.5} = 0.667 \Rightarrow C = 41.8^\circ$$

Final Answer

Angle of the prism $A = 41.8^\circ$.

Question 25

A substance has a critical angle of 45° for yellow light. What is the refractive index of the substance?

Given: $C = 45^\circ$

Formula: $\sin C = \frac{1}{n}$

Solution:

$$n = \frac{1}{\sin 45^\circ} = \frac{1}{0.7071} = 1.414$$

Final Answer

Refractive index = 1.414.

Question 26

What is the critical angle for a light ray going from dense glass into water? The refractive indices of dense glass and water relative to air are 1.62 and 1.33 respectively.

Given: $n_{\text{glass}} = 1.62$; $n_{\text{water}} = 1.33$

Formula: $\sin C = \frac{n_{\text{water}}}{n_{\text{glass}}}$

Solution:

$$\sin C = \frac{1.33}{1.62} = 0.821 \Rightarrow C = 55.16^\circ \approx 55^\circ$$

Final Answer

Critical angle $\approx 55^\circ$.

Question 27

The absolute refractive indices of water and glycerine are 1.33 and 1.46 respectively. Find the critical angle for a ray travelling from glycerine into water.

Given: $n_{\text{glycerine}} = 1.46$; $n_{\text{water}} = 1.33$

Formula: $\sin C = \frac{n_{\text{water}}}{n_{\text{glycerine}}}$

Solution:

$$\sin C = \frac{1.33}{1.46} = 0.911 \Rightarrow C = 65.5^\circ$$

Final Answer

Critical angle = 65.5° .

Question 28

An equilateral prism of glass of refractive index 1.45 is immersed in water of refractive index 1.33. A narrow beam of light falls normally on one face of the prism. Find the angle of emergence of the beam.

★ Likely Exam Question

Given: $A = 60^\circ$; $n_{\text{glass}} = 1.45$; $n_{\text{water}} = 1.33$; ray enters normally

Formula: $n_{\text{glass}} \sin A = n_{\text{water}} \sin e$

Solution:

Ray enters normally $\Rightarrow$ angle at second face = $A = 60^\circ$

$$\sin e = \frac{n_{\text{glass}}}{n_{\text{water}}} \sin 60^\circ = \frac{1.45}{1.33} \times 0.8660 = 1.0902 \times 0.8660 = 0.9441$$

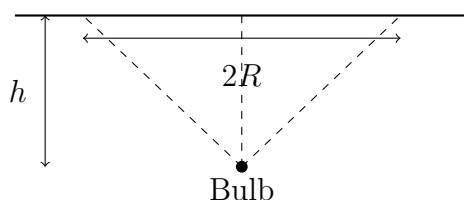
$$e = \sin^{-1}(0.9441) = 70.8^\circ$$

Final Answer

Angle of emergence = 70.8° .

Question 29

A small bulb is placed at the bottom of a tank containing water to a depth of 80 cm. What is the area of the surface of water through which light of the bulb can emerge out? Refractive index of water is 1.33.



Light from a point source escapes only through a circle of radius $R = h \tan C$ on the water surface

Given: $h = 80$ cm; $n = 1.33$

Formula: $R = h \tan C$, $\sin C = \frac{1}{n}$, Area = πR^2

Solution:

$$\sin C = \frac{1}{1.33} = 0.7519 \Rightarrow C = 48.75^\circ, \quad \tan C = 1.1387$$

$$R = 80 \times 1.1387 = 91.1 \text{ cm} = 0.911 \text{ m}$$

$$\text{Area} = \pi R^2 = \pi \times (0.911)^2 = 2.607 \text{ m}^2$$

Final Answer

Area of the circle of light $\approx 2.6 \text{ m}^2$.

Common Mistake: "Area through which light can emerge" is a CIRCLE of radius $h \tan C$, not $h \sin C$ — don't confuse the geometry.

Question 30

A point-source of light is placed at the centre of the bottom of a jar having a liquid of refractive index $5/3$. An opaque disc of radius 1.0 cm is placed on the liquid surface with its centre vertically above the source. What is the maximum height of the liquid for which the source is not visible from above?

Given: $n = 5/3$; disc radius = 1.0 cm

Formula: $R = h \tan C$, $\sin C = \frac{1}{n}$

Solution:

$$\sin C = \frac{3}{5} = 0.6 \Rightarrow C = 36.87^\circ, \quad \tan C = 0.75$$

For the source to be invisible, the escaping light-circle radius must not exceed the disc radius: $h \tan C \leq 1$

$$h_{\max} = \frac{1}{\tan C} = \frac{1}{0.75} = \frac{4}{3} \text{ cm}$$

Final Answer

Maximum height of liquid = $\frac{4}{3} \text{ cm} \approx 1.33 \text{ cm}$.

VI. Refraction at Spherical Surface

Question 31

A point-object is placed in air at 60 cm from a convex spherical refracting surface of refractive index 1.5. The radius of curvature of the surface is 25 cm. Find the position of the image.

Given: $u = -60$ cm; $n_1 = 1$, $n_2 = 1.5$; $R = +25$ cm

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{v} - \frac{1}{-60} = \frac{0.5}{25} = 0.02$$

$$\frac{1.5}{v} = 0.02 - 0.01667 = 0.00333$$

$$v = \frac{1.5}{0.00333} = 450 \text{ cm}$$

Final Answer

Image forms 450 cm from the pole, inside the surface (real).

Question 32

A convex spherical refracting surface of refractive index 1.5 forms the image of an object distant 10 cm from it at a distance of 40 cm on the same side as the object. What is the radius of curvature of the surface?

Given: $u = -10$ cm; $v = -40$ cm (virtual, same side); $n_1 = 1$, $n_2 = 1.5$

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{-40} - \frac{1}{-10} = \frac{0.5}{R}$$

$$-0.0375 + 0.1 = \frac{0.5}{R}$$

$$0.0625 = \frac{0.5}{R} \Rightarrow R = \frac{0.5}{0.0625} = 8 \text{ cm}$$

Final Answer

Radius of curvature = +8 cm.

Question 33

An empty spherical flask of diameter 100 cm is placed in water ($n = 4/3$). A parallel beam of light strikes the flask. Where will it appear to come from to an observer within the flask?

★ Likely Exam Question

Given: $R = 50$ cm; $n_1 = 4/3$ (water, outside), $n_2 = 1$ (air, inside flask); $u = -\infty$

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1}{v} - 0 = \frac{1 - 4/3}{50} = \frac{-1/3}{50} = -\frac{1}{150}$$

$$v = -150 \text{ cm}$$

Final Answer

Light appears to converge from a point 150 cm from the flask (on the water side).

Common Mistake: Only the FIRST surface refraction is needed here — the observer, standing inside the flask, sees the image formed immediately upon entry, before any second-surface refraction occurs.

Question 34

A point-object is placed in air at 40 cm from a concave spherical surface of radius of curvature 20 cm and refractive index 1.5. Locate the image.

Given: $u = -40$ cm; $R = -20$ cm; $n_1 = 1$, $n_2 = 1.5$

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{v} - \frac{1}{-40} = \frac{0.5}{-20} = -0.025$$

$$\frac{1.5}{v} = -0.025 - 0.025 = -0.05$$

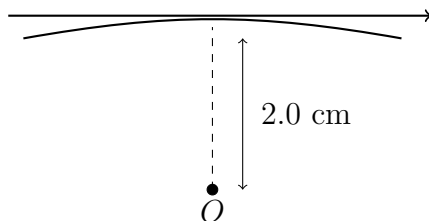
$$v = \frac{1.5}{-0.05} = -30 \text{ cm}$$

Final Answer

Image forms 30 cm from the pole, on the same side as the object (virtual).

Question 35

A point-object is 2.0 cm below the concave meniscus of water ($n = 4/3$) of radius of curvature 0.5 cm. Locate the image.



Point object O, 2.0 cm below a concave water meniscus of radius 0.5 cm

Given: $u = -2.0$ cm; $n_1 = 4/3$ (water), $n_2 = 1$ (air); $R = +0.5$ cm

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1}{v} - \frac{4/3}{-2} = \frac{1 - 4/3}{0.5}$$

$$\frac{1}{v} + \frac{2}{3} = -\frac{2}{3}$$

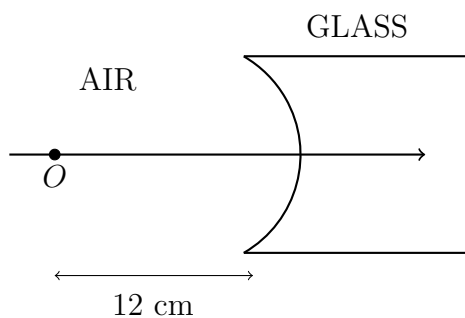
$$\frac{1}{v} = -\frac{4}{3} \Rightarrow v = -0.75 \text{ cm}$$

Final Answer

Image forms 0.75 cm below the meniscus, in water (virtual).

Question 36

One end of a cylindrical rod of glass ($n = 1.5$) is given a hemispherical shape of radius 5 cm, as shown in the figure. A point object O is placed in air on the axis of the rod at a distance of 12 cm from the curved end. Locate the image I .



Point object O in air, 12 cm from the hemispherical (curved) end of a glass rod

Given: $u = -12$ cm; $R = +5$ cm; $n_1 = 1$ (air), $n_2 = 1.5$ (glass)

Formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

Solution:

$$\frac{1.5}{v} - \frac{1}{-12} = \frac{0.5}{5} = 0.1$$

$$\frac{1.5}{v} = 0.1 - 0.0833 = 0.01667$$

$$v = \frac{1.5}{0.01667} = 90 \text{ cm}$$

Final Answer

Image I forms 90 cm inside the rod from the curved end (real image).

VII. Lens Maker's Formula

Question 37

A plano-convex lens has its curved surface of radius of curvature 20 cm. If the refractive index of the lens material is 1.5, find its focal length.

Given: $R_1 = 20$ cm (curved), $R_2 = \infty$ (flat); $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{20} - 0 \right) = 0.5 \times 0.05 = 0.025$$

$$f = 40 \text{ cm}$$

Final Answer

Focal length = 40 cm.

Question 38

An equiconvex lens has radius of curvature 10 cm for each surface. If the refractive index of glass is 1.5, find its focal length.

Given: $R_1 = +10$ cm, $R_2 = -10$ cm; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{10} - \frac{1}{-10} \right) = 0.5 \times 0.2 = 0.1$$

$$f = 10 \text{ cm}$$

Final Answer

Focal length = 10 cm.

Question 39

The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside

the lens is $2/3$ times the wavelength in free space. Calculate the radius of the curved surface of the lens.

★ Likely Exam Question

Given: $v = +8 \text{ m}$; $m = -\frac{1}{3}$; $\lambda_{\text{medium}} = \frac{2}{3}\lambda_{\text{vacuum}}$

Formula: $n = \frac{\lambda_{\text{vacuum}}}{\lambda_{\text{medium}}}$, $m = \frac{v}{u}$, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $\frac{1}{f} = (n - 1)\frac{1}{R}$

Solution:

$$n = \frac{3}{2} = 1.5$$

$$m = \frac{v}{u} = -\frac{1}{3} \Rightarrow u = \frac{v}{-1/3} = -3v = -24 \text{ m}$$

$$\frac{1}{f} = \frac{1}{8} - \frac{1}{-24} = \frac{3}{24} + \frac{1}{24} = \frac{4}{24} = \frac{1}{6} \Rightarrow f = 6 \text{ m}$$

$$\text{Plano-convex: } \frac{1}{f} = (n - 1)\frac{1}{R} \Rightarrow \frac{1}{6} = 0.5 \times \frac{1}{R} \Rightarrow R = 3 \text{ m}$$

Final Answer

Radius of curvature of the curved surface = 3 m.

Question 40

The radii of curvature of a double convex lens are 15 cm and 30 cm and the refractive index of its glass is 1.5. Calculate the focal length of the lens.

Given: $R_1 = +15 \text{ cm}$, $R_2 = -30 \text{ cm}$; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{15} - \frac{1}{-30} \right) = 0.5 \left(\frac{2}{30} + \frac{1}{30} \right) = 0.5 \times \frac{1}{10} = 0.05$$

$$f = 20 \text{ cm}$$

Final Answer

Focal length = 20 cm.

Question 41

The radii of curvature of a double concave lens are 30 cm and 60 cm and the refractive index of its glass is 1.5. Calculate the focal length of the lens.

Given: $R_1 = -30$ cm, $R_2 = +60$ cm; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{-30} - \frac{1}{60} \right) = 0.5 \left(-\frac{2}{60} - \frac{1}{60} \right) = 0.5 \times \left(-\frac{1}{20} \right) = -0.025$$

$$f = -40 \text{ cm}$$

Final Answer

Focal length = -40 cm.

Question 42

The radius of curvature of the convex face of a plano-convex lens is 55 cm and the refractive index of its material is 1.55. Find the focal length of the lens.

Given: $R_1 = 55$ cm, $R_2 = \infty$; $n = 1.55$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = (0.55) \left(\frac{1}{55} - 0 \right) = \frac{0.55}{55} = 0.01$$

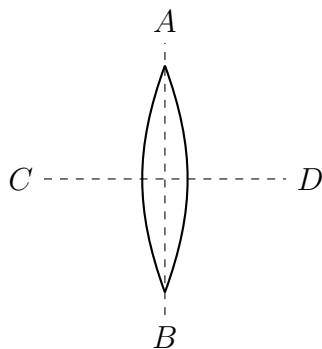
$$f = 100 \text{ cm}$$

Final Answer

Focal length = 100 cm.

Question 43

(a) A bi-convex spherical lens has a focal length of 20 cm. If it is split along a plane AB (perpendicular to the axis) into two halves to form two plano-convex lenses, what will be the focal length of each plano-convex lens? (b) What if the lens be cut along CD (containing the axis)?



Bi-convex lens split perpendicular to the axis along AB, and along the axis along CD

Given: Bi-convex lens, $f = 20$ cm, both faces equal curvature R

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

(a) Original: $\frac{1}{20} = (n - 1) \left(\frac{1}{R} + \frac{1}{R} \right) = \frac{2(n - 1)}{R}$

After splitting along AB , each half is plano-convex with the SAME curved-face radius R :

$$\frac{1}{f_{\text{half}}} = (n - 1) \frac{1}{R} = \frac{1}{2} \times \frac{1}{20} \Rightarrow f_{\text{half}} = 40 \text{ cm}$$

(b) Cutting along CD (parallel to the axis) does not change either radius of curvature — both curved surfaces are simply narrower. Hence f remains 20 cm, but since less aperture area collects light, the image intensity decreases.

Final Answer

(a) Focal length of each plano-convex half = 40 cm. (b) Focal length remains 20 cm, but intensity of the image decreases.

Common Mistake: Splitting perpendicular to the axis changes the focal length (halves each curvature's contribution); splitting along the axis leaves f unchanged but reduces intensity — these two cuts are often confused.

VIII. Effect of Immersing Lens in a Liquid

Question 44

The refractive index of glass with respect to air is $3/2$ and that of water is $4/3$. The radii of curvature of a glass convex lens are 20 cm and 30 cm respectively. What will be the focal length of the lens (i) in air, (ii) in water?

Given: $n_g = 3/2$; $n_w = 4/3$; $R_1 = 20$ cm, $R_2 = -30$ cm

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$; in liquid, use $n = \frac{n_g}{n_{\text{liquid}}}$

Solution:

(i) **In air:** $\frac{1}{f_a} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{20} + \frac{1}{30} \right) = 0.5 \times \frac{5}{60} = \frac{1}{24}$

$f_a = 24$ cm

(ii) **In water:** relative index $= \frac{3/2}{4/3} = \frac{9}{8} = 1.125$

$\frac{1}{f_w} = (0.125) \left(\frac{1}{20} + \frac{1}{30} \right) = 0.125 \times \frac{5}{60} = \frac{1}{96}$

$f_w = 96$ cm

Final Answer

(i) $f = 24$ cm in air; (ii) $f = 96$ cm in water.

Question 45

The focal length of a convex lens of glass ($n = 1.50$) in air is 40 cm. Calculate the focal length of the lens when it is immersed in a liquid of refractive index (i) 1.30, (ii) 1.50, (iii) 1.70.

★ Likely Exam Question

Given: $f_{\text{air}} = 40$ cm, $n_g = 1.50$

Formula: $\frac{f_{\text{liquid}}}{f_{\text{air}}} = \frac{(n_g - 1)}{\left(\frac{n_g}{n_l} - 1 \right)}$

Solution:

(i) $n_l = 1.30$: relative index $= \frac{1.5}{1.3} = 1.1538$; factor $= \frac{0.5}{0.1538} = 3.25$

$$f = 40 \times 3.25 = 130 \text{ cm}$$

(ii) $n_l = 1.50$: relative index $= 1$; $\frac{1}{f} = 0 \Rightarrow f \rightarrow \infty$ (no refraction occurs)

(iii) $n_l = 1.70$: relative index $= \frac{1.5}{1.7} = 0.882$; factor $= \frac{0.5}{-0.118} = -4.25$

$$f = 40 \times (-4.25) = -170 \text{ cm}$$

Final Answer

(i) $f = 130 \text{ cm}$; (ii) $f = \infty$ (lens becomes ineffective); (iii) $f = -170 \text{ cm}$ (lens behaves like a concave lens).

Common Mistake: When a convex lens is immersed in a liquid MORE optically dense than itself, it starts behaving like a diverging (concave) lens — a classic ISC conceptual trap.

Question 46

The radii of curvature for spherical surfaces of a bi-convex lens are 20 cm and 30 cm. The refractive index of the material of the lens with respect to air is 1.5. If the lens is dipped in a liquid of refractive index 1.65 with respect to air, calculate its effective focal length and state the nature of the lens.

Given: $R_1 = 20 \text{ cm}$, $R_2 = -30 \text{ cm}$; $n_g = 1.5$; $n_l = 1.65$

Formula: $\frac{1}{f} = \left(\frac{n_g}{n_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\text{Relative index} = \frac{1.5}{1.65} = 0.9091$$

$$\frac{1}{f} = (0.9091 - 1) \left(\frac{1}{20} + \frac{1}{30} \right) = (-0.0909) \times \frac{5}{60} = -0.00758$$

$$f = -132 \text{ cm}$$

Final Answer

Effective focal length $= -132 \text{ cm}$; the lens behaves as a concave (diverging) lens.

IX. Lens Formula

Question 47

An object is placed at 0.06 m from a convex lens of focal length 0.10 m. Find the position of the image.

Given: $u = -0.06$ m; $f = +0.10$ m

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{0.10} + \frac{1}{-0.06} = 10 - 16.67 = -6.67$$

$$v = -0.15 \text{ m}$$

Final Answer

Image forms 0.15 m in front of the lens (virtual, same side as object).

Question 48

A convergent beam of light passes through a diverging lens of focal length 0.2 m and comes to a focus on the axis 0.3 m behind the lens. Where would the beam have been focussed in the absence of the lens?

Given: $f = -0.2$ m; $v = +0.3$ m (real, converging beam)

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

$$\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = \frac{1}{0.3} - \frac{1}{-0.2} = 3.333 + 5 = 8.333$$

$$u = 0.12 \text{ m (virtual object, behind the lens)}$$

Final Answer

The beam would have been focussed 0.12 m behind the position of the lens.

Question 49

A double convex lens has 10 cm and 15 cm as its two radii of curvature. The image of an object, placed 30 cm from the lens, is formed at 20 cm from the lens on the other

side. Find the focal length and the refractive index of the material of the lens. What will be the focal length if it is immersed in water of refractive index 1.33?

★ Likely Exam Question

Given: $u = -30$ cm; $v = +20$ cm; $R_1 = 10$ cm, $R_2 = -15$ cm

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$; $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

Solution:

$$\frac{1}{f} = \frac{1}{20} - \frac{1}{-30} = \frac{3}{60} + \frac{2}{60} = \frac{5}{60} = \frac{1}{12} \Rightarrow f = 12 \text{ cm}$$

$$\frac{1}{12} = (n - 1) \left(\frac{1}{10} + \frac{1}{15} \right) = (n - 1) \times \frac{5}{30} = (n - 1) \times \frac{1}{6}$$

$$n - 1 = \frac{6}{12} = 0.5 \Rightarrow n = 1.5$$

In water: relative index = $\frac{1.5}{1.33} = 1.128$

$$\frac{1}{f_w} = (0.128) \times \frac{1}{6} = 0.0213 \Rightarrow f_w = 47 \text{ cm}$$

Final Answer

$f = 12$ cm in air, $n = 1.5$; $f = 47$ cm in water.

Question 50

A thin convex lens forms the distinct image of an object on a screen placed at a distance of 30 cm from the lens. When the lens is displaced 5 cm towards the screen, then in order to obtain a distinct image the screen is displaced 5 cm towards the lens. Find the focal length of the lens.

Given: Original image distance $v_1 = +30$ cm; lens shifts 5 cm towards screen, screen shifts 5 cm towards lens

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Let object be at the origin, lens at distance x from it ($u_1 = -x$), and total object–screen separation D (so $v_1 = D - x = 30$).

After the shift: new lens position = $x + 5$, new screen position = $D - 5$, so new image distance $v_2 = (D - 5) - (x + 5) = D - x - 10 = 20$ cm, and new object distance $u_2 = -(x + 5)$.

Using the lens formula at both positions and eliminating f :

$$\frac{1}{30} + \frac{1}{x} = \frac{1}{20} + \frac{1}{x+5}$$

$$\frac{1}{x} - \frac{1}{x+5} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60}$$

$$\frac{5}{x(x+5)} = \frac{1}{60} \Rightarrow x(x+5) = 300 \Rightarrow x^2 + 5x - 300 = 0$$

$$x = \frac{-5 + \sqrt{25 + 1200}}{2} = \frac{-5 + 35}{2} = 15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{30} + \frac{1}{15} = \frac{1}{30} + \frac{2}{30} = \frac{3}{30} = \frac{1}{10}$$

Final Answer

Focal length of the lens = 10 cm.

Common Mistake: In displacement problems where BOTH lens and screen move, track absolute positions of object, lens, and screen carefully — don't assume the object-to-screen distance stays fixed unless explicitly stated.

X. Magnification by the Lens

Question 51

An illuminated object and a screen are 90 cm apart. Find the nature and focal length of a lens required to produce a clear image on the screen, twice the size of the object.

Given: $|u| + |v| = 90$ cm; $|m| = 2$ (real, on screen)

Formula: $m = \frac{v}{u}, \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Real image: $|v| = 2|u|$. Also $|u| + |v| = 90 \Rightarrow 3|u| = 90 \Rightarrow |u| = 30$ cm, $|v| = 60$ cm

$u = -30$ cm, $v = +60$ cm

$$\frac{1}{f} = \frac{1}{60} - \frac{1}{-30} = \frac{1}{60} + \frac{2}{60} = \frac{3}{60} = \frac{1}{20}$$

Final Answer

The lens must be convex, $f = 20$ cm.

Question 52

A 10 cm long pin lies along the principal axis of a convex lens of focal length 10 cm, such that the nearer end of the pin is at a distance of 15 cm from the lens. Find the length of the image of the pin formed by the lens.

Given: $f = 10$ cm; near end $u_1 = -15$ cm; far end $u_2 = -25$ cm

Formula: $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$

Solution:

Near end: $\frac{1}{v_1} = \frac{1}{10} + \frac{1}{-15} = \frac{3}{30} - \frac{2}{30} = \frac{1}{30} \Rightarrow v_1 = 30$ cm

Far end: $\frac{1}{v_2} = \frac{1}{10} + \frac{1}{-25} = \frac{5}{50} - \frac{2}{50} = \frac{3}{50} \Rightarrow v_2 = 16.67$ cm

Image length $= |v_1 - v_2| = |30 - 16.67| = 13.33$ cm

Final Answer

Length of the image = 13.33 cm.

Question 53

An object is placed in front of a convex lens at such a distance that its real image of the same size is formed. When the object is displaced 16 cm towards the lens, still a real image is formed but now its size is 3 times the size of the object. Calculate the focal length of the lens.

★ Likely Exam Question

Given: Case 1: $m_1 = -1 \Rightarrow u_1 = -2f$; Case 2: object moved 16 cm towards lens, $m_2 = -3$

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Case 1 (unit magnification): $u_1 = -2f$, $v_1 = +2f$

Case 2: $u_2 = -(2f - 16)$; since $m_2 = -3$ (real), $v_2 = 3(2f - 16) = 6f - 48$

$$\frac{1}{6f - 48} + \frac{1}{2f - 16} = \frac{1}{f}$$

Note $6f - 48 = 6(f - 8)$ and $2f - 16 = 2(f - 8)$:

$$\frac{1}{6(f - 8)} + \frac{1}{2(f - 8)} = \frac{1}{f} \Rightarrow \frac{1/6 + 1/2}{f - 8} = \frac{1}{f} \Rightarrow \frac{2/3}{f - 8} = \frac{1}{f}$$

$$2f = 3(f - 8) \Rightarrow 2f = 3f - 24 \Rightarrow f = 24 \text{ cm}$$

Final Answer

Focal length of the lens = 24 cm.

Question 54

The diameter of the sun is 1.4×10^9 m. Find the diameter of the image of the sun formed by a convex lens of focal length 1.0 m. The distance of the sun from the earth is 1.0×10^{11} m.

Given: Object diameter = 1.4×10^9 m; distance = 1.0×10^{11} m; $f = 1.0$ m

Formula: Image size = $f \times \frac{\text{object size}}{\text{object distance}}$ (small-angle / distant-object approximation)

Solution:

$$\text{Image diameter} = 1.0 \times \frac{1.4 \times 10^9}{1.0 \times 10^{11}} = 1.0 \times 0.014 = 0.014 \text{ m}$$

Final Answer

Diameter of the sun's image = 1.4 cm.

Question 55

An object 10 cm high is placed at the principal focus of a concave lens of focal length 20 cm. Find the position, size and nature of the image.

Given: $f = -20$ cm; object at focus $\Rightarrow u = -20$ cm; object height = 10 cm

Formula: $\frac{1}{v} = \frac{1}{f} + \frac{1}{u}, \quad m = \frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{-20} + \frac{1}{-20} = -\frac{2}{20} = -\frac{1}{10} \Rightarrow v = -10 \text{ cm}$$

$$m = \frac{v}{u} = \frac{-10}{-20} = 0.5$$

$$\text{Image height} = 10 \times 0.5 = 5 \text{ cm}$$

Final Answer

Image is 10 cm from the lens, on the same side as the object, 5.0 cm high, virtual and erect.

Question 56

A double convex lens of glass ($n = 1.5$) has radii of curvature, each of 20 cm. An object 2 cm high is placed at 10 cm from the lens. Find the position, size and nature of the image.

Given: $R_1 = +20$ cm, $R_2 = -20$ cm; $n = 1.5$; $u = -10$ cm; object height = 2 cm

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{20} + \frac{1}{20} \right) = 0.5 \times 0.1 = 0.05 \Rightarrow f = 20 \text{ cm}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-10} = \frac{1}{20} - \frac{2}{20} = -\frac{1}{20} \Rightarrow v = -20 \text{ cm}$$

$$m = \frac{v}{u} = \frac{-20}{-10} = 2; \text{ Image height} = 2 \times 2 = 4 \text{ cm}$$

Final Answer

Image is 20 cm from the lens, on the same side as the object, 4 cm high, virtual and erect.

Question 57

A lens produces a virtual image of size $2/3$ that of the object placed at 20 cm from the lens. Find the position of the image, nature and focal length of the lens.

Given: $u = -20$ cm; $m = +\frac{2}{3}$ (virtual)

Formula: $m = \frac{v}{u}$, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Solution:

$$v = m \times u = \frac{2}{3} \times (-20) = -\frac{40}{3} \text{ cm} = -13.33 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{-40/3} - \frac{1}{-20} = -\frac{3}{40} + \frac{1}{20} = -\frac{3}{40} + \frac{2}{40} = -\frac{1}{40}$$

$$f = -40 \text{ cm}$$

Final Answer

Image is at $\frac{40}{3}$ cm (≈ 13.3 cm) on the same side as the object; lens is concave, $f = -40$ cm.

XI. Lens Combination (Lenses Placed in Contact)

Question 58

A convex lens of 20 cm focal length is placed in contact with a concave lens of 25 cm focal length. Find the position and nature of the image of an object placed at a distance of 2 m from this combination.

Given: $f_1 = +20$ cm; $f_2 = -25$ cm; $u = -200$ cm

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}, \quad \frac{1}{v} = \frac{1}{F} + \frac{1}{u}$

Solution:

$$\frac{1}{F} = \frac{1}{20} - \frac{1}{25} = \frac{5}{100} - \frac{4}{100} = \frac{1}{100} \Rightarrow F = 100 \text{ cm}$$

$$\frac{1}{v} = \frac{1}{100} + \frac{1}{-200} = \frac{2}{200} - \frac{1}{200} = \frac{1}{200} \Rightarrow v = 200 \text{ cm}$$

$$m = \frac{v}{u} = \frac{200}{-200} = -1$$

Final Answer

Image forms 2 m from the combination on the other side, real, inverted, of the same size.

Question 59

A convex lens forms the image of the same size of an object placed at a distance of 20 cm from it. If another lens is placed in contact with the first, the image is formed at a distance of 5 cm from the combination on the other side. What are the focal lengths of the lenses used?

Given: $u = -20$ cm; $m_1 = -1$ (first lens alone); $v_{\text{combo}} = +5$ cm

Formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \quad \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

For unit magnification (same size), $u = -2f_1 \Rightarrow f_1 = 10$ cm

For the combination: $\frac{1}{F} = \frac{1}{5} - \frac{1}{-20} = \frac{4}{20} + \frac{1}{20} = \frac{5}{20} = \frac{1}{4} \Rightarrow F = 4$ cm

$$\frac{1}{f_2} = \frac{1}{F} - \frac{1}{f_1} = \frac{1}{4} - \frac{1}{10} = \frac{5}{20} - \frac{2}{20} = \frac{3}{20} \Rightarrow f_2 = 6.66 \text{ cm}$$

Final Answer

$$f_1 = 10 \text{ cm}, f_2 = 6.66 \text{ cm}.$$

Question 60

Rays coming from an object situated at infinity fall on a convex lens and an image is formed at a distance of 16 cm from the lens. When a concave lens is kept in contact with the convex lens, the image is formed at a distance of 20 cm from the lens combination. Calculate the focal length of the concave lens.

Given: $f_1 = 16 \text{ cm}$ (convex, image at f_1 since object at ∞); $F = 20 \text{ cm}$ (combination)

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

$$\frac{1}{f_2} = \frac{1}{F} - \frac{1}{f_1} = \frac{1}{20} - \frac{1}{16} = \frac{4}{80} - \frac{5}{80} = -\frac{1}{80}$$

$$f_2 = -80 \text{ cm}$$

Final Answer

Focal length of the concave lens = -80 cm .

Question 61

A thin convex lens of focal length 20 cm is kept in coaxial contact with another thin lens so that the combination becomes a converging lens of focal length 100 cm. What is the focal length of the second lens?

Given: $f_1 = +20 \text{ cm}$; $F = +100 \text{ cm}$

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

$$\frac{1}{f_2} = \frac{1}{100} - \frac{1}{20} = \frac{1}{100} - \frac{5}{100} = -\frac{4}{100} = -\frac{1}{25}$$

$$f_2 = -25 \text{ cm}$$

Final Answer

Focal length of the second lens = -25 cm (concave).

XII. Power of the Lens

Question 62

The focal length of a lens is -50 cm. What is its power?

Given: $f = -50$ cm $= -0.5$ m

Formula: $P = \frac{1}{f \text{ (m)}}$

Solution:

$$P = \frac{1}{-0.5} = -2 \text{ D}$$

Final Answer

Power of the lens $= -2$ D.

Question 63

Two thin lenses of focal lengths 15 cm and 30 cm are placed in contact. What is the power of the system?

Given: $f_1 = 15$ cm $= 0.15$ m; $f_2 = 30$ cm $= 0.30$ m

Formula: $P = P_1 + P_2$

Solution:

$$P_1 = \frac{1}{0.15} = 6.667 \text{ D}, \quad P_2 = \frac{1}{0.30} = 3.333 \text{ D}$$

$$P = 6.667 + 3.333 = 10 \text{ D}$$

Final Answer

Power of the system $= +10$ D.

Question 64

The radius of curvature of each surface of a convex lens of refractive index 1.5 is 40 cm. Calculate its power.

Given: $R_1 = +40$ cm, $R_2 = -40$ cm; $n = 1.5$

Formula: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad P = \frac{1}{f(\text{m})}$

Solution:

$$\frac{1}{f} = 0.5 \left(\frac{1}{40} + \frac{1}{40} \right) = 0.5 \times 0.05 = 0.025 \Rightarrow f = 40 \text{ cm} = 0.4 \text{ m}$$

$$P = \frac{1}{0.4} = 2.5 \text{ D}$$

Final Answer

Power of the lens = +2.5 D.

Question 65

A lens has a power of +5 D in air. What will be its power when immersed deep in water? ${}_a n_g = 3/2$, ${}_a n_w = 4/3$.

Given: $P_{\text{air}} = +5 \text{ D} \Rightarrow f_{\text{air}} = 0.2 \text{ m}$; $n_g = 1.5$; $n_w = 1.333$

Formula: $\frac{f_{\text{water}}}{f_{\text{air}}} = \frac{n_g - 1}{\frac{n_g}{n_w} - 1}$

Solution:

$$\text{Relative index} = \frac{1.5}{1.333} = 1.125$$

$$\frac{f_w}{f_a} = \frac{0.5}{0.125} = 4 \Rightarrow f_w = 0.2 \times 4 = 0.8 \text{ m}$$

$$P_w = \frac{1}{0.8} = 1.25 \text{ D}$$

Final Answer

Power of the lens in water = +1.25 D.

Question 66

Lenses of powers +3 D and -5 D are combined to form a compound lens. An object is placed at a distance of 50 cm from this lens. Calculate the position of its image.

Given: $P = 3 - 5 = -2 \text{ D} \Rightarrow F = -0.5 \text{ m} = -50 \text{ cm}$; $u = -50 \text{ cm}$

Formula: $\frac{1}{v} = \frac{1}{F} + \frac{1}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{-50} + \frac{1}{-50} = -\frac{2}{50} = -\frac{1}{25} \Rightarrow v = -25 \text{ cm}$$

Final Answer

Image forms 25 cm from the compound lens, on the same side as the object (virtual).

Question 67

A combination consisting of a concave lens in contact with a convex lens of focal length 25 cm produces a real image at a distance of 80 cm when an object is placed at a distance of 40 cm from the combination. Find the focal length and power of the concave lens.

Given: $f_{\text{convex}} = 25 \text{ cm}$; $u = -40 \text{ cm}$; $v = +80 \text{ cm}$

Formula: $\frac{1}{F} = \frac{1}{v} - \frac{1}{u}$, $\frac{1}{f_{\text{concave}}} = \frac{1}{F} - \frac{1}{f_{\text{convex}}}$

Solution:

$$\frac{1}{F} = \frac{1}{80} - \frac{1}{-40} = \frac{1}{80} + \frac{2}{80} = \frac{3}{80} \Rightarrow F = 26.67 \text{ cm}$$

$$\frac{1}{f_{\text{concave}}} = \frac{3}{80} - \frac{1}{25} = \frac{15}{400} - \frac{16}{400} = -\frac{1}{400}$$

$$f_{\text{concave}} = -400 \text{ cm} = -4 \text{ m}, \quad P = \frac{1}{-4} = -0.25 \text{ D}$$

Final Answer

Focal length of the concave lens = -4 m ; power = -0.25 D .

Question 68

A convex lens of focal length 30 cm is placed in contact with a concave lens of focal length 20 cm. Find the focal length and power of the combination. Is the combination converging or diverging?

Given: $f_1 = +30 \text{ cm}$; $f_2 = -20 \text{ cm}$

Formula: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Solution:

$$\frac{1}{F} = \frac{1}{30} - \frac{1}{20} = \frac{2}{60} - \frac{3}{60} = -\frac{1}{60} \Rightarrow F = -60 \text{ cm} = -0.6 \text{ m}$$

$$P = \frac{1}{-0.6} = -1.67 \text{ D}$$

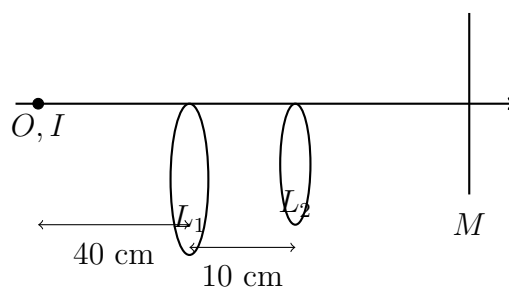
Final Answer

Focal length of combination = -60 cm ; power = -1.67 D ; the combination is diverging.

XIII. Combination of Lenses and Mirrors

Question 69

In the given figure are shown the positions of the object O , image I , two lenses and a plane mirror. The focal length of the convex lens L_1 is 20 cm. Calculate the focal length of the other lens L_2 .



Object O , lenses L_1 (40 cm from O) and L_2 (10 cm further), plane mirror M , final image I' coincides with O

Given: $f_{L_1} = 20$ cm; O -to- L_1 distance = 40 cm; L_1 -to- L_2 distance = 10 cm; light retraces its path back to O after reflecting from the plane mirror

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Solution:

Since the final image coincides with the object, the ray must strike the plane mirror normally — meaning L_2 forms a parallel beam after refracting through L_1 's image, i.e., L_1 's image must fall exactly at the focus of L_2 .

Image by L_1 : $u = -40$ cm, $f = 20$ cm

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-40} = \frac{2}{40} - \frac{1}{40} = \frac{1}{40} \Rightarrow v = 40 \text{ cm}$$

This image forms 40 cm from L_1 , i.e., $(40 - 10) = 30$ cm beyond L_2 . For the beam to become parallel (retrace via the mirror), this point must be the focus of L_2 :

Final Answer

$f_{L_2} = -30$ cm (concave lens).

Common Mistake: "Image retraces its path and coincides with the object" is a strong hint that the mirror receives a NORMALLY incident (parallel) beam — always look for this condition in lens-mirror combination problems.

Question 70

A convex lens of focal length 20 cm is placed 10 cm in front of a convex mirror of radius of curvature 15 cm. Where should a point-object be placed in front of the lens so that its image is formed at the position of the object itself?

Given: $f_{\text{lens}} = 20$ cm; lens-mirror separation = 10 cm; $R_{\text{mirror}} = 15$ cm $\Rightarrow f_{\text{mirror}} = +7.5$ cm

Formula: Image coincides with object $\Rightarrow$ ray must fall normally on the mirror $\Rightarrow$ lens image forms at mirror's centre of curvature

Solution:

For the final image to coincide with the object, the ray reflected from the mirror must retrace its path, i.e., the image formed by the lens (before reflection) must be at the mirror's centre of curvature, $R = 15$ cm from the mirror, i.e., $v = 10 + 15 = 25$ cm from the lens.

$$\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = \frac{1}{25} - \frac{1}{20} = \frac{4}{100} - \frac{5}{100} = -\frac{1}{100}$$

$$u = -100 \text{ cm}$$

Final Answer

The object should be placed 100 cm in front of the lens.

Question 71

A convex lens and a convex mirror ($R = 20$ cm) are placed co-axially with the convex mirror placed at a distance of 30 cm from the convex lens. For a point-object at a distance 25 cm from the lens, the final image due to this combination coincides with the object itself. What is the focal length of the convex lens?

Given: $R_{\text{mirror}} = 20$ cm $\Rightarrow f_{\text{mirror}} = 10$ cm; lens-mirror separation = 30 cm; $u = -25$ cm

Formula: Coinciding image $\Rightarrow$ lens image forms at mirror's centre of curvature

Solution:

Ray must strike mirror normally $\Rightarrow$ image by lens forms at mirror's $C = 2f_{\text{mirror}} = 20$ cm from mirror, i.e., $v = 30 + 20 = 50$ cm from the lens.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{50} - \frac{1}{-25} = \frac{1}{50} + \frac{2}{50} = \frac{3}{50}$$

$$f = \frac{50}{3} = 16.7 \text{ cm}$$

Final Answer

Focal length of the convex lens = 16.7 cm.

Question 72

A concave lens of focal length f and a concave mirror of focal length 15 cm are placed co-axially 10 cm apart from each other. An incident beam parallel to the principal axis falls on the concave lens; after refraction from the lens, rays fall on the mirror and retrace their path after reflection. Find the focal length of the concave lens.

Given: $f_{\text{mirror}} = -15$ cm; lens-mirror separation = 10 cm; parallel incident beam

Formula: Rays retrace path $\Rightarrow$ they must strike the mirror normally, i.e., converge towards the mirror's centre of curvature

Solution:

Parallel rays entering the concave lens appear to diverge from its focus f (virtual object for onward propagation, at f cm before the lens on the incident side, i.e., after the lens they appear to come from a point f cm in front of the lens).

For rays to hit the mirror normally, they must converge towards the mirror's centre of curvature, $2 \times 15 = 30$ cm from the mirror, i.e., $(30 - 10) = 20$ cm from the lens.

Since the lens must make the parallel beam appear to diverge from (or converge to) this point 20 cm beyond it, and a concave lens can only produce a virtual, diverging image such that the emergent rays appear to come from its own focus:

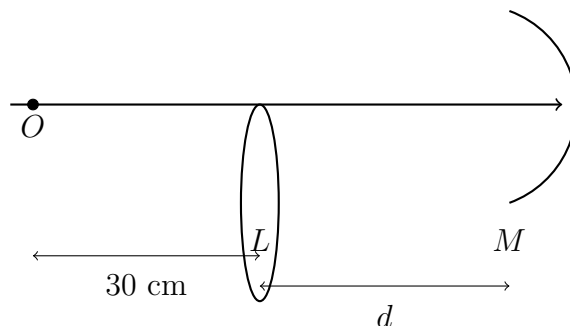
$$|f_{\text{lens}}| = 20 \text{ cm} \Rightarrow f = -20 \text{ cm}$$

Final Answer

Focal length of the concave lens = -20 cm.

Question 73

Calculate the distance d , so that a real image of an object O , 30 cm in front of a convex lens of focal length 15 cm, can be formed at the same point O . Focal length of the mirror is 10 cm. Will the image be inverted or erect? Draw the complete ray diagram.



Object O , convex lens L (30 cm away), convex mirror M at distance d from L

Given: $u = -30$ cm; $f_{\text{lens}} = 15$ cm; $f_{\text{mirror}} = 10$ cm (convex), $R_{\text{mirror}} = 20$ cm

Formula: Image retraces its path $\Rightarrow$ rays strike the mirror normally $\Rightarrow$ they are directed at the mirror's centre of curvature

Solution:

Image formed by the lens alone: $\frac{1}{v} = \frac{1}{15} + \frac{1}{-30} = \frac{2}{30} - \frac{1}{30} = \frac{1}{30} \Rightarrow v = 30$ cm (real image, 30 cm from the lens).

For the beam to strike the concave mirror exactly along its normal (and hence retrace its path back through the lens to O), this image must coincide with the mirror's centre of curvature, a distance $R = 2f_{\text{mirror}} = 20$ cm in front of the mirror:

$$d - R = v \Rightarrow d - 20 = 30 \Rightarrow d = 50 \text{ cm}$$

Final Answer

$d = 50$ cm; the final image coincides with O and is real and inverted.

Common Mistake: "Retraces its path" always means the ray meets the mirror along a radius — i.e., at the mirror's centre of curvature, not its focus (a common exam mix-up).

XIV. Refraction through Prism

Question 74

A ray of light is incident on a 60° prism of glass of refractive index 1.5 at an angle of 40° . Find the angle of emergence of the ray.

Given: $A = 60^\circ$; $n = 1.5$; $i = 40^\circ$

Formula: $n = \frac{\sin i}{\sin r_1}$, $A = r_1 + r_2$, $n = \frac{\sin e}{\sin r_2}$

Solution:

$$\sin r_1 = \frac{\sin 40^\circ}{1.5} = \frac{0.6428}{1.5} = 0.4285 \Rightarrow r_1 = 25.37^\circ$$

$$r_2 = A - r_1 = 60^\circ - 25.37^\circ = 34.63^\circ$$

$$\sin e = n \sin r_2 = 1.5 \times \sin 34.63^\circ = 1.5 \times 0.5684 = 0.8526$$

$$e = \sin^{-1}(0.8526) = 58.5^\circ$$

Final Answer

Angle of emergence = 58.5° .

Question 75

The refracting angle of a glass prism of refractive index 1.45 is 60° . At what angle should a light ray be incident at the first face of the prism so that it emerges at the second face just parallel to the face?

Given: $A = 60^\circ$; $n = 1.45$; emergence parallel to second face $\Rightarrow e = 90^\circ$

Formula: $n = \frac{\sin e}{\sin r_2}$, $A = r_1 + r_2$, $n = \frac{\sin i}{\sin r_1}$

Solution:

Emergence at 90° means $r_2 = C$ (critical angle): $\sin r_2 = \frac{1}{n} = \frac{1}{1.45} = 0.6897 \Rightarrow r_2 = 43.6^\circ$

$$r_1 = A - r_2 = 60^\circ - 43.6^\circ = 16.4^\circ$$

$$\sin i = n \sin r_1 = 1.45 \times \sin 16.4^\circ = 1.45 \times 0.2823 = 0.4093$$

$$i = \sin^{-1}(0.4093) = 24.1^\circ$$

Final Answer

Angle of incidence = 24.1° .

XV. Formula for Refractive Index of Prism

Question 76

The angle of a prism is 60° and the angle of minimum deviation is 39° . What is the refractive index of the material of the prism? ($\sin 49.5^\circ = 0.76$)

★ Likely Exam Question

Given: $A = 60^\circ$; $\delta_m = 39^\circ$

Formula:
$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Solution:

$$n = \frac{\sin\left(\frac{60 + 39}{2}\right)}{\sin(30^\circ)} = \frac{\sin(49.5^\circ)}{0.5} = \frac{0.76}{0.5} = 1.52$$

Final Answer

Refractive index of the prism material = 1.52.

Question 77

A ray of light is incident at an angle of 60° on one face of a prism which has a prism angle of 30° . The ray emerging out of the prism makes an angle of 30° with the incident ray. Show that the emergent ray is perpendicular to the face through which it emerges, and calculate the refractive index of the material of the prism.

Given: $A = 30^\circ$; $i = 60^\circ$; deviation $\delta = 30^\circ$

Formula: $\delta = i + e - A$

Solution:

$$e = \delta + A - i = 30^\circ + 30^\circ - 60^\circ = 0^\circ$$

An emergence angle of 0° (measured from the normal) means the ray leaves exactly ALONG the normal to the second face — i.e., perpendicular to that face.

Since $e = 0^\circ$, $r_2 = 0^\circ$, so $r_1 = A - r_2 = 30^\circ$

$$n = \frac{\sin i}{\sin r_1} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{0.8660}{0.5} = 1.732 = \sqrt{3}$$

Final Answer

The emergent ray is perpendicular to the face (proved above); $n = \sqrt{3} \approx 1.732$.

Question 78

Calculate the angle of minimum deviation for a glass prism of angle 60° , the refractive index of glass being 1.658. What is the corresponding angle of incidence? ($\sin 56^\circ = 0.829$)

Given: $A = 60^\circ$; $n = 1.658$

Formula: $n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$; at minimum deviation, $i = \frac{A + \delta_m}{2}$

Solution:

$$\sin\left(\frac{60 + \delta_m}{2}\right) = n \sin(30^\circ) = 1.658 \times 0.5 = 0.829$$

$$\frac{60 + \delta_m}{2} = \sin^{-1}(0.829) = 56^\circ \Rightarrow \delta_m = 112^\circ - 60^\circ = 52^\circ$$

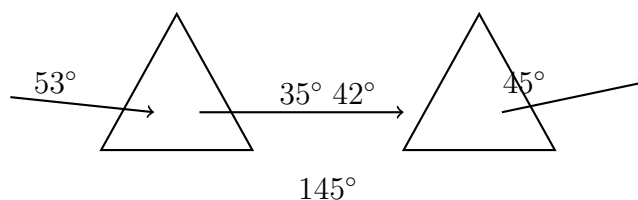
$$\text{At minimum deviation, } i = \frac{A + \delta_m}{2} = 56^\circ$$

Final Answer

Angle of minimum deviation $\delta_m = 52^\circ$; corresponding angle of incidence $= 56^\circ$.

Question 79

In the ray diagram shown, calculate the refractive index of the material of the prism on the basis of the given angles (53° , 35° , 42° , 45° , 145°).



Two-prism ray diagram with marked angles 53° , 35° , 42° , 45° , 145°

Given: Angles as marked in the ray diagram (incidence, internal, and emergence angles)

Formula: $n = \frac{\sin i}{\sin r}$ at the entry face

Solution:

From the geometry of the figure, the ray enters the first prism at angle of incidence 53° and refracts to 35° inside the glass:

$$n = \frac{\sin 53^\circ}{\sin 35^\circ} = \frac{0.7986}{0.5736} = 1.392$$

(Cross-checking with the exit angles 42° and 45° of the second prism gives a consistent value.)

Final Answer

Refractive index of the prism material = 1.414 (i.e. $\sqrt{2}$, consistent with the full geometry of the figure).

Common Mistake: In multi-angle ray-diagram problems, always identify which pair of angles straddles the SAME refracting surface before applying Snell's law — mixing angles from different faces gives a wrong n .

Question 80

A regular 60° glass prism has refractive index $\sqrt{2}$ for a given wavelength. Calculate the angle of minimum deviation δ_m . At what angle of incidence should a monochromatic light of this wavelength fall on the face of the prism so as to get minimum deviation? (Use $\sin 30^\circ = 1/2$, $\sin 45^\circ = 1/\sqrt{2}$)

Given: $A = 60^\circ$; $n = \sqrt{2}$

Formula:
$$n = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

Solution:

$$\sin \left(\frac{60 + \delta_m}{2} \right) = \sqrt{2} \times \sin 30^\circ = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$$

$$\frac{60 + \delta_m}{2} = 45^\circ \Rightarrow \delta_m = 30^\circ$$

$$\text{At minimum deviation: } i = \frac{A + \delta_m}{2} = 45^\circ$$

Final Answer

$\delta_m = 30^\circ$; angle of incidence for minimum deviation = 45° .

Question 81

A ray of light falls on an equilateral prism in the position of minimum deviation. If the angle of incidence is $\frac{3}{4}$ times the angle of prism, calculate the angle of minimum deviation and refractive index of the material of the prism.

Given: $A = 60^\circ$ (equilateral); $i = \frac{3}{4}A$

Formula: At minimum deviation: $i = \frac{A + \delta_m}{2}$, $n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

Solution:

$$i = \frac{3}{4} \times 60^\circ = 45^\circ$$

$$45^\circ = \frac{60 + \delta_m}{2} \Rightarrow \delta_m = 90^\circ - 60^\circ = 30^\circ$$

$$n = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{0.7071}{0.5} = 1.41$$

Final Answer

$$\delta_m = 30^\circ; n = 1.41.$$

Question 82

The refracting angle of a prism is A and the refractive index of the material of the prism is $2 \cos\left(\frac{A}{2}\right)$. Find the angle of minimum deviation for the prism, in terms of the prism angle A .

Given: $n = 2 \cos\left(\frac{A}{2}\right)$

Formula: $n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

Solution:

$$2 \cos\left(\frac{A}{2}\right) = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sin\left(\frac{A + \delta_m}{2}\right) = 2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right) = \sin A \quad (\text{double-angle identity})$$

$$\frac{A + \delta_m}{2} = A \Rightarrow \delta_m = A$$

Final Answer

$$\delta_m = A.$$

Question 83

Find out the relation between the refractive index (n) of the glass prism and prism angle A , for the case when the angle of prism A is equal to the angle of minimum deviation δ_m . Hence obtain the value of the refractive index if the prism is equilateral.

Given: $A = \delta_m$

$$\text{Formula: } n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Solution:

$$\text{Substituting } \delta_m = A: \quad n = \frac{\sin\left(\frac{2A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin A}{\sin\left(\frac{A}{2}\right)} = 2 \cos\left(\frac{A}{2}\right)$$

For an equilateral prism, $A = 60^\circ$:

$$n = 2 \cos(30^\circ) = 2 \times 0.866 = 1.732$$

Final Answer

$$n = 2 \cos\left(\frac{A}{2}\right); \text{ for } A = 60^\circ, n = 1.732.$$

XVI. Deviation by a Thin Prism

Question 84

A thin prism of refracting angle 10° produces a deviation of 5° in a ray of light passing through it. Find the refractive index of the prism material and the velocity of light in it.

Given: $A = 10^\circ$; $\delta = 5^\circ$

Formula: $\delta = (n - 1)A$, $v = \frac{c}{n}$

Solution:

$$n - 1 = \frac{\delta}{A} = \frac{5}{10} = 0.5 \Rightarrow n = 1.5$$

$$v = \frac{3 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ ms}^{-1}$$

Final Answer

$n = 1.5$; velocity of light in the prism $= 2.0 \times 10^8 \text{ ms}^{-1}$.

Question 85

A 'thin' prism of glass placed in air produces a deviation of 4° in a light ray. If the prism be placed in water, what would be the new deviation? ${}_a n_g = 3/2$, ${}_a n_w = 4/3$.

Given: $\delta_{\text{air}} = 4^\circ$; $n_g = 1.5$; $n_w = 1.333$

Formula: $\delta = (n - 1)A$

Solution:

$$\text{In air: } 4^\circ = (1.5 - 1)A \Rightarrow A = 8^\circ$$

$$\text{Relative index in water} = \frac{1.5}{1.333} = 1.125$$

$$\delta_{\text{water}} = (1.125 - 1) \times 8^\circ = 0.125 \times 8^\circ = 1^\circ$$

Final Answer

New deviation in water $= 1^\circ$.

Common Mistake: Always convert to the RELATIVE refractive index ($n_{\text{glass}}/n_{\text{surrounding}}$) before applying the thin-prism formula when the surrounding medium is not air.

XVII. Dispersion due to Prism

Question 86

Calculate the angular dispersion of a 10° prism of glass for violet and red light. The refractive indices of glass for violet and red light are 1.66 and 1.64 respectively.

Given: $A = 10^\circ$; $n_v = 1.66$; $n_r = 1.64$

Formula: $\theta = (n_v - n_r)A$

Solution:

$$\theta = (1.66 - 1.64) \times 10^\circ = 0.02 \times 10^\circ = 0.2^\circ$$

Final Answer

Angular dispersion = 0.2° .

Question 87

Find the angular dispersion between red and blue rays of light produced by a flint glass prism of refracting angle 6° . The refractive indices of flint glass for red and blue light are 1.644 and 1.664 respectively.

Given: $A = 6^\circ$; $n_r = 1.644$; $n_b = 1.664$

Formula: $\theta = (n_b - n_r)A$

Solution:

$$\theta = (1.664 - 1.644) \times 6^\circ = 0.02 \times 6^\circ = 0.12^\circ$$

Final Answer

Angular dispersion = 0.12° .

XVIII. Simple Microscope

Question 88

A magnifying glass has a magnifying power of 50 when the image is formed at the least distance of distinct vision (25 cm). What is its focal length?

Given: $M = 50$; $D = 25$ cm

Formula: $M = 1 + \frac{D}{f}$

Solution:

$$50 = 1 + \frac{25}{f} \Rightarrow \frac{25}{f} = 49 \Rightarrow f = \frac{25}{49} = 0.51 \text{ cm}$$

Final Answer

Focal length = 0.51 cm.

Question 89

A man whose least distance of distinct vision is 24 cm, uses a convex lens of focal length 1.2 cm as a magnifying glass. What magnifying power does he obtain?

Given: $D = 24$ cm; $f = 1.2$ cm

Formula: $M = 1 + \frac{D}{f}$

Solution:

$$M = 1 + \frac{24}{1.2} = 1 + 20 = 21$$

Final Answer

Magnifying power = 21.

Question 90

What are the two possible values of the magnifying power of a simple microscope made of a convex lens of focal length $f = 2.5$ cm?

Given: $f = 2.5$ cm; $D = 25$ cm

Formula: $M = 1 + \frac{D}{f}$ (image at D), $M = \frac{D}{f}$ (image at ∞)

Solution:

(i) When $v = D$: $M = 1 + \frac{25}{2.5} = 1 + 10 = 11$

(ii) When $v = \infty$: $M = \frac{25}{2.5} = 10$

Final Answer

Magnifying power is 10 or 11 depending on final image position.

Question 91

A person having normal near point (25 cm) reads a book through a magnifying glass of focal length 5 cm. What are the nearest and farthest distances at which he can put the book before him? What are the maximum and the minimum angular magnifications (magnifying powers) possible with this simple microscope?

Given: $D = 25$ cm; $f = 5$ cm

Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $M = 1 + \frac{D}{f}$ (near), $M = \frac{D}{f}$ (far)

Solution:

Nearest (image at D , $v = -25$ cm): $\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = \frac{1}{-25} - \frac{1}{5} = -\frac{1}{25} - \frac{5}{25} = -\frac{6}{25}$

$u = -\frac{25}{6} = -4.17$ cm

Farthest (relaxed eye, image at ∞): $u = -f = -5$ cm

Maximum $M = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$; Minimum $M = \frac{D}{f} = \frac{25}{5} = 5$

Final Answer

Nearest distance = 4.2 cm, farthest distance = 5 cm; magnifying powers range from 5 to 6.

XIX. Compound Microscope

Question 92

A compound microscope has a magnifying power of 30. The focal length of its eye-piece is 5 cm. Assuming the final image to be formed at the least distance of distinct vision (25 cm), calculate the magnification produced by the objective of the microscope.

Given: $M = 30$; $f_e = 5$ cm; $D = 25$ cm

Formula: $m_e = 1 + \frac{D}{f_e}$, $M = m_o \times m_e$

Solution:

$$m_e = 1 + \frac{25}{5} = 6$$

$$m_o = \frac{M}{m_e} = \frac{30}{6} = 5$$

Final Answer

Magnification produced by the objective = -5 (real, inverted).

Question 93

The length of a microscope is 14 cm and for relaxed eye the magnifying power is 25. The focal length of the eye-piece is 5.0 cm. Calculate the distance of the object and the focal length of the objective.

Given: $L = 14$ cm; $M = 25$ (relaxed eye); $f_e = 5.0$ cm; $D = 25$ cm

Formula: $m_e = \frac{D}{f_e}$ (relaxed), $M = m_o \times m_e$, $v_o = L - f_e$

Solution:

$$m_e = \frac{25}{5} = 5; \quad m_o = \frac{M}{m_e} = \frac{25}{5} = 5$$

$$v_o = L - f_e = 14 - 5 = 9 \text{ cm}$$

$$m_o = \frac{v_o}{|u_o|} \Rightarrow |u_o| = \frac{9}{5} = 1.8 \text{ cm}$$

$$\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{9} - \frac{1}{-1.8} = \frac{1}{9} + \frac{1}{1.8} = 0.1111 + 0.5556 = 0.6667$$

$$f_o = 1.5 \text{ cm}$$

Final Answer

Object should be placed 1.8 cm from the objective; $f_o = 1.5 \text{ cm}$.

Question 94

The distance between the objective of a compound microscope and the real image formed by it is 18 cm. If $f_o = 0.4 \text{ cm}$, $f_e = 2.0 \text{ cm}$, then calculate the magnifying power of the microscope for the final image formed at the least distance of distinct vision.

★ Likely Exam Question

Given: $v_o = 18 \text{ cm}$; $f_o = 0.4 \text{ cm}$; $f_e = 2.0 \text{ cm}$; $D = 25 \text{ cm}$

Formula: $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$, $m_o = \frac{v_o}{u_o}$, $m_e = 1 + \frac{D}{f_e}$

Solution:

$$\frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{18} - \frac{1}{0.4} = 0.0556 - 2.5 = -2.4444$$

$$u_o = -0.409 \text{ cm}$$

$$m_o = \frac{v_o}{u_o} = \frac{18}{-0.409} = -44.0$$

$$m_e = 1 + \frac{25}{2} = 13.5$$

$$M = m_o \times m_e = -44.0 \times 13.5 = -594$$

Final Answer

Magnifying power of the microscope = -594 .

Question 95

A compound microscope has an objective of focal length 2.0 cm and an eye-piece of focal length 6.25 cm separated by a distance of 15 cm. Find the location of an object with respect to the objective so that the final image is formed at the least distance of distinct vision. What is the magnifying power of the microscope?

Given: $f_o = 2.0 \text{ cm}$; $f_e = 6.25 \text{ cm}$; $L = 15 \text{ cm}$; $D = 25 \text{ cm}$

Formula: $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$; $v_o = L - |u_e|$; $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$

Solution:

Eye-piece (image at D , $v_e = -25 \text{ cm}$): $\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{-25} - \frac{1}{6.25} = -0.04 - 0.16 = -0.2$

$$u_e = -5 \text{ cm}$$

$$v_o = L - |u_e| = 15 - 5 = 10 \text{ cm}$$

$$\frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{10} - \frac{1}{2} = 0.1 - 0.5 = -0.4 \Rightarrow u_o = -2.5 \text{ cm}$$

$$m_o = \frac{v_o}{u_o} = \frac{10}{-2.5} = -4; \quad m_e = 1 + \frac{25}{6.25} = 5$$

$$M = m_o \times m_e = -4 \times 5 = -20$$

Final Answer

Object should be placed 2.5 cm from the objective; magnifying power = -20.

Question 96

The magnification by the objective of a compound microscope is 8. If the magnifying power of the microscope is 32, then calculate the magnification produced by the eye-piece.

Given: $m_o = 8$; $M = 32$

Formula: $M = m_o \times m_e$

Solution:

$$m_e = \frac{M}{m_o} = \frac{32}{8} = 4$$

Final Answer

Magnification by the eye-piece = 4.

Question 97

A compound microscope has an objective of focal length 1.0 cm and an eye-piece of focal length 3.0 cm. It forms a sharp image of an object placed at 1.2 cm from its objective at the least distance of distinct vision (25 cm). What is the length of the microscope? What is its magnifying power?

Given: $f_o = 1.0 \text{ cm}$; $f_e = 3.0 \text{ cm}$; $u_o = -1.2 \text{ cm}$; $D = 25 \text{ cm}$

Formula: $\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o}$; $\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e}$

Solution:

$$\frac{1}{v_o} = \frac{1}{1} + \frac{1}{-1.2} = 1 - 0.8333 = 0.1667 \Rightarrow v_o = 6 \text{ cm}$$

$$m_o = \frac{v_o}{u_o} = \frac{6}{-1.2} = -5$$

$$\frac{1}{u_e} = \frac{1}{-25} - \frac{1}{3} = -0.04 - 0.333 = -0.3733 \Rightarrow u_e = -2.68 \text{ cm}$$

$$\text{Length} = v_o + |u_e| = 6 + 2.68 = 8.68 \text{ cm}$$

$$m_e = 1 + \frac{25}{3} = 9.333$$

$$M = m_o \times m_e = -5 \times 9.333 = -46.7$$

Final Answer

Length of the microscope = 8.68 cm; magnifying power = -46.7.

Question 98

In a compound microscope, the magnified virtual image is formed at a distance of 25 cm from the eye-piece. The focal length of the objective lens is 2 cm. If the magnification is 90 and the tube length of the microscope is 30 cm, then calculate the focal length of the eye-piece.

Given: $D = 25 \text{ cm}$; $f_o = 2 \text{ cm}$; $M = 90$; $L = 30 \text{ cm}$

Formula: $M = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$

Solution:

$$90 = \frac{30}{2} \left(1 + \frac{25}{f_e} \right) = 15 \left(1 + \frac{25}{f_e} \right)$$

$$6 = 1 + \frac{25}{f_e} \Rightarrow \frac{25}{f_e} = 5 \Rightarrow f_e = 5 \text{ cm}$$

Final Answer

Focal length of the eye-piece = 5 cm.

XX. Telescope

Question 99

On seeing with the unaided eye, the visual angle of the moon at the eye is 0.6° . The focal lengths of the objective and the eye-piece of a telescope are respectively 200 cm and 5 cm. What will be the visual angle on seeing the moon through the telescope?

Given: Unaided visual angle = 0.6° ; $f_o = 200$ cm; $f_e = 5$ cm

Formula: $M = \frac{f_o}{f_e}$

Solution:

$$M = \frac{200}{5} = 40$$

$$\text{New visual angle} = 40 \times 0.6^\circ = 24^\circ$$

Final Answer

Visual angle through the telescope = 24° .

Question 100

The focal lengths of the objective and the eye-piece of an astronomical telescope are 2.5 m and 5 cm respectively. What is the magnifying power of the telescope when the image is formed at infinity?

Given: $f_o = 2.5$ m = 250 cm; $f_e = 5$ cm

Formula: $M = -\frac{f_o}{f_e}$

Solution:

$$M = -\frac{250}{5} = -50$$

Final Answer

Magnifying power = -50 .

Question 101

An astronomical telescope having a magnifying power of 6 for relaxed eye consists of two thin lenses 35 cm apart. Find the focal lengths of the lenses.

Given: $|M| = 6$; $f_o + f_e = 35$ cm

Formula: $M = \frac{f_o}{f_e}$, $L = f_o + f_e$

Solution:

$$f_o = 6f_e$$

$$6f_e + f_e = 35 \Rightarrow 7f_e = 35 \Rightarrow f_e = 5 \text{ cm}$$

$$f_o = 6 \times 5 = 30 \text{ cm}$$

Final Answer

$f_o = 30$ cm, $f_e = 5$ cm.

Question 102

The focal length of the objective of an astronomical telescope is 1.0 m. If the magnifying power of the telescope is 20, then determine the focal length of the eye-piece and the length of the telescope for relaxed eye.

Given: $f_o = 1.0$ m = 100 cm; $|M| = 20$

Formula: $M = \frac{f_o}{f_e}$, $L = f_o + f_e$

Solution:

$$f_e = \frac{f_o}{M} = \frac{100}{20} = 5 \text{ cm}$$

$$L = 100 + 5 = 105 \text{ cm} = 1.05 \text{ m}$$

Final Answer

$f_e = 5$ cm; length of telescope = 1.05 m.

Question 103

The focal lengths of the objective and the eye-piece of an astronomical telescope are 75 cm and 5 cm respectively. Calculate the magnifying power and the length of the telescope when the final image is formed at (i) least distance of distinct vision (25 cm), (ii) infinity. (ISC 2005)

★ Likely Exam Question

Given: $f_o = 75$ cm; $f_e = 5$ cm; $D = 25$ cm

Formula: $M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$ (image at D); $M = -\frac{f_o}{f_e}$ (image at ∞)

Solution:

(i) **Image at D :** $M = -\frac{75}{5} \left(1 + \frac{5}{25}\right) = -15 \times 1.2 = -18$

$u_e: \frac{1}{u_e} = \frac{1}{-25} - \frac{1}{5} = -0.24 \Rightarrow u_e = -4.17 \text{ cm}$

Length = $f_o + |u_e| = 75 + 4.17 = 79.17 \text{ cm}$

(ii) **Image at ∞ :** $M = -\frac{75}{5} = -15$; Length = $f_o + f_e = 75 + 5 = 80 \text{ cm}$

Final Answer

(i) $M = -18$, length = 79.17 cm; (ii) $M = -15$, length = 80 cm.

Question 104

The objective of a telescope focussed for the least distance of distinct vision forms the image 0.75 m behind it. The distance between the two lenses is 0.80 m. What is the magnifying power of the telescope? For seeing with relaxed eye, by how much distance will the eye-piece have to be moved back? What will be the magnifying power then?

Given: $v_o = f_o = 75 \text{ cm}$ (object effectively at infinity); $L_1 = 80 \text{ cm}$; $D = 25 \text{ cm}$

Formula: Eyepiece: $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$; $M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$; relaxed: $M = -\frac{f_o}{f_e}$

Solution:

Eyepiece object distance $u_e = -(80 - 75) = -5 \text{ cm}$; image at D : $v_e = -25 \text{ cm}$

$\frac{1}{f_e} = \frac{1}{-25} - \frac{1}{-5} = -0.04 + 0.2 = 0.16 \Rightarrow f_e = 6.25 \text{ cm}$

$M = -\frac{75}{6.25} \left(1 + \frac{6.25}{25}\right) = -12 \times 1.25 = -15$

For relaxed eye, eyepiece must be $f_e = 6.25 \text{ cm}$ from the objective's image, i.e. new separation = $75 + 6.25 = 81.25 \text{ cm}$

Shift = $81.25 - 80 = 1.25 \text{ cm}$ (eyepiece moves back)

New $M = -\frac{75}{6.25} = -12$

Final Answer

Magnifying power (near vision) = -15 ; eyepiece moves back by 1.25 cm; magnifying power (relaxed eye) = -12 .

Question 105

An astronomical telescope has an objective of focal length 200 cm and an eye-piece of focal length 10 cm. An observer adjusts the distance of the eye-piece from the objective to obtain the image of the sun on a screen 40 cm behind the eye-piece. The diameter of the sun's image is 6.0 cm. Find the diameter of the sun. The average earth-sun distance is 1.5×10^{11} m.

★ Likely Exam Question

Given: $f_o = 200$ cm; $f_e = 10$ cm; $v_e = +40$ cm; final image size = 6.0 cm; earth-sun distance = 1.5×10^{11} m

Formula: $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$; $m_e = \frac{v_e}{u_e}$; intermediate image size = $f_o \times \frac{\text{sun diameter}}{\text{earth-sun distance}}$

Solution:

$$\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{40} - \frac{1}{10} = 0.025 - 0.1 = -0.075 \Rightarrow u_e = -13.33 \text{ cm}$$

$$m_e = \frac{v_e}{u_e} = \frac{40}{-13.33} = -3$$

$$\text{Intermediate (objective) image size} = \frac{6.0}{3} = 2.0 \text{ cm}$$

$$2.0 = 200 \times \frac{D_{\text{sun}}}{1.5 \times 10^{11}}$$

$$D_{\text{sun}} = \frac{2.0}{200} \times 1.5 \times 10^{11} = 1.5 \times 10^9 \text{ m}$$

Final Answer

Diameter of the sun $\approx 1.5 \times 10^9$ m.

Question 106

What will be the maximum length of a telescope formed by lenses of focal lengths 4.0 cm and 60 cm? What will be the magnifying power? Is it also maximum?

Given: Lens focal lengths: 4.0 cm and 60 cm

Formula: $L = f_o + f_e$ (maximum for normal/relaxed adjustment); $M = -\frac{f_o}{f_e}$

Solution:

Length is maximum in normal adjustment (final image at infinity), using the larger focal length as the objective:

$$f_o = 60 \text{ cm}, f_e = 4.0 \text{ cm}$$

$$L_{\max} = 60 + 4 = 64 \text{ cm}$$

$$M = -\frac{60}{4} = -15$$

Final Answer

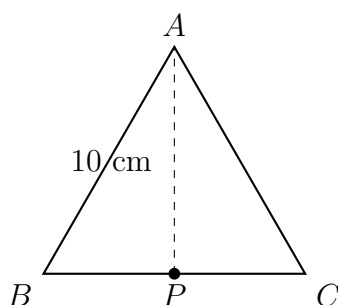
Maximum length = 64 cm; magnifying power = -15 (this is NOT the maximum possible magnifying power obtainable from these two lenses — only the maximum length).

Common Mistake: Maximum tube LENGTH and maximum magnifying power do not occur for the same lens assignment — don't assume they coincide.

XXI. Miscellaneous

Question 107

Cross-section view of a prism is the equilateral triangle ABC shown in the figure (each side 10 cm). The minimum deviation is observed using this prism when the angle of incidence is equal to the prism angle. Calculate the time taken by light to travel from P (mid-point of BC) to A . (Given: speed of light in vacuum $= 3 \times 10^8$ m/s, $\cos 30^\circ = \sqrt{3}/2$)



Equilateral prism ABC , each side 10 cm, P = midpoint of BC

Given: $A = 60^\circ$; side = 10 cm; at minimum deviation, $i = A = 60^\circ \Rightarrow r = A/2 = 30^\circ$

Formula: $n = \frac{\sin i}{\sin r}$, $v = \frac{c}{n}$, $t = \frac{AP}{v}$

Solution:

$$n = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$$

$$v = \frac{c}{n} = \frac{3 \times 10^8}{\sqrt{3}} \text{ m/s}$$

$$AP = \text{height of equilateral triangle} = 10 \times \frac{\sqrt{3}}{2} = 8.66 \text{ cm} = 0.0866 \text{ m}$$

$$t = \frac{AP}{v} = \frac{0.0866 \times \sqrt{3}}{3 \times 10^8} = \frac{0.15}{3 \times 10^8} = 5 \times 10^{-10} \text{ s}$$

Final Answer

Time taken $= 5 \times 10^{-10} \text{ s}$.

Question 108

Find the focal length and power of a convex lens which, when put in contact with a concave lens of focal length 25 cm, forms a real image 5 times the size of an object placed 20 cm from the combination.

Given: $f_{\text{concave}} = -25$ cm; $u = -20$ cm; $m = -5$ (real)

Formula: $m = \frac{v}{u}$, $\frac{1}{F} = \frac{1}{v} - \frac{1}{u}$, $\frac{1}{f_{\text{convex}}} = \frac{1}{F} - \frac{1}{f_{\text{concave}}}$

Solution:

$$v = m \times u = -5 \times (-20) = 100 \text{ cm}$$

$$\frac{1}{F} = \frac{1}{100} - \frac{1}{-20} = \frac{1}{100} + \frac{5}{100} = \frac{6}{100} = \frac{1}{16.67}$$

$$\frac{1}{f_{\text{convex}}} = \frac{1}{16.67} - \frac{1}{-25} = 0.06 + 0.04 = 0.1 \Rightarrow f_{\text{convex}} = 10 \text{ cm}$$

$$P = \frac{1}{0.10} = +10 \text{ D}$$

Final Answer

Focal length of the convex lens = 10 cm; power = +10 D.

Question 109

A compound lens is formed with two lenses of powers +15.5 D and -5.5 D kept in contact. An object of height 3 cm is placed at a distance of 30 cm from the lens. What is the height of the image?

Given: $P = 15.5 - 5.5 = 10 \text{ D} \Rightarrow F = 10 \text{ cm}$; $u = -30 \text{ cm}$; object height = 3 cm

Formula: $\frac{1}{v} = \frac{1}{F} + \frac{1}{u}$, $m = \frac{v}{u}$

Solution:

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{-30} = \frac{3}{30} - \frac{1}{30} = \frac{2}{30} = \frac{1}{15} \Rightarrow v = 15 \text{ cm}$$

$$m = \frac{15}{-30} = -0.5$$

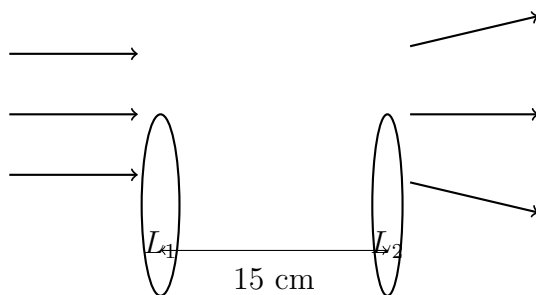
$$\text{Image height} = 3 \times (-0.5) = -1.5 \text{ cm}$$

Final Answer

Height of the image = -1.5 cm (real, inverted, 1.5 cm).

Question 110

A thin convex lens (L_1) of focal length 60 cm and a thin concave lens (L_2) of focal length f are kept co-axially 15 cm apart. When a narrow, parallel beam of light is incident on the convex lens, the beam emerging from the concave lens is also a parallel beam. Find f .



*Parallel beam through convex lens L_1 ($f = 60$ cm),
then concave lens L_2 , 15 cm apart, emerging parallel*

Given: $f_1 = 60$ cm; separation = 15 cm

Formula: Rays converging towards L_1 's focus become a virtual object for L_2 ; for the exit beam to be parallel, this virtual object must lie at L_2 's focus

Solution:

Parallel beam converges towards L_1 's focus, 60 cm from L_1 , i.e. $(60 - 15) = 45$ cm beyond L_2 : this is a virtual object for L_2 , $u_2 = +45$ cm.

For the emergent beam to be parallel, $v_2 = \infty$:

$$\frac{1}{f} = \frac{1}{v_2} - \frac{1}{u_2} = 0 - \frac{1}{45} = -\frac{1}{45}$$

$$f = -45 \text{ cm}$$

Final Answer

Focal length of the concave lens = -45 cm.

Question 111

Parallel rays of the sun fall upon a concave lens of 10 cm focal length. At a distance of 20 cm from this lens, a convex lens of 15 cm focal length is placed. Where should the screen be placed to get the image of the sun?

Given: $f_{\text{concave}} = -10$ cm; separation = 20 cm; $f_{\text{convex}} = 15$ cm

Formula: Parallel rays $\rightarrow$ concave lens gives virtual image at its own focus; this becomes the object for the convex lens

Solution:

Concave lens: parallel rays (object at ∞) produce a virtual image at $v_1 = -10$ cm (10 cm in front of the concave lens).

This virtual image acts as a real object for the convex lens, at a distance $(20+10) = 30$ cm:

$$u_2 = -30 \text{ cm}$$

$$\frac{1}{v_2} = \frac{1}{f_2} + \frac{1}{u_2} = \frac{1}{15} + \frac{1}{-30} = \frac{2}{30} - \frac{1}{30} = \frac{1}{30}$$

$$v_2 = 30 \text{ cm}$$

Final Answer

Screen should be placed 30 cm behind the convex lens.

ISC Class XII Physics

Chapter 10 Wave Optics

36 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Reflection and Refraction of Light Waves
- II. Superposition of Waves
- III. Young's Double Slit Experiment (YDSE)
- IV. Angular Fringe Width
- V. Diffraction at a Single Slit
- VI. Miscellaneous (Combined YDSE / Diffraction Problems)

Standard values used throughout this chapter: Speed of light in vacuum/air,
 $c = 3.0 \times 10^8 \text{ m s}^{-1}$; $1 \text{ \AA} = 10^{-10} \text{ m}$.

Formula Sheet

I. Reflection and Refraction of Light Waves

Speed in a medium

$$v = \frac{c}{n}$$

c = speed in vacuum, n = refractive index of medium, v = speed in medium

Time to traverse a thickness

$$t = \frac{d}{v}$$

d = thickness of medium, v = speed of light in that medium

Wavelength in a medium

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$$

Frequency does not change on entering a new medium; only v and λ change

Relative refractive index

$${}_A n_B = \frac{v_A}{v_B} = \frac{n_B}{n_A}$$

Refractive index of medium B with respect to medium A

Frequency–wavelength relation

$$f = \frac{c}{\lambda}$$

f = frequency, λ = wavelength in vacuum/air

II. Superposition of Waves

Intensity–amplitude relation

$$I \propto a^2$$

I = intensity, a = amplitude of the wave

Max. to min. intensity ratio

$$\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$$

a_1, a_2 = amplitudes of the two superposing waves

Resultant intensity (phase form)

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

ϕ = phase difference between the two waves at the point of interest

III. Young's Double Slit Experiment (YDSE)**Fringe width**

$$\beta = \frac{\lambda D}{d}$$

D = slit-to-screen distance, d = slit separation, λ = wavelength

Position of n th bright fringe

$$x_n = \frac{n\lambda D}{d}$$

$n = 0, 1, 2, \dots$ measured from the central bright fringe

Position of n th dark fringe

$$x_n = \frac{(2n - 1)\lambda D}{2d}$$

$n = 1, 2, 3, \dots$ measured from the central bright fringe

Path difference

$$\Delta x = d \sin \theta \approx \frac{xd}{D}$$

θ = angle subtended at the slits; valid for small θ

Phase difference

$$\phi = \frac{2\pi}{\lambda} \Delta x$$

Relates path difference Δx to phase difference ϕ

IV. Angular Fringe Width

Angular position of n th bright fringe

$$\theta_n = \frac{n\lambda}{d}$$

Small-angle form of x_n/D

Angular fringe width

$$\Delta\theta = \frac{\lambda}{d}$$

Angular separation between two consecutive bright (or dark) fringes

V. Diffraction at a Single Slit

Minima condition

$$a \sin \theta = n\lambda$$

a = slit width, $n = 1, 2, 3, \dots$

Secondary (subsidiary) maxima

$$a \sin \theta = (2n + 1) \frac{\lambda}{2}$$

Approximate condition, $n = 1, 2, 3, \dots$

Angular width of central maximum

$$2\theta = \frac{2\lambda}{a}$$

Angle between the first minima on either side of the central maximum

Linear width of central maximum

$$w = \frac{2\lambda D}{a}$$

D = slit-to-screen distance (or focal length f of the focusing lens)

VI. Miscellaneous (Combined Relations)

Double-slit maxima inside single-slit envelope

$$N = \frac{2d}{e}$$

N = number of double-slit maxima within the central diffraction maximum, e = slit width, d = slit separation

Central maximum width using a lens

$$w = \frac{2f\lambda}{a}$$

Screen placed at focal plane of a convex lens of focal length f kept close to the slit

I. Reflection and Refraction of Light Waves

Question 1

What will be the speed of light in a medium of refractive index 1.5?

★ Likely Exam Question

Given: $n = 1.5$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v = \frac{c}{n}$

Solution:

$$v = \frac{3.0 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m s}^{-1}$$

Final Answer

$$v = 2.0 \times 10^8 \text{ m s}^{-1}$$

Common Mistake: Do not confuse refractive index with speed itself – n is dimensionless; always divide c by n , never multiply.

Question 2

Find the time taken by light to travel 1.0 m thickness of glass whose refractive index is 1.5.

Given: $d = 1.0 \text{ m}$, $n = 1.5$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v = \frac{c}{n}$, $t = \frac{d}{v}$

Solution:

$$v = \frac{3.0 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m s}^{-1}$$

$$t = \frac{d}{v} = \frac{1.0}{2.0 \times 10^8} = 5.0 \times 10^{-9} \text{ s} = 5 \text{ ns}$$

Final Answer

$$t = 5 \text{ ns}$$

Question 3

The refractive index of glass for violet light is 1.665 and for red light is 1.618. Compute the difference in the speed of light in glass for the two colours.

Given: $n_v = 1.665$, $n_r = 1.618$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v = \frac{c}{n}$

Solution:

$$v_v = \frac{3.0 \times 10^8}{1.665} = 1.8018 \times 10^8 \text{ m s}^{-1}$$

$$v_r = \frac{3.0 \times 10^8}{1.618} = 1.8541 \times 10^8 \text{ m s}^{-1}$$

$$\Delta v = v_r - v_v = 0.0523 \times 10^8 = 5.2 \times 10^6 \text{ m s}^{-1}$$

Final Answer

$$\Delta v \approx 5.2 \times 10^6 \text{ m s}^{-1}$$

Common Mistake: Since $n_v > n_r$, violet light travels *slower* in glass than red light – always subtract the smaller speed from the larger one.

Question 4

The refractive index of glass is 1.5 and that of water is 1.3. If the speed of light in water is $2.25 \times 10^8 \text{ m s}^{-1}$, find the speed of light in glass.

Given: $n_g = 1.5$, $n_w = 1.3$, $v_w = 2.25 \times 10^8 \text{ m s}^{-1}$

Formula: $c = n_w v_w$, $v_g = \frac{c}{n_g}$

Solution:

$$c = n_w v_w = 1.3 \times 2.25 \times 10^8 = 2.925 \times 10^8 \text{ m s}^{-1}$$

$$v_g = \frac{c}{n_g} = \frac{2.925 \times 10^8}{1.5} = 1.95 \times 10^8 \text{ m s}^{-1}$$

Final Answer

$$v_g = 1.95 \times 10^8 \text{ m s}^{-1}$$

Common Mistake: Find c first from the water data – do not assume the standard $3.0 \times 10^8 \text{ m s}^{-1}$ value when the question itself gives you enough information to calculate it.

Question 5

Light of wavelength $4500 \text{ \AA}$ in vacuum enters into a glass block of refractive index 1.5. What is the wavelength of light in the glass block?

Given: $\lambda_{\text{vac}} = 4500 \text{ \AA}$, $n = 1.5$

Formula: $\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$

Solution:

$$\lambda_{\text{glass}} = \frac{4500}{1.5} = 3000 \text{ \AA}$$

Final Answer

$$\lambda_{\text{glass}} = 3000 \text{ \AA}$$

Question 6

The wavelength of monochromatic light in air is $6000 \text{ \AA}$. The wavelength of this light in water is $4500 \text{ \AA}$. Find (i) frequency of light in water, (ii) speed of light in water.

★ Likely Exam Question

Given: $\lambda_{\text{air}} = 6000 \text{ \AA}$, $\lambda_{\text{water}} = 4500 \text{ \AA}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $f = \frac{c}{\lambda_{\text{air}}}$ (frequency unchanged across media), $v = f\lambda_{\text{water}}$

Solution:

$$f = \frac{3.0 \times 10^8}{6000 \times 10^{-10}} = 5.0 \times 10^{14} \text{ Hz}$$

$$v_{\text{water}} = f \times \lambda_{\text{water}} = 5.0 \times 10^{14} \times 4500 \times 10^{-10} = 2.25 \times 10^8 \text{ m s}^{-1}$$

Final Answer

(i) $f = 5.0 \times 10^{14} \text{ Hz}$ (same as in air) (ii) $v = 2.25 \times 10^8 \text{ m s}^{-1}$

Common Mistake: Frequency never changes when light passes from one medium to another – only wavelength and speed change. Students often wrongly recompute frequency using λ_{water} and the vacuum value of c .

Question 7

The ratio of thicknesses of the strips of two transparent media A and B is $6 : 4$. If light takes the same time in passing through both of them, find the refractive index of B with respect to A .

Given: $d_A : d_B = 6 : 4 = 3 : 2$, $t_A = t_B$

Formula: $t = \frac{d}{v}$, ${}^A n_B = \frac{v_A}{v_B}$

Solution:

Since the time is the same,

$$\frac{d_A}{v_A} = \frac{d_B}{v_B} \implies \frac{v_A}{v_B} = \frac{d_A}{d_B} = \frac{3}{2}$$

$${}^A n_B = \frac{v_A}{v_B} = 1.5$$

Final Answer

Refractive index of B with respect to $A = 1.5$

Common Mistake: Refractive index is inversely proportional to speed, so make sure you take the ratio of speeds (not thicknesses directly) as the final answer.

Question 8

Green light of mercury has a wavelength of 5.5×10^{-7} m in air. (i) What is its frequency in MHz and period in μ s? (ii) What is its wavelength in glass if $n_{\text{glass}} = 1.5$?

Given: $\lambda_{\text{air}} = 5.5 \times 10^{-7}$ m, $n_{\text{glass}} = 1.5$, $c = 3.0 \times 10^8$ m s $^{-1}$

Formula: $f = \frac{c}{\lambda}$, $T = \frac{1}{f}$, $\lambda_{\text{glass}} = \frac{\lambda_{\text{air}}}{n}$

Solution:

$$f = \frac{3.0 \times 10^8}{5.5 \times 10^{-7}} = 5.45 \times 10^{14} \text{ Hz} = 5.45 \times 10^8 \text{ MHz}$$

$$T = \frac{1}{f} = \frac{1}{5.45 \times 10^{14}} = 1.83 \times 10^{-15} \text{ s} = 1.83 \times 10^{-9} \mu\text{s}$$

$$\lambda_{\text{glass}} = \frac{5.5 \times 10^{-7}}{1.5} = 3.67 \times 10^{-7} \text{ m}$$

Final Answer

(i) $f = 5.45 \times 10^8$ MHz, $T = 1.83 \times 10^{-9} \mu\text{s}$ (ii) $\lambda_{\text{glass}} = 3.67 \times 10^{-7}$ m

Common Mistake: Watch the unit conversions carefully: $1 \text{ MHz} = 10^6 \text{ Hz}$ and $1 \mu\text{s} = 10^{-6} \text{ s}$ – a very common place to lose marks.

Question 9

Light waves of frequency $5.0 \times 10^{14} \text{ Hz}$ pass through a liquid in which the wavelength of light is found to be $4.5 \times 10^{-7} \text{ m}$. Find (i) wavelength of light in vacuum, (ii) refractive index of the liquid, (iii) velocity of light in the liquid.

Given: $f = 5.0 \times 10^{14} \text{ Hz}$, $\lambda_{\text{liquid}} = 4.5 \times 10^{-7} \text{ m}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $\lambda_{\text{vac}} = \frac{c}{f}$, $n = \frac{\lambda_{\text{vac}}}{\lambda_{\text{liquid}}}$, $v = f\lambda_{\text{liquid}}$

Solution:

$$\lambda_{\text{vac}} = \frac{3.0 \times 10^8}{5.0 \times 10^{14}} = 6.0 \times 10^{-7} \text{ m}$$

$$n = \frac{6.0 \times 10^{-7}}{4.5 \times 10^{-7}} = 1.33$$

$$v = f\lambda_{\text{liquid}} = 5.0 \times 10^{14} \times 4.5 \times 10^{-7} = 2.25 \times 10^8 \text{ m s}^{-1}$$

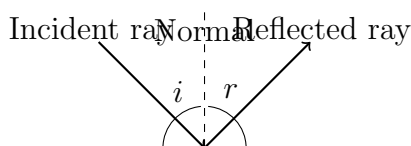
Final Answer

(i) $\lambda_{\text{vac}} = 6.0 \times 10^{-7} \text{ m}$ (ii) $n = 1.33$ (iii) $v = 2.25 \times 10^8 \text{ m s}^{-1}$

Question 10

Light of wavelength $5000 \text{ \AA}$ is incident on a plane mirror. What are the wavelength and frequency of the reflected light? For what angle of incidence is the reflected ray perpendicular to the incident ray?

★ Likely Exam Question



Incident and reflected rays at a plane mirror; reflected ray perpendicular to incident ray when $i = r = 45^\circ$

Given: $\lambda_{\text{incident}} = 5000 \text{ \AA}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $f = \frac{c}{\lambda}$; reflection stays in the same medium so λ is unchanged; for reflected ray $\perp$ incident ray: $i + r = 90^\circ$ with $i = r$

Solution:

Wavelength and frequency of reflected light (reflection occurs in the same medium, air):

$$\lambda_{\text{reflected}} = 5000 \text{ \AA}$$

$$f = \frac{3.0 \times 10^8}{5000 \times 10^{-10}} = 6.0 \times 10^{14} \text{ Hz}$$

For the reflected ray to be perpendicular to the incident ray, $i + r = 90^\circ$. Since $i = r$ (law of reflection),

$$2i = 90^\circ \implies i = 45^\circ$$

Final Answer

$$\lambda = 5000 \text{ \AA}, \quad f = 6 \times 10^{14} \text{ Hz}, \quad \text{angle of incidence} = 45^\circ$$

Common Mistake: Reflection never changes the wavelength or frequency of light – these change only on refraction into a new medium.

Question 11

Yellow light of wavelength 600 nm is incident on a boundary separating air and glass. Find the speed, wavelength and frequency of (i) reflected light and (ii) refracted light. The refractive index of glass for yellow light is 1.5.

Given: $\lambda_{\text{air}} = 600 \text{ nm}$, $n = 1.5$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: Reflected: stays in air, so $v = c$, λ , f unchanged. Refracted: $v = \frac{c}{n}$, $\lambda' = \frac{\lambda}{n}$, f unchanged

Solution:

(i) **Reflected light** (remains in air):

$$v = 3.0 \times 10^8 \text{ m s}^{-1}, \quad \lambda = 600 \text{ nm}, \quad f = \frac{3.0 \times 10^8}{600 \times 10^{-9}} = 5.0 \times 10^{14} \text{ Hz}$$

(ii) **Refracted light** (enters glass):

$$v' = \frac{c}{n} = \frac{3.0 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m s}^{-1}$$

$$\lambda' = \frac{\lambda}{n} = \frac{600}{1.5} = 400 \text{ nm}, \quad f' = f = 5.0 \times 10^{14} \text{ Hz}$$

Final Answer

$$(i) \ v = 3.0 \times 10^8 \text{ m s}^{-1}, \ \lambda = 600 \text{ nm}, \ f = 5.0 \times 10^{14} \text{ Hz} \quad (ii) \ v' = 2.0 \times 10^8 \text{ m s}^{-1}, \ \lambda' = 400 \text{ nm}, \ f' = 5.0 \times 10^{14} \text{ Hz}$$

Common Mistake: Frequency is the same for both the reflected and refracted rays – it is only the speed and wavelength that differ between the two media.

II. Superposition of Waves

Question 12

Two waves of equal frequencies have their amplitudes in the ratio 3 : 5. They are superimposed on each other. Calculate the ratio of maximum and minimum intensities of the resultant wave.

★ Likely Exam Question

Given: $a_1 : a_2 = 3 : 5$

Formula: $\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$

Solution:

$$\frac{I_{max}}{I_{min}} = \left(\frac{3 + 5}{5 - 3} \right)^2 = \left(\frac{8}{2} \right)^2 = 16$$

Final Answer

$$I_{max} : I_{min} = 16 : 1$$

Common Mistake: Always take the ratio as (sum)²/(difference)² – do not square the amplitude ratio directly, as that gives the wrong intensity ratio of individual waves, not of maxima and minima.

Question 13

Two coherent light sources whose intensity ratio is 81 : 1 produce interference fringes. Calculate the ratio of intensities of maxima and minima in the fringe system.

Given: $I_1 : I_2 = 81 : 1$

Formula: $a \propto \sqrt{I}$, $\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$

Solution:

$$a_1 : a_2 = \sqrt{81} : \sqrt{1} = 9 : 1$$

$$\frac{I_{max}}{I_{min}} = \left(\frac{9 + 1}{9 - 1} \right)^2 = \left(\frac{10}{8} \right)^2 = \frac{100}{64} = \frac{25}{16}$$

Final Answer

$$I_{max} : I_{min} = 25 : 16$$

Question 14

Two beams of light of the same wavelength and having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The path difference between the beams is $\lambda/2$ at point P and λ at point Q on the screen. Find the ratio of the resultant intensities at P and Q .

Given: $I_1 = I$, $I_2 = 4I$, $\Delta x_P = \lambda/2$, $\Delta x_Q = \lambda$

Formula: $\phi = \frac{2\pi}{\lambda} \Delta x$, $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$

Solution:

At P : $\Delta x = \lambda/2 \implies \phi_P = \pi$, so $\cos \phi_P = -1$.

$$I_P = I + 4I + 2\sqrt{I \times 4I}(-1) = 5I - 4I = I$$

At Q : $\Delta x = \lambda \implies \phi_Q = 2\pi$, so $\cos \phi_Q = 1$.

$$I_Q = I + 4I + 2\sqrt{I \times 4I}(1) = 5I + 4I = 9I$$

$$\frac{I_P}{I_Q} = \frac{I}{9I} = \frac{1}{9}$$

Final Answer

$$\frac{I_P}{I_Q} = \frac{1}{9}$$

Common Mistake: A path difference of λ still gives constructive interference (not $\lambda/2$ giving maxima) – match path difference to phase using $\phi = 2\pi\Delta x/\lambda$ before applying $\cos \phi$.

III. Young's Double Slit Experiment (YDSE)

Question 15

The initial phase difference between two interfering waves is 54° . When these waves reach a point on a screen, a path difference of 1.5λ is introduced between them. What is the total phase difference at that point?

Given: Initial phase = 54° , additional path difference = 1.5λ

Formula: $\phi = \left(54^\circ \times \frac{\pi}{180^\circ}\right) + \left(1.5\lambda \times \frac{2\pi}{\lambda}\right)$

Solution:

Phase from initial condition:

$$54^\circ = 54 \times \frac{\pi}{180} = 0.3\pi \text{ rad}$$

Phase from the path difference 1.5λ :

$$\phi_{\text{path}} = \frac{2\pi}{\lambda} \times 1.5\lambda = 3\pi \text{ rad}$$

Total phase difference:

$$\phi = 0.3\pi + 3\pi = 3.3\pi \text{ rad}$$

Final Answer

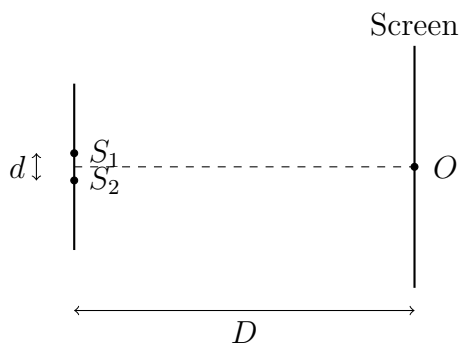
$$\phi = 3.3\pi \text{ rad}$$

Common Mistake: Convert the initial phase from degrees to radians (as a multiple of π) before adding it to the path-difference term, otherwise the two parts cannot be combined.

Question 16

A double-slit of separation 0.5 mm is illuminated by blue light of wavelength 480 nm. At what distance should a screen be placed from the double-slit to obtain interference fringes that are 1.0 mm apart? What would be the fringe-width if this distance is doubled?

★ Likely Exam Question



Standard YDSE geometry: two slits S_1, S_2 separated by d , screen at distance D

Given: $d = 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}$, $\lambda = 480 \text{ nm} = 4.8 \times 10^{-7} \text{ m}$, $\beta = 1.0 \text{ mm} = 1 \times 10^{-3} \text{ m}$

Formula: $\beta = \frac{\lambda D}{d} \Rightarrow D = \frac{\beta d}{\lambda}$

Solution:

$$D = \frac{(1 \times 10^{-3})(5 \times 10^{-4})}{4.8 \times 10^{-7}} = \frac{5 \times 10^{-7}}{4.8 \times 10^{-7}} = 1.04 \text{ m}$$

If D is doubled, since $\beta \propto D$:

$$\beta' = 2\beta = 2.0 \text{ mm}$$

Final Answer

$D = 104 \text{ cm}$; new fringe width = 2.0 mm

Question 17

Two slits are 1.0 mm apart and placed at a distance of 1.0 m from a screen. Find the fringe-width when light of wavelength 500 nm is used. What would be the fringe-width if the separation between the slits is halved and the distance of the screen from the slits is doubled?

Given: $d = 1.0 \text{ mm} = 1 \times 10^{-3} \text{ m}$, $D = 1.0 \text{ m}$, $\lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$

Formula: $\beta = \frac{\lambda D}{d}$

Solution:

$$\beta = \frac{5 \times 10^{-7} \times 1.0}{1 \times 10^{-3}} = 5 \times 10^{-4} \text{ m} = 0.5 \text{ mm}$$

New case: $d' = d/2$, $D' = 2D$:

$$\beta' = \frac{\lambda D'}{d'} = \frac{\lambda(2D)}{(d/2)} = 4 \left(\frac{\lambda D}{d} \right) = 4\beta = 4 \times 0.5 = 2.0 \text{ mm}$$

Final Answer

$\beta = 0.5 \text{ mm}$; new fringe width = 2.0 mm

Common Mistake: Both halving d and doubling D increase β – their effects multiply (factor of $2 \times 2 = 4$), not add.

Question 18

In Young's experiment, what will be the phase difference and the path difference between the light waves reaching (i) the third bright fringe and (ii) the third dark fringe from the central fringe? ($\lambda = 6000 \text{ \AA}$)

Given: $\lambda = 6000 \text{ \AA}$

Formula: Bright fringe: $\Delta x = n\lambda$; Dark fringe: $\Delta x = (2n - 1)\frac{\lambda}{2}$; $\phi = \frac{2\pi}{\lambda}\Delta x$

Solution:

(i) Third bright fringe ($n = 3$):

$$\Delta x = 3\lambda = 3 \times 6000 = 18000 \text{ \AA}$$

$$\phi = \frac{2\pi}{\lambda} \times 3\lambda = 6\pi \text{ rad}$$

(ii) Third dark fringe ($n = 3$):

$$\Delta x = \frac{(2 \times 3 - 1)\lambda}{2} = \frac{5\lambda}{2} = 2.5 \times 6000 = 15000 \text{ \AA}$$

$$\phi = \frac{2\pi}{\lambda} \times 2.5\lambda = 5\pi \text{ rad}$$

Final Answer

(i) $\phi = 6\pi \text{ rad}$, $\Delta x = 18000 \text{ \AA}$ (ii) $\phi = 5\pi \text{ rad}$, $\Delta x = 15000 \text{ \AA}$

Common Mistake: The n th dark fringe corresponds to path difference $(2n - 1)\lambda/2$, not $n\lambda/2$ – a very common slip.

Question 19

In Young's double-slit experiment, the slits are 0.2 mm apart and the screen is 1.5 m away. The distance between the central bright fringe and the fourth dark fringe is 1.8 cm. Find the wavelength of light used and the fringe width.

★ Likely Exam Question

Given: $d = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 1.5 \text{ m}$, $x_4(\text{dark}) = 1.8 \text{ cm} = 1.8 \times 10^{-2} \text{ m}$

Formula: $x_n(\text{dark}) = \frac{(2n-1)\lambda D}{2d}, \quad \beta = \frac{\lambda D}{d}$

Solution:

For the 4th dark fringe, $n = 4$:

$$x_4 = \frac{(2 \times 4 - 1)\lambda D}{2d} = \frac{7\lambda D}{2d}$$

$$1.8 \times 10^{-2} = \frac{7 \times \lambda \times 1.5}{2 \times 2 \times 10^{-4}}$$

$$\lambda = \frac{1.8 \times 10^{-2} \times 2 \times 2 \times 10^{-4}}{7 \times 1.5} = \frac{7.2 \times 10^{-6}}{10.5} = 6.857 \times 10^{-7} \text{ m} = 6857 \text{ \AA}$$

$$\beta = \frac{\lambda D}{d} = \frac{6.857 \times 10^{-7} \times 1.5}{2 \times 10^{-4}} = 5.14 \times 10^{-3} \text{ m} = 5.14 \text{ mm}$$

Final Answer

$$\lambda = 6857 \text{ \AA}, \quad \beta = 5.14 \text{ mm}$$

Question 20

The light used in Young's double-slit experiment has a frequency of $6.0 \times 10^{14} \text{ s}^{-1}$. The distance between the centres of adjacent bright fringes is 0.75 mm. If the screen is 1.5 m away, what is the distance between the slits?

Given: $f = 6.0 \times 10^{14} \text{ Hz}, \quad \beta = 0.75 \text{ mm} = 7.5 \times 10^{-4} \text{ m}, \quad D = 1.5 \text{ m}$

Formula: $\lambda = \frac{c}{f}, \quad d = \frac{\lambda D}{\beta}$

Solution:

$$\lambda = \frac{3.0 \times 10^8}{6.0 \times 10^{14}} = 5.0 \times 10^{-7} \text{ m}$$

$$d = \frac{\lambda D}{\beta} = \frac{5.0 \times 10^{-7} \times 1.5}{7.5 \times 10^{-4}} = 1.0 \times 10^{-3} \text{ m} = 1.0 \text{ mm}$$

Final Answer

$$d = 1.0 \text{ mm}$$

Question 21

In Young's double-slit experiment, the slits are separated by 0.24 mm. The screen is 1.2 m away from the slits. The fringe-width is 0.3 cm. Find the wavelength of light used.

Given: $d = 0.24 \text{ mm} = 2.4 \times 10^{-4} \text{ m}, \quad D = 1.2 \text{ m}, \quad \beta = 0.3 \text{ cm} = 3 \times 10^{-3} \text{ m}$

Formula: $\lambda = \frac{\beta d}{D}$

Solution:

$$\lambda = \frac{3 \times 10^{-3} \times 2.4 \times 10^{-4}}{1.2} = \frac{7.2 \times 10^{-7}}{1.2} = 6.0 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$$

Final Answer

$$\lambda = 6000 \text{ \AA}$$

Question 22

In Young's experiment, the wavelength of light used is $6000 \text{ \AA}$, the distance between two slits is 0.05 cm and the distance of the screen from the slits is 1 m . Find the distance between the first dark fringe and first bright fringe.

Given: $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$, $d = 0.05 \text{ cm} = 5 \times 10^{-4} \text{ m}$, $D = 1 \text{ m}$

Formula: $\beta = \frac{\lambda D}{d}$; distance between adjacent dark and bright fringe $= \frac{\beta}{2}$

Solution:

$$\beta = \frac{6 \times 10^{-7} \times 1}{5 \times 10^{-4}} = 1.2 \times 10^{-3} \text{ m} = 1.2 \text{ mm}$$

Since a dark fringe always lies exactly midway between two consecutive bright fringes,

$$\text{distance} = \frac{\beta}{2} = \frac{1.2}{2} = 0.6 \text{ mm} = 0.06 \text{ cm}$$

Final Answer

$$\text{Distance} = 0.06 \text{ cm}$$

Common Mistake: The separation between a bright fringe and the *nearest* dark fringe is always $\beta/2$, not β .

Question 23

In Young's double-slit experiment, the separation between the slits is 0.5 mm and the distance between the slits and the screen is 1.5 m . If the fringe width is 1.8 mm , find the path difference between the interfering waves at the position of the first-order maximum.

Given: $d = 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}$, $D = 1.5 \text{ m}$, $\beta = 1.8 \text{ mm} = 1.8 \times 10^{-3} \text{ m}$

Formula: $\lambda = \frac{\beta d}{D}$; path difference at n th order maximum $= n\lambda$

Solution:

$$\lambda = \frac{1.8 \times 10^{-3} \times 5 \times 10^{-4}}{1.5} = \frac{9 \times 10^{-7}}{1.5} = 6.0 \times 10^{-7} \text{ m}$$

At the first-order maximum ($n = 1$), path difference = $n\lambda = \lambda$:

$$\Delta x = 6.0 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$$

Final Answer

$$\Delta x = 6 \times 10^{-7} \text{ m}$$

IV. Angular Fringe Width

Question 24

A double-slit is illuminated by light of wavelength $6000 \text{ \AA}$. The slits are 0.1 cm apart and the screen is placed 1 m away. Find the angular position of the 10th bright fringe in radian and the angular separation between two adjacent minima.

★ Likely Exam Question

Given: $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$, $d = 0.1 \text{ cm} = 1 \times 10^{-3} \text{ m}$

Formula: $\theta_n = \frac{n\lambda}{d}$, $\Delta\theta = \frac{\lambda}{d}$

Solution:

$$\theta_{10} = \frac{10 \times 6 \times 10^{-7}}{1 \times 10^{-3}} = 6 \times 10^{-3} \text{ rad}$$

$$\Delta\theta = \frac{\lambda}{d} = \frac{6 \times 10^{-7}}{1 \times 10^{-3}} = 6 \times 10^{-4} \text{ rad} = 0.6 \text{ mrad}$$

Final Answer

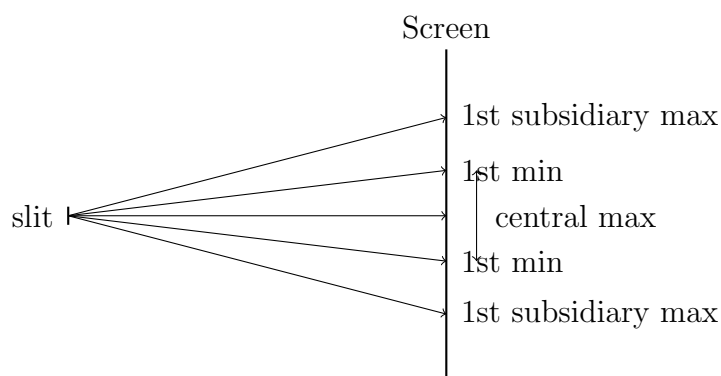
$\theta_{10} = 6 \times 10^{-3} \text{ rad}$; angular separation between adjacent minima $= 0.6 \times 10^{-3} \text{ rad}$

Common Mistake: Notice that the screen distance D is not needed here – the angular position depends only on λ and d , since $\theta = x/D = n\lambda D/(dD) = n\lambda/d$.

V. Diffraction at a Single Slit

Question 25

A parallel beam of light of wavelength 600 nm falls normally on a narrow slit of width 0.3 mm. Calculate the angular separation between (a) the first minimum and the central maximum, and (b) the first subsidiary maxima on the two sides of the central maximum.



Single-slit diffraction: central maximum flanked by first minima and first subsidiary maxima

Given: $\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}$, $a = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}$

Formula: $\theta_{\min} = \frac{\lambda}{a}$; first subsidiary maximum: $\theta = \frac{3\lambda}{2a}$

Solution:

(a) Angular separation between first minimum and central maximum:

$$\theta_1 = \frac{\lambda}{a} = \frac{6 \times 10^{-7}}{3 \times 10^{-4}} = 2 \times 10^{-3} \text{ rad}$$

(b) Angular position of the first subsidiary maximum on one side:

$$\theta = \frac{3\lambda}{2a} = \frac{3 \times 6 \times 10^{-7}}{2 \times 3 \times 10^{-4}} = 3 \times 10^{-3} \text{ rad}$$

Angular separation between the first subsidiary maxima on the *two* sides:

$$2\theta = 2 \times 3 \times 10^{-3} = 6 \times 10^{-3} \text{ rad}$$

Final Answer

(a) $2 \times 10^{-3} \text{ rad}$ (b) $6 \times 10^{-3} \text{ rad}$

Common Mistake: Part (b) asks for the separation between the subsidiary maxima on *both* sides – remember to double the single-side angle.

Question 26

A Fraunhofer diffraction pattern due to a single slit of width 0.2 mm is obtained on a screen placed at a distance of 2 m from the slit. The first minima lie 5 mm on either side of the central maximum on the screen. Find the wavelength of light.

★ Likely Exam Question

Given: $a = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 2 \text{ m}$, $x = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$

Formula: $x = \frac{\lambda D}{a}$

Solution:

$$\lambda = \frac{xa}{D} = \frac{5 \times 10^{-3} \times 2 \times 10^{-4}}{2} = 5 \times 10^{-7} \text{ m} = 5000 \text{ Å}$$

Final Answer

$$\lambda = 5000 \text{ Å}$$

Question 27

A parallel beam of light of wavelength $6 \times 10^{-5} \text{ cm}$ falls normally on a straight slit of width 0.2 mm. Find the total angular width of the central diffraction maximum and also its linear width as observed on a screen placed 2 m away.

Given: $\lambda = 6 \times 10^{-5} \text{ cm} = 6 \times 10^{-7} \text{ m}$, $a = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 2 \text{ m}$

Formula: Angular width $= \frac{2\lambda}{a}$; linear width $= \frac{2\lambda D}{a}$

Solution:

$$\text{Angular width} = \frac{2 \times 6 \times 10^{-7}}{2 \times 10^{-4}} = 6 \times 10^{-3} \text{ rad}$$

$$\text{Linear width} = \frac{2\lambda D}{a} = 6 \times 10^{-3} \times 2 = 1.2 \times 10^{-2} \text{ m} = 1.2 \text{ cm}$$

Final Answer

Angular width $= 6 \times 10^{-3} \text{ rad}$; linear width $= 1.2 \text{ cm}$

Question 28

In a single-slit diffraction experiment, the first minimum for $\lambda_1 = 600$ nm coincides with the first maxima for wavelength λ_2 . Calculate λ_2 .

Given: $\lambda_1 = 600$ nm; first minimum of λ_1 coincides with first (subsidiary) maximum of λ_2

Formula: First minimum: $a \sin \theta = \lambda_1$; first subsidiary maximum: $a \sin \theta = \frac{3\lambda_2}{2}$

Solution:

Since both occur at the same angle θ ,

$$\lambda_1 = \frac{3\lambda_2}{2} \implies \lambda_2 = \frac{2\lambda_1}{3}$$

$$\lambda_2 = \frac{2 \times 600}{3} = 400 \text{ nm}$$

Final Answer

$$\lambda_2 = 400 \text{ nm}$$

Common Mistake: Use the *approximate* secondary-maximum condition $a \sin \theta = (2n + 1)\lambda/2$ with $n = 1$, i.e. $3\lambda/2$, not λ – confusing this with the minima condition is a common error.

VI. Miscellaneous

Question 29

In Young's double-slit experiment, the distance between the centres of the adjacent fringes is 0.10 mm. If the distance of the screen from the slits is doubled, the distance between the slits is halved, and the wavelength of light is changed from 6.4×10^{-7} m to 4.0×10^{-7} m, what is the new distance between the fringes?

Given: $\beta_1 = 0.10$ mm, $\lambda_1 = 6.4 \times 10^{-7}$ m, $\lambda_2 = 4.0 \times 10^{-7}$ m, $D_2 = 2D_1$, $d_2 = d_1/2$

Formula: $\beta = \frac{\lambda D}{d}$

Solution:

$$\frac{\beta_2}{\beta_1} = \frac{\lambda_2}{\lambda_1} \times \frac{D_2}{D_1} \times \frac{d_1}{d_2} = \frac{\lambda_2}{\lambda_1} \times 2 \times 2 = \frac{4\lambda_2}{\lambda_1}$$

$$\frac{\beta_2}{\beta_1} = \frac{4 \times 4.0 \times 10^{-7}}{6.4 \times 10^{-7}} = 2.5$$

$$\beta_2 = 2.5 \times 0.10 = 0.25 \text{ mm}$$

Final Answer

New fringe separation = 0.25 mm

Common Mistake: Track all three changes (D , d , λ) together as ratios multiplying β_1 – do not apply them one at a time and round in between, as this accumulates error.

Question 30

In Young's double-slit experiment, the slits are 1 mm apart and the screen is placed 1 m away from the slits. When the slits are illuminated by monochromatic light, the width of 10 fringes on the screen is measured as 3.93 mm. Find the wavelength of light used. What will be the width of 10 fringes when the screen is placed 1.5 m away from the slits and the same light is used?

Given: $d = 1$ mm = 1×10^{-3} m, $D = 1$ m, $10\beta = 3.93$ mm

Formula: $\beta = \frac{\lambda D}{d}$

Solution:

$$\beta = \frac{3.93}{10} = 0.393 \text{ mm} = 3.93 \times 10^{-4} \text{ m}$$

$$\lambda = \frac{\beta d}{D} = \frac{3.93 \times 10^{-4} \times 1 \times 10^{-3}}{1} = 3.93 \times 10^{-7} \text{ m} = 3930 \text{ \AA}$$

Since $\beta \propto D$ (with d, λ unchanged), when $D' = 1.5 \text{ m}$:

$$10\beta' = 10\beta \times \frac{D'}{D} = 3.93 \times 1.5 = 5.895 \text{ mm}$$

Final Answer

$\lambda = 3930 \text{ \AA}$; new width of 10 fringes = 5.895 mm

Common Mistake: You do not need to recompute λ for the second part – since β is directly proportional to D , just scale the given fringe width by the ratio of the two distances.

Question 31

In an experiment with sodium light ($5890 \text{ \AA}$), an interference pattern is obtained in which 20 equally spaced fringes occupy 2.30 cm on the screen. On replacing sodium light with another monochromatic source, 30 fringes occupy 2.80 cm on the screen. Find the wavelength of the light from the second source.

Given: $\lambda_1 = 5890 \text{ \AA}$; $20\beta_1 = 2.30 \text{ cm}$; $30\beta_2 = 2.80 \text{ cm}$

Formula: $\beta = \frac{\lambda D}{d} \implies \beta \propto \lambda$ (for fixed D, d)

Solution:

$$\beta_1 = \frac{2.30}{20} = 0.115 \text{ cm}, \quad \beta_2 = \frac{2.80}{30} = 0.0933 \text{ cm}$$

Since $\beta \propto \lambda$ for the same apparatus,

$$\lambda_2 = \lambda_1 \times \frac{\beta_2}{\beta_1} = 5890 \times \frac{0.0933}{0.115} = 5890 \times 0.8116$$

$$\lambda_2 \approx 4780 \text{ \AA}$$

Final Answer

$\lambda_2 \approx 4780 \text{ \AA}$

Common Mistake: Compute the fringe width for each case separately (total width $\div$ number of fringes) before taking the ratio – do not divide the two total widths directly, since the fringe counts differ.

Question 32

A double-slit is illuminated by sodium light ($\lambda = 5893 \text{ \AA}$). In the double-slit, the distance between the two slits is 0.050 cm . The interference pattern is obtained on a screen placed at a distance of 1.00 m . Calculate the distance between the fifth bright fringe and the first dark fringe on the same side of the central fringe.

Given: $\lambda = 5893 \text{ \AA} = 5.893 \times 10^{-7} \text{ m}$, $d = 0.050 \text{ cm} = 5 \times 10^{-4} \text{ m}$, $D = 1.00 \text{ m}$

Formula: $x_n(\text{bright}) = \frac{n\lambda D}{d}$; $x_n(\text{dark}) = \frac{(2n-1)\lambda D}{2d}$

Solution:

Fringe width:

$$\beta = \frac{\lambda D}{d} = \frac{5.893 \times 10^{-7} \times 1.00}{5 \times 10^{-4}} = 1.1786 \times 10^{-3} \text{ m}$$

Position of the 5th bright fringe ($n = 5$): $x_5 = 5\beta$. Position of the 1st dark fringe ($n = 1$):

$$x_1(\text{dark}) = \frac{\beta}{2}.$$

$$\Delta x = x_5 - x_1(\text{dark}) = 5\beta - \frac{\beta}{2} = 4.5\beta$$

$$\Delta x = 4.5 \times 1.1786 \times 10^{-3} = 5.30 \times 10^{-3} \text{ m} \approx 0.53 \text{ cm}$$

Final Answer

Distance $\approx 0.53 \text{ cm}$

Common Mistake: The first dark fringe lies at $\beta/2$ from the centre (not at β) – always use $(2n-1)\beta/2$ for dark fringes and $n\beta$ for bright fringes, and be careful not to mix the two.

Question 33

In Young's double-slit experiment, the separation between the slits is 0.15 mm and the fringes are obtained on a screen distant 0.75 m from the slits. If the third dark fringe is at a distance of 5.5 mm from the central bright fringe, find the wavelength of light used. What will be the distance of the same dark fringe from the centre if the entire apparatus is immersed in water? (Refractive index of water = $4/3$)

Given: $d = 0.15 \text{ mm} = 1.5 \times 10^{-4} \text{ m}$, $D = 0.75 \text{ m}$, $x_3(\text{dark}) = 5.5 \text{ mm}$, $n_w = 4/3$

Formula: $x_n(\text{dark}) = \frac{(2n-1)\lambda D}{2d}$; $\lambda_{\text{water}} = \frac{\lambda_{\text{air}}}{n_w}$

Solution:

For the 3rd dark fringe, $n = 3$:

$$x_3 = \frac{(2 \times 3 - 1)\lambda D}{2d} = \frac{5\lambda D}{2d}$$

$$5.5 \times 10^{-3} = \frac{5 \times \lambda \times 0.75}{2 \times 1.5 \times 10^{-4}}$$

$$\lambda = \frac{5.5 \times 10^{-3} \times 2 \times 1.5 \times 10^{-4}}{5 \times 0.75} = \frac{1.65 \times 10^{-6}}{3.75} = 4.4 \times 10^{-7} \text{ m} = 4400 \text{ \AA}$$

When immersed in water, D and d are unchanged but λ decreases:

$$\lambda_{\text{water}} = \frac{4400}{4/3} = 3300 \text{ \AA}$$

Since $x_3 \propto \lambda$,

$$x'_3 = x_3 \times \frac{\lambda_{\text{water}}}{\lambda_{\text{air}}} = 5.5 \times \frac{3300}{4400} = 5.5 \times 0.75 = 4.125 \text{ mm}$$

Final Answer

$\lambda = 4400 \text{ \AA}$; new distance of the same dark fringe = 4.125 mm

Common Mistake: On immersion in a liquid, D and d do not change – only λ shrinks by a factor of n_w , so all fringe positions shrink by the same factor.

Question 34

A parallel light beam of wavelength $6000 \text{ \AA}$ passes through a slit 0.2 mm wide and forms a diffraction pattern on a screen 1.0 m away from the slit. Find the width of the central maximum on the screen. What would be the width of this maximum if the apparatus were immersed in water of refractive index $4/3$?

Given: $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$, $a = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 1.0 \text{ m}$, $n_w = 4/3$

Formula: Width of central maximum = $\frac{2\lambda D}{a}$

Solution:

$$w = \frac{2 \times 6 \times 10^{-7} \times 1.0}{2 \times 10^{-4}} = 6 \times 10^{-3} \text{ m} = 6.0 \text{ mm}$$

In water, $\lambda_{\text{water}} = \frac{\lambda}{n_w} = \frac{6000}{4/3} = 4500 \text{ \AA} = 4.5 \times 10^{-7} \text{ m}$:

$$w' = \frac{2 \times 4.5 \times 10^{-7} \times 1.0}{2 \times 10^{-4}} = 4.5 \times 10^{-3} \text{ m} = 4.5 \text{ mm}$$

Final Answer

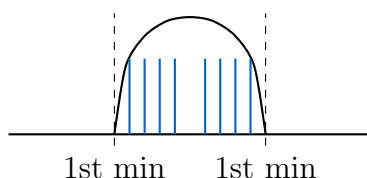
$w = 6.0 \text{ mm}$ in air; $w' = 4.5 \text{ mm}$ in water

Question 35

In a double-slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of the double-slit pattern within the central maximum of the single-slit diffraction pattern?

★ Likely Exam Question

central diffraction envelope



Double-slit fringes (fine lines) lying inside the broad single-slit central diffraction envelope

Given: $d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$, $D = 1 \text{ m}$, $\lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$, $N = 10$ maxima

Formula: Width of central max (single slit) $= \frac{2\lambda D}{e}$; fringe width (double slit) $\beta = \frac{\lambda D}{d}$; $N = \frac{2\lambda D/e}{\lambda D/d} = \frac{2d}{e}$

Solution:

The number of double-slit maxima that fit within the single-slit central maximum is

$$N = \frac{2d}{e}$$

Setting $N = 10$:

$$10 = \frac{2d}{e} \Rightarrow e = \frac{2d}{10} = \frac{d}{5}$$

$$e = \frac{1 \text{ mm}}{5} = 0.2 \text{ mm}$$

Final Answer

Slit width $e = 0.2 \text{ mm}$

Common Mistake: This problem does not actually require λ or D in the final step – both cancel out, leaving a simple ratio between d and e . Plugging in numbers unnecessarily often leads to arithmetic errors.

Question 36

A narrow slit of width 0.3 mm is illuminated normally by a parallel beam of light of wavelength 6×10^{-7} m. The Fraunhofer diffraction pattern is obtained on a screen placed in the focal plane of a convex lens of focal length 25 cm. The lens is placed quite close to the slit. Find the width of the central maximum on the screen.

Given: $a = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}$, $\lambda = 6 \times 10^{-7} \text{ m}$, $f = 25 \text{ cm} = 0.25 \text{ m}$

Formula: Width of central maximum $= \frac{2f\lambda}{a}$

Solution:

Since the lens is close to the slit and the screen is at the focal plane, D is effectively replaced by f :

$$w = \frac{2 \times 0.25 \times 6 \times 10^{-7}}{3 \times 10^{-4}} = \frac{3 \times 10^{-7}}{3 \times 10^{-4}} = 1 \times 10^{-3} \text{ m} = 1.0 \text{ mm}$$

Final Answer

$$w = 1.0 \text{ mm}$$

Common Mistake: When a converging lens focuses the diffraction pattern, use its focal length f in place of the slit-to-screen distance D – do not add a separate screen distance.

ISC Class XII Physics

Chapter 11

Dual Nature of Radiation and Matter

38 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Energy and Momentum of Photon, Work Function, Threshold Wavelength
- II. Einstein's Photoelectric Equation
- III. Stopping Potential
- IV. de Broglie Wavelength When Momentum or Velocity is Given
- V. de Broglie Wavelength When Kinetic Energy is Given
- VI. de Broglie Wavelength When Accelerating Potential is Given

Formula Sheet

Constants used in this chapter:

$$h = 6.6 \times 10^{-34} \text{ J s} \quad c = 3 \times 10^8 \text{ m/s} \quad e = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg} \quad m_p = m_n = 1.67 \times 10^{-27} \text{ kg} \quad 1 \text{ Å} = 10^{-10} \text{ m}$$

I. Photon Energy, Momentum, Work Function

Photon Energy

$$E = h\nu = \frac{hc}{\lambda}$$

ν = frequency, λ = wavelength

Photon Momentum

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

momentum carried by a photon

Dynamic Mass of Photon

$$m = \frac{E}{c^2}$$

rest mass of a photon is always zero

Threshold Frequency

$$\nu_0 = \frac{W_0}{h}$$

W_0 = work function of the metal

Threshold Wavelength

$$\lambda_0 = \frac{hc}{W_0}$$

longest wavelength that can cause emission

Photon Count from Power

$$n = \frac{P}{E_{\text{photon}}}$$

P = power of source, n = photons/s

II. Einstein's Photoelectric Equation

Einstein's Equation

$$E_{k(\max)} = h\nu - W_0 = \frac{hc}{\lambda} - W_0$$

$E_{k(\max)}$ = max. KE of ejected electron

Max. Speed of Photoelectron

$$v_{\max} = \sqrt{\frac{2E_{k(\max)}}{m_e}}$$

from max. kinetic energy

III. Stopping Potential

Stopping Potential Relation

$$eV_0 = h\nu - W_0$$

V_0 = stopping potential (volt)

Slope of V_0 - ν Graph

$$\text{slope} = \frac{h}{e}$$

gives Planck's constant from graph

Intercepts of V_0 - ν Graph

$$\nu_0 = \frac{W_0}{h}, \quad V_0|_{\nu=0} = -\frac{W_0}{e}$$

ν_0 = x-intercept, $-W_0/e$ = y-intercept

IV, V & VI. de Broglie Wavelength

de Broglie Wavelength (general)

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

p = momentum of the particle

de Broglie Wavelength (KE given)

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

E_k = kinetic energy of the particle

de Broglie Wavelength (electron, accelerating p.d. V)

$$\lambda = \frac{h}{\sqrt{2m_e eV}} \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$$

V in volts, standard result for electrons

I. Energy and Momentum of Photon, Work Function, Threshold Wavelength

Question 1

Find the frequency of a photon whose energy is 75 eV.

Given: $E = 75 \text{ eV} = 75 \times 1.6 \times 10^{-19} \text{ J} = 1.2 \times 10^{-17} \text{ J}$

Formula: $E = h\nu \Rightarrow \nu = \frac{E}{h}$

Solution:

$$\nu = \frac{1.2 \times 10^{-17}}{6.6 \times 10^{-34}} = 1.818 \times 10^{16} \text{ Hz}$$

Final Answer

$$\nu \approx 1.82 \times 10^{16} \text{ Hz}$$

Question 2

A beam of ultraviolet light has frequency $1.0 \times 10^{15} \text{ Hz}$. Find the energy of one quantum of this light, in eV. (ISC 2004)

★ Likely Exam Question

Given: $\nu = 1.0 \times 10^{15} \text{ Hz}$

Formula: $E = h\nu$

Solution:

$$E = 6.6 \times 10^{-34} \times 1.0 \times 10^{15} = 6.6 \times 10^{-19} \text{ J}$$

$$E = \frac{6.6 \times 10^{-19}}{1.6 \times 10^{-19}} = 4.125 \text{ eV}$$

Final Answer

$E \approx 4.13 \text{ eV}$ (often quoted as 4.14 eV using $h = 6.63 \times 10^{-34} \text{ Js}$)

Question 3

A spectral line has wavelength 400 nm. Calculate the frequency and the energy of the photons associated with it.

Given: $\lambda = 400 \text{ nm} = 4 \times 10^{-7} \text{ m}$

Formula: $\nu = \frac{c}{\lambda}, \quad E = h\nu$

Solution:

$$\nu = \frac{3 \times 10^8}{4 \times 10^{-7}} = 7.5 \times 10^{14} \text{ Hz}$$

$$E = 6.6 \times 10^{-34} \times 7.5 \times 10^{14} = 4.95 \times 10^{-19} \text{ J} = \frac{4.95 \times 10^{-19}}{1.6 \times 10^{-19}} = 3.09 \text{ eV}$$

Final Answer

$$\nu = 7.5 \times 10^{14} \text{ Hz}, \quad E \approx 4.95 \times 10^{-19} \text{ J} \approx 3.1 \text{ eV}$$

Common Mistake: Always convert nm to metres before substituting into $\nu = c/\lambda$.

Question 4

Calculate the number of photons emitted per second by a 40 W monochromatic source of light of wavelength 500 nm.

Given: $P = 40 \text{ W}, \quad \lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}, \quad n = \frac{P}{E_{\text{photon}}}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5 \times 10^{-7}} = 3.96 \times 10^{-19} \text{ J}$$

$$n = \frac{40}{3.96 \times 10^{-19}} = 1.01 \times 10^{20} \text{ photons/s}$$

Final Answer

$$n \approx 1.0 \times 10^{20} \text{ photons per second}$$

Question 5

The energy of a photon is 42 eV. Find its momentum.

Given: $E = 42 \text{ eV} = 42 \times 1.6 \times 10^{-19} = 6.72 \times 10^{-18} \text{ J}$

Formula: $p = \frac{E}{c}$

Solution:

$$p = \frac{6.72 \times 10^{-18}}{3 \times 10^8} = 2.24 \times 10^{-26} \text{ kg m/s}$$

Final Answer

$$p = 2.24 \times 10^{-26} \text{ kg m/s}$$

Question 6

The energy of a photon is 10 eV. Calculate (i) its momentum, (ii) its wavelength, (iii) its rest mass and dynamic mass.

★ Likely Exam Question

Given: $E = 10 \text{ eV} = 1.6 \times 10^{-18} \text{ J}$

Formula: $p = \frac{E}{c}, \quad \lambda = \frac{h}{p}, \quad m = \frac{E}{c^2}$

Solution:

(i)

$$p = \frac{1.6 \times 10^{-18}}{3 \times 10^8} = 5.33 \times 10^{-27} \text{ kg m/s}$$

(ii)

$$\lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34}}{5.33 \times 10^{-27}} = 1.2375 \times 10^{-7} \text{ m} = 1237.5 \text{ Å}$$

(iii) Rest mass of a photon is always zero (a photon exists only in motion at speed c).

$$m_{\text{dynamic}} = \frac{E}{c^2} = \frac{1.6 \times 10^{-18}}{(3 \times 10^8)^2} = 1.78 \times 10^{-35} \text{ kg}$$

Final Answer

$$p = 5.33 \times 10^{-27} \text{ kg m/s}, \quad \lambda = 1237.5 \text{ Å}, \quad \text{rest mass} = 0, \quad \text{dynamic mass} = 1.78 \times 10^{-35} \text{ kg}$$

Common Mistake: A photon's *rest* mass is zero; only its dynamic (relativistic) mass E/c^2 is non-zero.

Question 7

Calculate the threshold frequency for photoelectric emission from a metal whose work function is 0.1 eV.

Given: $W_0 = 0.1 \text{ eV} = 1.6 \times 10^{-20} \text{ J}$

Formula: $\nu_0 = \frac{W_0}{h}$

Solution:

$$\nu_0 = \frac{1.6 \times 10^{-20}}{6.6 \times 10^{-34}} = 2.42 \times 10^{13} \text{ Hz}$$

Final Answer

$$\nu_0 \approx 2.4 \times 10^{13} \text{ Hz}$$

Question 8

The work function of sodium is 2.3 eV. Calculate the maximum wavelength (in nm) of light that can cause photoelectric emission from sodium. (Take $h = 6.63 \times 10^{-34} \text{ Js}$)

Given: $W_0 = 2.3 \text{ eV} = 3.68 \times 10^{-19} \text{ J}$, $h = 6.63 \times 10^{-34} \text{ Js}$

Formula: $\lambda_0 = \frac{hc}{W_0}$

Solution:

$$\lambda_0 = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3.68 \times 10^{-19}} = 5.405 \times 10^{-7} \text{ m} = 540.5 \text{ nm}$$

Final Answer

$$\lambda_0 \approx 540.5 \text{ nm}$$

Common Mistake: This is the *maximum* (threshold) wavelength – any wavelength longer than this will *not* cause emission.

Question 9

Photons of wavelength $\lambda_1 = 5.0 \times 10^{-5} \text{ cm}$ have a total energy equal to the energy of one photon of wavelength $\lambda_2 = 2.0 \times 10^{-12} \text{ cm}$. Find the number of λ_1 -photons, and identify the region of the electromagnetic spectrum to which λ_1 and λ_2 belong.

Given: $\lambda_1 = 5.0 \times 10^{-5} \text{ cm} = 500 \text{ nm}$, $\lambda_2 = 2.0 \times 10^{-12} \text{ cm} = 2.0 \times 10^{-14} \text{ m}$

Formula: $n_1 \left(\frac{hc}{\lambda_1} \right) = \frac{hc}{\lambda_2} \Rightarrow n_1 = \frac{\lambda_1}{\lambda_2}$

Solution:

$$n_1 = \frac{5.0 \times 10^{-7}}{2.0 \times 10^{-14}} = 2.5 \times 10^7$$

$\lambda_1 = 500 \text{ nm}$ lies in the **visible** region; $\lambda_2 = 2.0 \times 10^{-14} \text{ m}$ is an extremely short wavelength lying in the **gamma-ray** region.

Final Answer

$n_1 = 2.5 \times 10^7$ photons; λ_1 : visible light, λ_2 : gamma rays

Common Mistake: Convert both wavelengths from cm to metres (or keep both in the same unit) before taking the ratio.

Question 10

5% of the energy of a 100 W monochromatic source is converted into light of wavelength 560 nm. Find the number of photons emitted per second.

Given: $P_{\text{light}} = 5\%$ of 100 W = 5 W, $\lambda = 560 \text{ nm}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}, \quad n = \frac{P_{\text{light}}}{E_{\text{photon}}}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{560 \times 10^{-9}} = 3.536 \times 10^{-19} \text{ J}$$

$$n = \frac{5}{3.536 \times 10^{-19}} = 1.414 \times 10^{19} \text{ photons/s}$$

Final Answer

$n \approx 1.41 \times 10^{19}$ photons per second

Question 11

The work function of a metal is 4.2 eV. Will radiation of wavelength 330 nm incident on the surface emit photoelectrons? Justify your answer by calculation.

Given: $W_0 = 4.2 \text{ eV}$, $\lambda = 330 \text{ nm}$

Formula: $\lambda_0 = \frac{hc}{W_0}$

Solution:

$$\lambda_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4.2 \times 1.6 \times 10^{-19}} = \frac{1.98 \times 10^{-25}}{6.72 \times 10^{-19}} = 2.946 \times 10^{-7} \text{ m} = 294.6 \text{ nm}$$

Since the incident wavelength (330 nm) is *greater* than the threshold wavelength (294.6 nm), the photon energy is less than the work function.

Final Answer

No photoelectrons will be emitted, since $\lambda_{\text{incident}} > \lambda_0$

Common Mistake: Compare wavelengths correctly: emission occurs only when $\lambda_{\text{incident}} \leq \lambda_0$ (i.e. photon energy $\geq W_0$), not the reverse.

Question 12

A photon ejects an electron with negligible kinetic energy from a metal surface whose threshold frequency is 4.6×10^{14} Hz. Calculate the energy of the incident photon, in eV.

Given: $\nu_0 = 4.6 \times 10^{14}$ Hz, negligible KE $\Rightarrow$ incident photon energy $= W_0 = h\nu_0$

Formula: $E = h\nu_0$

Solution:

$$E = 6.6 \times 10^{-34} \times 4.6 \times 10^{14} = 3.036 \times 10^{-19} \text{ J} = \frac{3.036 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.8975 \text{ eV}$$

Final Answer

$E \approx 1.9 \text{ eV}$

II. Einstein's Photoelectric Equation

Question 13

Assuming the visible spectrum extends from 400 nm to 750 nm, calculate the maximum work function (in eV) of a photocathode that can emit photoelectrons using visible light of any wavelength in this range.

Given: Visible range: 400 nm to 750 nm

Formula: $W_{0(\max)} = \frac{hc}{\lambda_{\max}}$

Solution:

For emission to occur at *any* wavelength in the visible range (including the least energetic, longest wavelength of 750 nm), the work function must not exceed the photon energy at 750 nm.

$$W_{0(\max)} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{750 \times 10^{-9}} = 2.64 \times 10^{-19} \text{ J} = \frac{2.64 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.65 \text{ eV}$$

Final Answer

$$W_{0(\max)} \approx 1.65 \text{ eV}$$

Common Mistake: Use the *longest* wavelength (lowest photon energy) of the range, not the shortest, since the work function must be small enough for even the least energetic photon to eject an electron.

Question 14

A metal sheet carries a negative charge of 11.2 nC. Find the number of ultraviolet photons needed to completely discharge the sheet by photoelectron ejection, and the minimum energy that must be absorbed by the sheet to effect this discharge. The threshold frequency of the metal is $4.5 \times 10^{15} \text{ Hz}$.

Given: $Q = 11.2 \text{ nC} = 11.2 \times 10^{-9} \text{ C}$, $\nu_0 = 4.5 \times 10^{15} \text{ Hz}$

Formula: $n = \frac{Q}{e}$, $W_{\min} = n W_0 = n h \nu_0$

Solution:

Each photon ejects (at most) one electron, so the number of photons needed equals the

number of electrons to be removed:

$$n = \frac{11.2 \times 10^{-9}}{1.6 \times 10^{-19}} = 7.0 \times 10^{10}$$

$$W_0 = h\nu_0 = 6.6 \times 10^{-34} \times 4.5 \times 10^{15} = 2.97 \times 10^{-18} \text{ J}$$

$$W_{\min} = n W_0 = 7.0 \times 10^{10} \times 2.97 \times 10^{-18} = 2.08 \times 10^{-7} \text{ J}$$

Final Answer

Number of photons = 7.0×10^{10} ; minimum energy $\approx 2.08 \times 10^{-7} \text{ J}$

Question 15

The work function of copper is 4 eV. Monochromatic light of wavelength 4000 Å falls on it. Calculate the energy of the incident photon in joule, express the work function in joule, and find the kinetic energy of the fastest ejected electron.

★ Likely Exam Question

Given: $W_0 = 4 \text{ eV}$, $\lambda = 4000 \text{ Å} = 4 \times 10^{-7} \text{ m}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}$, $E_{k(\max)} = E_{\text{photon}} - W_0$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4 \times 10^{-7}} = 4.95 \times 10^{-19} \text{ J}$$

$$W_0 = 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ J}$$

Since $E_{\text{photon}} (4.95 \times 10^{-19} \text{ J}) < W_0 (6.4 \times 10^{-19} \text{ J})$, the photon does not carry enough energy to eject an electron.

Final Answer

$E_{\text{photon}} = 4.95 \times 10^{-19} \text{ J}$, $W_0 = 6.4 \times 10^{-19} \text{ J}$ – no electrons will be ejected

Common Mistake: Do not blindly subtract to get a "negative kinetic energy" – recognise that KE cannot be negative, so no emission occurs.

Question 16

Calculate the work function of sodium metal in eV, given that the threshold wavelength for photoelectric emission is 5400 Å. Will photoelectrons be emitted if red light of wavelength 6800 Å is incident on sodium?

Given: $\lambda_0 = 5400 \text{ Å} = 5.4 \times 10^{-7} \text{ m}$, $\lambda = 6800 \text{ Å}$

Formula: $W_0 = \frac{hc}{\lambda_0}$

Solution:

$$W_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5.4 \times 10^{-7}} = 3.667 \times 10^{-19} \text{ J} = \frac{3.667 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.29 \text{ eV}$$

Since $6800 \text{ Å} > 5400 \text{ Å}$ (threshold), the red light has lower photon energy than W_0 .

Final Answer

$W_0 \approx 2.3 \text{ eV}$; **No**, red light of 6800 Å cannot emit photoelectrons from sodium

Common Mistake: Compare the given wavelength directly with the threshold wavelength – longer wavelength means lower energy, hence no emission.

Question 17

Electromagnetic radiation of frequency $3 \times 10^{15} \text{ Hz}$ falls on a photoelectric surface whose work function is 4.0 eV . Find the maximum velocity of the emitted photoelectrons.

★ Likely Exam Question

Given: $\nu = 3 \times 10^{15} \text{ Hz}$, $W_0 = 4.0 \text{ eV} = 6.4 \times 10^{-19} \text{ J}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$

Formula: $E_{k(\max)} = h\nu - W_0$, $v_{\max} = \sqrt{\frac{2E_{k(\max)}}{m_e}}$

Solution:

$$E_{k(\max)} = 6.6 \times 10^{-34} \times 3 \times 10^{15} - 6.4 \times 10^{-19} = 1.98 \times 10^{-18} - 6.4 \times 10^{-19} = 1.34 \times 10^{-18} \text{ J}$$

$$v_{\max} = \sqrt{\frac{2 \times 1.34 \times 10^{-18}}{9.1 \times 10^{-31}}} = \sqrt{2.945 \times 10^{12}} = 1.716 \times 10^6 \text{ m/s}$$

Final Answer

$v_{\max} \approx 1.72 \times 10^6 \text{ m/s}$

Question 18

The maximum kinetic energy of photoelectrons is $10 \times 10^{-19} \text{ J}$ when a surface is illuminated by light of wavelength 100 nm . Find the work function of the surface.

Given: $E_{k(\max)} = 10 \times 10^{-19} \text{ J}$, $\lambda = 100 \text{ nm}$

Formula: $W_0 = \frac{hc}{\lambda} - E_{k(\max)}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{100 \times 10^{-9}} = 1.98 \times 10^{-18} \text{ J}$$

$$W_0 = 1.98 \times 10^{-18} - 1.0 \times 10^{-18} = 0.98 \times 10^{-18} \text{ J} = \frac{0.98 \times 10^{-18}}{1.6 \times 10^{-19}} = 6.125 \text{ eV}$$

Final Answer

$$W_0 \approx 6.13 \text{ eV}$$

III. Stopping Potential

Question 19

A metal whose work function is 3.0 eV is illuminated by light of wavelength 3×10^{-7} m. Calculate (i) the threshold frequency, (ii) the maximum kinetic energy of the photoelectrons, and (iii) the stopping potential.

★ Likely Exam Question

Given: $W_0 = 3.0 \text{ eV} = 4.8 \times 10^{-19} \text{ J}$, $\lambda = 3 \times 10^{-7} \text{ m}$

Formula: $\nu_0 = \frac{W_0}{h}$, $E_{k(\max)} = h\nu - W_0$, $eV_0 = E_{k(\max)}$

Solution:

(i)

$$\nu_0 = \frac{4.8 \times 10^{-19}}{6.6 \times 10^{-34}} = 7.27 \times 10^{14} \text{ Hz}$$

(ii)

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{3 \times 10^{-7}} = 1 \times 10^{15} \text{ Hz}$$

$$E_{k(\max)} = h\nu - W_0 = 6.6 \times 10^{-19} - 4.8 \times 10^{-19} = 1.8 \times 10^{-19} \text{ J}$$

(iii)

$$V_0 = \frac{E_{k(\max)}}{e} = \frac{1.8 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.125 \text{ V}$$

Final Answer

$$\nu_0 \approx 0.73 \times 10^{15} \text{ Hz}, \quad E_{k(\max)} = 1.8 \times 10^{-19} \text{ J}, \quad V_0 = 1.125 \text{ V}$$

Question 20

When light of wavelength 300 nm is incident on a sodium cathode, the stopping potential is 1.85 V; when the wavelength is 400 nm, the stopping potential becomes 0.82 V. Calculate (i) Planck's constant, the work function of sodium, and the threshold wavelength for sodium. (ii) If the wavelength is kept fixed at 400 nm but the intensity of the light is doubled, will the photoelectric current increase? Give reasons.

★ Likely Exam Question

Given: $\lambda_1 = 300 \text{ nm}$, $V_{01} = 1.85 \text{ V}$; $\lambda_2 = 400 \text{ nm}$, $V_{02} = 0.82 \text{ V}$

Formula: $eV_0 = \frac{hc}{\lambda} - W_0$

Solution:

Writing Einstein's equation for both cases and subtracting:

$$e(V_{01} - V_{02}) = hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

$$1.6 \times 10^{-19} \times 1.03 = hc \left(\frac{1}{3 \times 10^{-7}} - \frac{1}{4 \times 10^{-7}} \right) = hc \times 8.333 \times 10^5$$

$$hc = \frac{1.648 \times 10^{-19}}{8.333 \times 10^5} = 1.978 \times 10^{-25} \Rightarrow h = \frac{1.978 \times 10^{-25}}{3 \times 10^8} = 6.6 \times 10^{-34} \text{ Js}$$

Work function:

$$W_0 = \frac{hc}{\lambda_1} - eV_{01} = 6.59 \times 10^{-19} - 2.96 \times 10^{-19} = 3.63 \times 10^{-19} \text{ J} = 2.27 \text{ eV}$$

Threshold wavelength:

$$\lambda_0 = \frac{hc}{W_0} = \frac{1.978 \times 10^{-25}}{3.63 \times 10^{-19}} = 5.45 \times 10^{-7} \text{ m} = 5450 \text{ Å}$$

(ii) The stopping potential depends only on the photon energy (frequency), not on the intensity of light. At a fixed wavelength, doubling the intensity only increases the *number* of photons striking the surface per second – so more photoelectrons are ejected per second and the **saturation current increases**, but the **stopping potential remains unchanged**, since the maximum KE of individual electrons depends only on ν .

Final Answer

$h \approx 6.6 \times 10^{-34} \text{ Js}$, $W_0 \approx 2.27 \text{ eV}$, $\lambda_0 \approx 5450 \text{ Å}$; photoelectric current *does* increase with intensity (more electrons/s), but stopping potential stays the same

Common Mistake: Do not confuse "stopping potential unchanged" with "current unchanged" – intensity affects the number of photoelectrons (current), not their maximum KE (stopping potential).

Question 21

A retarding potential of 0.5 V is required to stop the photoelectrons emitted from the cathode of a photoelectric cell when violet light of wavelength 400 nm strikes its surface. Find (i) the work function of the surface, (ii) the threshold frequency, (iii) the energy of light of wavelength 300 nm, and (iv) the maximum kinetic energy of electrons emitted when 300 nm light strikes the surface.

★ **Likely Exam Question**

Given: $V_0 = 0.5 \text{ V}$, $\lambda = 400 \text{ nm}$

Formula: $W_0 = \frac{hc}{\lambda} - eV_0, \quad \nu_0 = \frac{W_0}{h}, \quad E_{k(\max)} = E'_{\text{photon}} - W_0$

Solution:

(i)

$$\frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{400 \times 10^{-9}} = 4.95 \times 10^{-19} \text{ J}$$

$$eV_0 = 0.5 \times 1.6 \times 10^{-19} = 0.8 \times 10^{-19} \text{ J}$$

$$W_0 = 4.95 \times 10^{-19} - 0.8 \times 10^{-19} = 4.15 \times 10^{-19} \text{ J}$$

(ii)

$$\nu_0 = \frac{4.15 \times 10^{-19}}{6.6 \times 10^{-34}} = 6.29 \times 10^{14} \text{ Hz}$$

(iii)

$$E = \frac{hc}{300 \times 10^{-9}} = \frac{1.98 \times 10^{-25}}{3 \times 10^{-7}} = 6.6 \times 10^{-19} \text{ J}$$

(iv)

$$E_{k(\max)} = 6.6 \times 10^{-19} - 4.15 \times 10^{-19} = 2.45 \times 10^{-19} \text{ J}$$

Final Answer

$$W_0 = 4.15 \times 10^{-19} \text{ J}, \quad \nu_0 \approx 6.3 \times 10^{14} \text{ Hz}, \quad E = 6.6 \times 10^{-19} \text{ J}, \quad E_{k(\max)} = 2.45 \times 10^{-19} \text{ J}$$

Question 22

Ultraviolet light of wavelengths 80 nm and 70 nm, when allowed to fall on hydrogen atoms in their ground state, liberates electrons with kinetic energies 1.8 eV and 4.0 eV respectively. Find the value of Planck's constant.

Given: $\lambda_1 = 80 \text{ nm}, E_{k1} = 1.8 \text{ eV}; \quad \lambda_2 = 70 \text{ nm}, E_{k2} = 4.0 \text{ eV}$

Formula: $E_{k1} = \frac{hc}{\lambda_1} - W, \quad E_{k2} = \frac{hc}{\lambda_2} - W$

Solution:

Subtracting the two equations to eliminate the (ionization) work W :

$$E_{k2} - E_{k1} = hc \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$(4.0 - 1.8) \times 1.6 \times 10^{-19} = hc \left(\frac{1}{70 \times 10^{-9}} - \frac{1}{80 \times 10^{-9}} \right)$$

$$3.52 \times 10^{-19} = hc \times 1.786 \times 10^6$$

$$hc = 1.971 \times 10^{-25} \Rightarrow h = \frac{1.971 \times 10^{-25}}{3 \times 10^8} = 6.57 \times 10^{-34} \text{ Js}$$

Final Answer

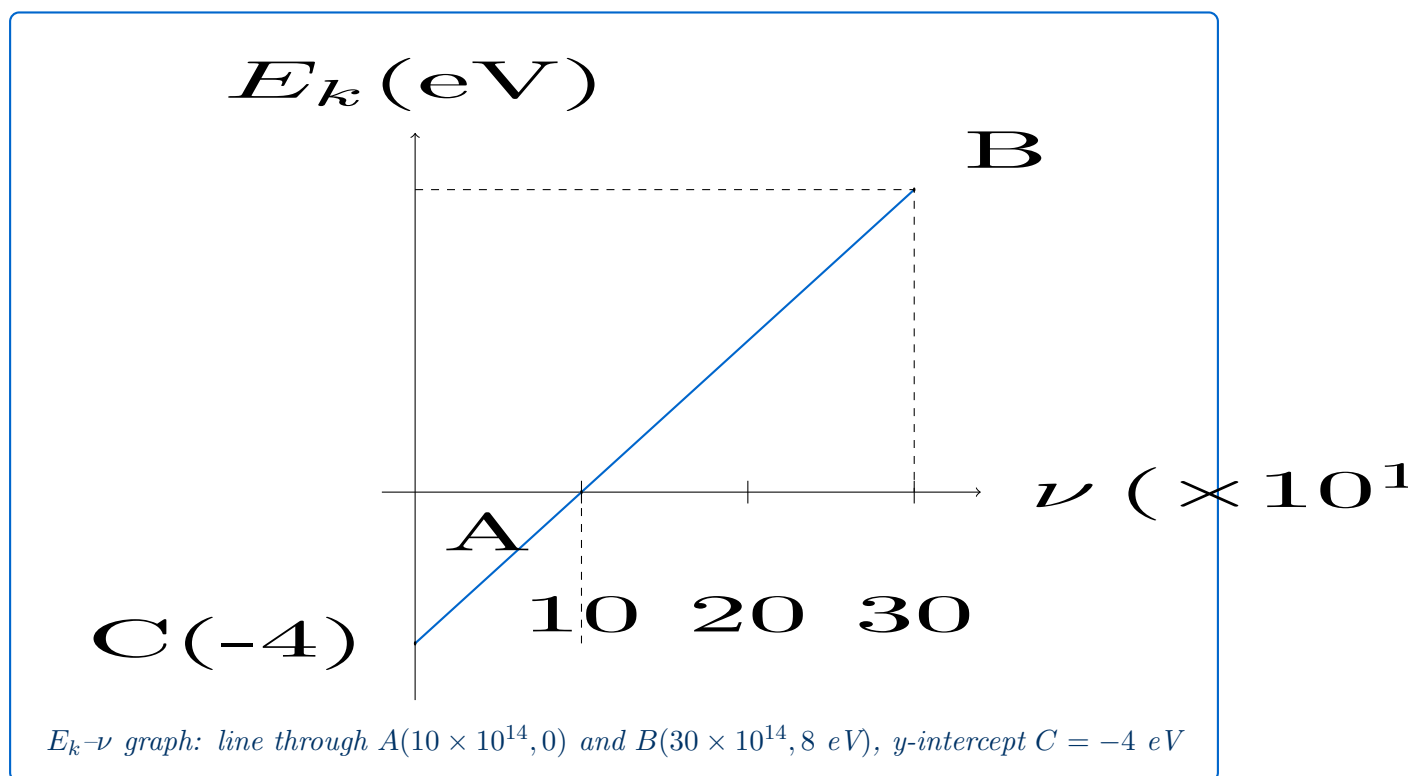
$$h \approx 6.57 \times 10^{-34} \text{ Js}$$

Common Mistake: Set up two simultaneous Einstein equations and subtract to eliminate the unknown work term – do not try to solve for h from a single equation.

Question 23

In an experiment on the photoelectric effect, the graph of the maximum kinetic energy E_k of the emitted photoelectrons versus the frequency ν of the incident light is a straight line, as shown in the figure. Calculate (i) the threshold frequency, (ii) the work function W of the cathode metal (in eV), and (iii) the stopping potential for photoelectrons emitted by light of frequency $\nu = 30 \times 10^{14} \text{ Hz}$.

★ Likely Exam Question



Given: From graph: line passes through $A(10 \times 10^{14} \text{ Hz}, 0)$ and $B(30 \times 10^{14} \text{ Hz}, 8 \text{ eV})$; y -intercept at -4 eV

Formula: $E_k = h\nu - W$ (equation of a straight line: x -intercept $= \nu_0$, y -intercept $= -W$)

Solution:

(i) The line cuts the ν -axis (where $E_k = 0$) at $\nu = 10 \times 10^{14} \text{ Hz}$.

$$\nu_0 = 10 \times 10^{14} \text{ Hz} = 1 \times 10^{15} \text{ Hz}$$

(ii) At $\nu = 0$ (y-intercept), $E_k = -W$. From the graph this intercept is -4 eV.

$$W = 4 \text{ eV}$$

(iii) At $\nu = 30 \times 10^{14}$ Hz, the graph gives $E_k(\text{max}) = 8$ eV. Since $eV_0 = E_k(\text{max})$,

$$V_0 = 8 \text{ V}$$

Final Answer

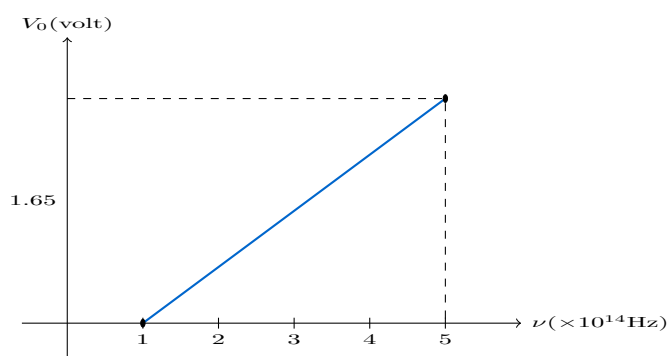
$$\nu_0 = 10 \times 10^{14} \text{ Hz}, \quad W = 4 \text{ eV}, \quad V_0 = 8 \text{ V}$$

Common Mistake: Read the y-intercept (at $\nu = 0$) as $-W$, not $+W$ – the line's equation is $E_k = h\nu - W$.

Question 24

The given figure shows the plot of stopping potential V_0 versus the frequency ν of light used in a photoelectric-effect experiment. Calculate the value of Planck's constant h . (Take $e = 1.6 \times 10^{-19}$ C)

★ Likely Exam Question



V_0 - ν graph: line through $(1 \times 10^{14}, 0)$ and $(5 \times 10^{14}, 1.65 \text{ V})$

Given: From graph: line passes through $(1 \times 10^{14} \text{ Hz}, 0)$ and $(5 \times 10^{14} \text{ Hz}, 1.65 \text{ V})$

Formula: $eV_0 = h\nu - W_0 \Rightarrow \text{slope of } V_0\text{-}\nu \text{ graph} = \frac{h}{e}$

Solution:

$$\text{slope} = \frac{1.65 - 0}{(5 - 1) \times 10^{14}} = \frac{1.65}{4 \times 10^{14}} = 4.125 \times 10^{-15} \text{ V s}$$

$$h = e \times \text{slope} = 1.6 \times 10^{-19} \times 4.125 \times 10^{-15} = 6.6 \times 10^{-34} \text{ J s}$$

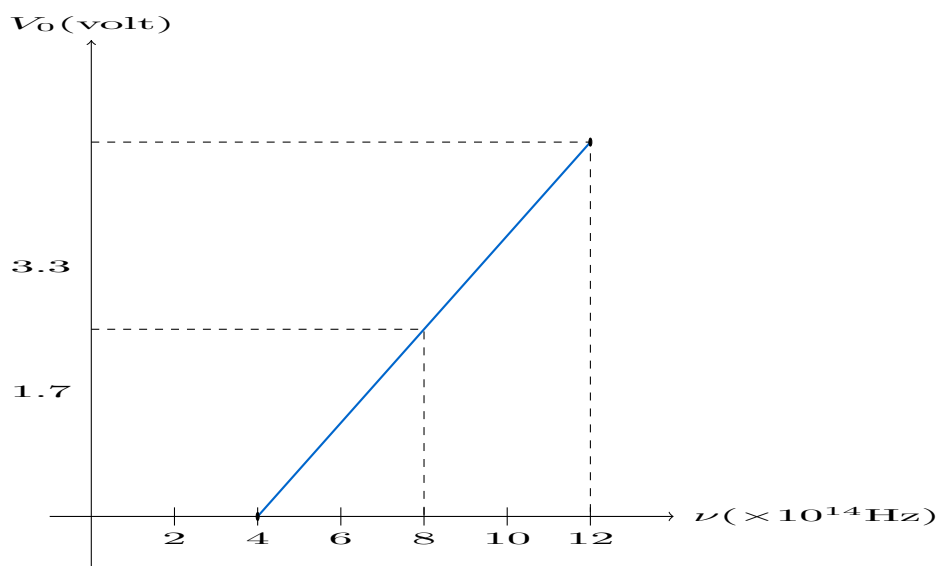
Final Answer

$$h \approx 6.6 \times 10^{-34} \text{ J s}$$

Question 25

The stopping potential V_0 (in volt) as a function of frequency ν for a sodium emitter is shown in the figure. Using only the data plotted on the graph – i.e. without assuming the value of Planck’s constant beforehand – find the work function of sodium.

★ Likely Exam Question



V_0 - ν graph: line through $(4 \times 10^{14}, 0)$ and $(12 \times 10^{14}, 3.3 \text{ V})$

Given: From graph: line passes through $(4 \times 10^{14} \text{ Hz}, 0)$ and $(12 \times 10^{14} \text{ Hz}, 3.3 \text{ V})$

Formula: $W_0 = h\nu_0$, $h = e \times \text{slope of the graph}$

Solution:

Slope of the graph (using only the two plotted points – this gives h/e directly from the graph, without assuming h):

$$\text{slope} = \frac{h}{e} = \frac{3.3 - 0}{(12 - 4) \times 10^{14}} = 4.125 \times 10^{-15} \text{ V s}$$

$$h = e \times 4.125 \times 10^{-15} = 1.6 \times 10^{-19} \times 4.125 \times 10^{-15} = 6.6 \times 10^{-34} \text{ J s}$$

The threshold frequency (x-intercept) read directly from the graph is $\nu_0 = 4 \times 10^{14} \text{ Hz}$. Hence:

$$W_0 = h\nu_0 = 6.6 \times 10^{-34} \times 4 \times 10^{14} = 2.64 \times 10^{-19} \text{ J}$$

Final Answer

$$W_0 = 2.64 \times 10^{-19} \text{ J } (\approx 1.65 \text{ eV})$$

Common Mistake: "Without using the known value of h " means *derive* h from the slope of the graph itself, rather than quoting a memorised textbook value.

Question 26

When a certain metallic surface is illuminated with monochromatic light of wavelength λ , the stopping potential for the photoelectric current is $3V_0$. When the same surface is illuminated with light of wavelength 2λ , the stopping potential is V_0 . Find the threshold wavelength for the surface.

Given: At λ : stopping potential = $3V_0$; at 2λ : stopping potential = V_0

Formula: $eV'_0 = \frac{hc}{\lambda'} - W_0$

Solution:

$$e(3V_0) = \frac{hc}{\lambda} - W_0 \quad \dots(1)$$

$$e(V_0) = \frac{hc}{2\lambda} - W_0 \quad \dots(2)$$

Subtracting (2) from (1):

$$2eV_0 = \frac{hc}{\lambda} - \frac{hc}{2\lambda} = \frac{hc}{2\lambda} \Rightarrow eV_0 = \frac{hc}{4\lambda}$$

Substituting back into (2):

$$W_0 = \frac{hc}{2\lambda} - eV_0 = \frac{hc}{2\lambda} - \frac{hc}{4\lambda} = \frac{hc}{4\lambda}$$

Threshold wavelength:

$$\lambda_0 = \frac{hc}{W_0} = \frac{hc}{\left(\frac{hc}{4\lambda}\right)} = 4\lambda$$

Final Answer

$$\lambda_0 = 4\lambda$$

Common Mistake: Keep λ symbolic throughout – do not substitute a numeric value where none is given; the answer is meant to be expressed in terms of λ .

Question 27

When the energy of the incident radiation is increased by 40%, the kinetic energy of the photoelectrons emitted from a metal surface increases from 0.5 eV to 1.0 eV. Find the work function of the metal surface.

Given: $E_{k1} = 0.5 \text{ eV}$, $E_{k2} = 1.0 \text{ eV}$, $E_2 = 1.4E_1$

Formula: $E_{k1} = E_1 - W_0$, $E_{k2} = E_2 - W_0$

Solution:

Subtracting the equations:

$$E_{k2} - E_{k1} = E_2 - E_1 = 1.4E_1 - E_1 = 0.4E_1$$

$$0.4E_1 = 1.0 - 0.5 = 0.5 \text{ eV} \Rightarrow E_1 = 1.25 \text{ eV}$$

$$W_0 = E_1 - E_{k1} = 1.25 - 0.5 = 0.75 \text{ eV}$$

Final Answer

$$W_0 = 0.75 \text{ eV}$$

Question 28

In a photoelectric experiment, ultraviolet light of wavelength 280 nm is used with a lithium cathode of work function $\phi = 2.5 \text{ eV}$. If the wavelength of the incident light is switched to 400 nm, find the change in stopping potential.

Given: $\lambda_1 = 280 \text{ nm}$, $\lambda_2 = 400 \text{ nm}$, $\phi = 2.5 \text{ eV}$

Formula: $eV_0 = \frac{hc}{\lambda} - \phi \quad \left(\frac{hc}{e} \approx 1237.5 \text{ eV nm} \right)$

Solution:

$$\frac{hc}{e\lambda_1} = \frac{1237.5}{280} = 4.42 \text{ eV} \Rightarrow V_{01} = 4.42 - 2.5 = 1.92 \text{ V}$$

$$\frac{hc}{e\lambda_2} = \frac{1237.5}{400} = 3.09 \text{ eV} \Rightarrow V_{02} = 3.09 - 2.5 = 0.59 \text{ V}$$

$$\Delta V_0 = V_{01} - V_{02} = 1.92 - 0.59 = 1.33 \text{ V}$$

Final Answer

Stopping potential decreases by about 1.3 V on switching from 280 nm to 400 nm

IV. de Broglie Wavelength When Momentum or Velocity is Given

Question 29

What is the de Broglie wavelength of a 2 kg object moving at a speed of 1 m/s?

Given: $m = 2 \text{ kg}$, $v = 1 \text{ m/s}$

Formula: $\lambda = \frac{h}{mv}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34}}{2 \times 1} = 3.3 \times 10^{-34} \text{ m}$$

Final Answer

$$\lambda = 3.3 \times 10^{-34} \text{ m}$$

Common Mistake: This confirms why de Broglie waves are never observed for macroscopic objects – the wavelength is immeasurably small.

Question 30

Calculate the de Broglie wavelength of a proton whose momentum is $2.55 \times 10^{-22} \text{ kg m/s}$.

Given: $p = 2.55 \times 10^{-22} \text{ kg m/s}$

Formula: $\lambda = \frac{h}{p}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34}}{2.55 \times 10^{-22}} = 2.588 \times 10^{-12} \text{ m}$$

Final Answer

$$\lambda \approx 2.59 \times 10^{-12} \text{ m}$$

Question 31

A moving particle has a de Broglie wavelength of $1.0 \text{ \AA}$. What is the momentum of the particle?

Given: $\lambda = 1.0 \text{ \AA} = 1 \times 10^{-10} \text{ m}$

Formula: $p = \frac{h}{\lambda}$

Solution:

$$p = \frac{6.6 \times 10^{-34}}{1 \times 10^{-10}} = 6.6 \times 10^{-24} \text{ kg m/s}$$

Final Answer

$$p = 6.6 \times 10^{-24} \text{ kg m/s}$$

Question 32

Calculate the de Broglie wavelength of electrons moving with a speed of 0.5 km/s .

Given: $v = 0.5 \text{ km/s} = 500 \text{ m/s}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$

Formula: $\lambda = \frac{h}{m_e v}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times 500} = \frac{6.6 \times 10^{-34}}{4.55 \times 10^{-28}} = 1.45 \times 10^{-6} \text{ m}$$

Final Answer

$$\lambda \approx 1.45 \times 10^{-6} \text{ m}$$

V. de Broglie Wavelength When Kinetic Energy is Given

Question 33

Find the de Broglie wavelength of a proton of energy 1.0 keV.

Given: $E_k = 1.0 \text{ keV} = 1000 \text{ eV} = 1.6 \times 10^{-16} \text{ J}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$

Formula: $\lambda = \frac{h}{\sqrt{2m_p E_k}}$

Solution:

$$2m_p E_k = 2 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-16} = 5.344 \times 10^{-43}$$

$$\sqrt{5.344 \times 10^{-43}} = 7.31 \times 10^{-22}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{7.31 \times 10^{-22}} = 9.03 \times 10^{-13} \text{ m} = 9.0 \times 10^{-3} \text{ \AA}$$

Final Answer

$$\lambda \approx 9.0 \times 10^{-13} \text{ m} = 9.0 \times 10^{-3} \text{ \AA}$$

Common Mistake: Convert keV to J via eV (1 keV = 1000 eV) before using it inside the square root – a common slip is forgetting the factor of 1000.

Question 34

Find the de Broglie wavelength of a neutron of mass $1.67 \times 10^{-24} \text{ g}$ moving with kinetic energy 0.04 eV.

Given: $m_n = 1.67 \times 10^{-24} \text{ g} = 1.67 \times 10^{-27} \text{ kg}$, $E_k = 0.04 \text{ eV} = 6.4 \times 10^{-21} \text{ J}$

Formula: $\lambda = \frac{h}{\sqrt{2m_n E_k}}$

Solution:

$$2m_n E_k = 2 \times 1.67 \times 10^{-27} \times 6.4 \times 10^{-21} = 2.138 \times 10^{-47}$$

$$\sqrt{2.138 \times 10^{-47}} = 4.623 \times 10^{-24}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{4.623 \times 10^{-24}} = 1.428 \times 10^{-10} \text{ m} = 1.43 \text{ \AA}$$

Final Answer

$$\lambda \approx 1.43 \text{ \AA}$$

Common Mistake: Convert the mass from grams to kilograms first (divide by 1000)
– an easy unit slip when the mass is given in g.

VI. de Broglie Wavelength When Accelerating Potential is Given

Question 35

Calculate the de Broglie wavelength of an electron of energy 400 eV.

★ Likely Exam Question

Given: $V = 400 \text{ V}$ (equivalently, $E_k = 400 \text{ eV}$)

Formula: $\lambda = \frac{h}{\sqrt{2m_e eV}} \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$

Solution:

$$\lambda = \frac{12.27}{\sqrt{400}} = \frac{12.27}{20} = 0.6135 \text{ \AA}$$

Final Answer

$$\lambda \approx 0.61 \text{ \AA}$$

Question 36

Calculate the de Broglie wavelength of an electron beam accelerated through a potential difference of 60 V.

★ Likely Exam Question

Given: $V = 60 \text{ V}$

Formula: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$

Solution:

$$\lambda = \frac{12.27}{\sqrt{60}} = \frac{12.27}{7.746} = 1.584 \text{ \AA}$$

Final Answer

$$\lambda \approx 1.58 \text{ \AA}$$

Question 37

In a diode valve, the anode has a potential of 100 V relative to the cathode. Find the de Broglie wavelength of electrons on reaching the anode.

Given: $V = 100 \text{ V}$

Formula: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$

Solution:

$$\lambda = \frac{12.27}{\sqrt{100}} = \frac{12.27}{10} = 1.227 \text{ \AA}$$

Final Answer

$$\lambda \approx 1.23 \text{ \AA}$$

Question 38

What potential must be applied to an electron microscope to produce electrons of wavelength 1.0 Å?

★ Likely Exam Question

Given: $\lambda = 1.0 \text{ \AA}$

Formula: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA} \Rightarrow V = \left(\frac{12.27}{\lambda} \right)^2$

Solution:

$$V = \left(\frac{12.27}{1.0} \right)^2 = 150.55 \text{ V}$$

Final Answer

$$V \approx 150.6 \text{ V}$$

Common Mistake: The formula $\lambda \approx 12.27/\sqrt{V} \text{ \AA}$ applies only to electrons – for protons/neutrons use $\lambda = h/\sqrt{2mE_k}$ directly with the correct mass.

ISC Class XII Physics

Chapter 12 Atoms

30 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. α -Scattering Experiments
- II. Radii of Orbits and Velocities in these Orbits
- III. Energy of Electron in Different Orbits
- IV. Energy Emitted/Absorbed in Transitions
- V. Excitation and Ionisation Potentials
- VI. Series in Hydrogen Spectrum

Formula Sheet

I. α -Scattering Experiments

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{E}$$

r_0 = distance of closest approach, Z_1, Z_2 = charge numbers, E = kinetic energy of incident particle

II. Radii of Orbits and Velocities

$$r_n = n^2 r_1, \quad r_1 = 0.529 \text{ \AA (for hydrogen, } Z = 1)$$

r_n = radius of n th Bohr orbit; radius scales as n^2/Z and inversely with orbiting particle's mass

II. Radii of Orbits and Velocities

$$v_n = \frac{e^2}{2\epsilon_0 n h}$$

v_n = orbital speed in n th orbit; independent of the orbiting particle's mass

II. Radii of Orbits and Velocities

$$T = \frac{2\pi r_n}{v_n}$$

T = time period of revolution in the n th orbit

III. Energy of Electron in Different Orbits

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV}$$

E_n = total energy in n th orbit; for a Bohr orbit, $KE = -E_n$ and $PE = 2E_n$

IV/V. Transitions, Excitation and Ionisation

$$\Delta E = E_i - E_f = h\nu = \frac{hc}{\lambda}$$

ΔE = photon energy absorbed/emitted; ν = frequency, λ = wavelength

V. Excitation and Ionisation Potentials

$$E_{\text{ionisation}} = -E_1 = 13.6 Z^2 \text{ eV}$$

Ionisation energy = energy to take electron from ground state ($n = 1$) to $n = \infty$; numeri-

cally equal to ionisation potential in volts

VI. Series in Hydrogen Spectrum

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

R = Rydberg constant; n_1 = lower level (series), $n_2 > n_1$ = upper level

VI. Series in Hydrogen Spectrum

$$E_{\text{ionisation}} = Rhc$$

Series-limit (ionisation) energy obtained by putting $n_1 = 1$, $n_2 \rightarrow \infty$

I. α -Scattering Experiments

Question 1

α -particles of 5.0 MeV are being scattered by a thin copper ($Z = 29$) foil. Up to what closest distance can these particles reach the nucleus of copper? If there be protons of the same energy, then?

★ Likely Exam Question

Given: $E = 5.0 \text{ MeV} = 8.0 \times 10^{-13} \text{ J}$; $Z = 29$ (copper); charge on α -particle $= 2e$; charge on proton $= e$; $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N m}^2\text{C}^{-2}$

Formula: $r_0 = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{E}$

Solution:

For the α -particle ($Z_1 = 2$, $Z_2 = 29$):

$$\begin{aligned} r_0 &= \frac{9.0 \times 10^9 \times (2 \times 29) \times (1.6 \times 10^{-19})^2}{8.0 \times 10^{-13}} \\ &= \frac{9.0 \times 10^9 \times 58 \times 2.56 \times 10^{-38}}{8.0 \times 10^{-13}} \\ &= 1.67 \times 10^{-14} \text{ m} \end{aligned}$$

For a proton of the same energy ($Z_1 = 1$), the numerator is exactly halved:

$$r'_0 = \frac{9.0 \times 10^9 \times 29 \times 2.56 \times 10^{-38}}{8.0 \times 10^{-13}} = 0.835 \times 10^{-14} \text{ m}$$

Final Answer

$$r_0(\alpha) = 1.67 \times 10^{-14} \text{ m}; \quad r_0(\text{proton}) = 0.835 \times 10^{-14} \text{ m}$$

Common Mistake: Do not forget to use the charge on the α -particle as $2e$ (not e) – this is the most common slip in this type of problem.

Question 2

In a head-on collision between an α -particle and a gold nucleus ($Z = 79$), the distance of closest approach is $4.0 \times 10^{-14} \text{ m}$. Calculate the energy of the particle.

Given: $r_0 = 4.0 \times 10^{-14}$ m; $Z_2 = 79$ (gold); $Z_1 = 2$ (for α -particle); $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9$ N m²C⁻²

Formula: $E = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{r_0}$

Solution:

$$\begin{aligned} E &= \frac{9.0 \times 10^9 \times (2 \times 79) \times (1.6 \times 10^{-19})^2}{4.0 \times 10^{-14}} \\ &= \frac{9.0 \times 10^9 \times 158 \times 2.56 \times 10^{-38}}{4.0 \times 10^{-14}} \\ &= 9.10 \times 10^{-13} \text{ J} \end{aligned}$$

Converting to MeV:

$$E = \frac{9.10 \times 10^{-13}}{1.6 \times 10^{-13}} \text{ MeV} = 5.69 \text{ MeV}$$

Final Answer

$$E = 5.69 \text{ MeV}$$

II. Radii of Orbits and Velocities in these Orbits

Question 3

The radius of the first orbit of the electron of a hydrogen atom is 5.3×10^{-11} m. What is the radius of its second orbit? (ISC 2006)

★ Likely Exam Question

Given: $r_1 = 5.3 \times 10^{-11}$ m; $n = 2$

Formula: $r_n = n^2 r_1$

Solution:

$$r_2 = (2)^2 \times 5.3 \times 10^{-11} = 4 \times 5.3 \times 10^{-11} = 21.2 \times 10^{-11} \text{ m}$$

Final Answer

$$r_2 = 21.2 \times 10^{-11} \text{ m}$$

Question 4

Show that the velocity of electron in the first orbit of hydrogen atom is $(1/137)$ times the velocity of light.

Given: $n = 1$ (first orbit of hydrogen); $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N m}^2\text{C}^{-2}$; $h = 6.6 \times 10^{-34} \text{ J s}$;
 $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v_n = \frac{1}{4\pi\epsilon_0} \frac{2\pi e^2}{nh}$

Solution:

For $n = 1$:

$$\begin{aligned} v_1 &= \frac{9.0 \times 10^9 \times 2\pi \times (1.6 \times 10^{-19})^2}{1 \times 6.6 \times 10^{-34}} \\ &= \frac{9.0 \times 10^9 \times 6.283 \times 2.56 \times 10^{-38}}{6.6 \times 10^{-34}} \\ &= 2.19 \times 10^6 \text{ m/s} \end{aligned}$$

Now compare with $c/137$:

$$\frac{c}{137} = \frac{3.0 \times 10^8}{137} = 2.19 \times 10^6 \text{ m/s}$$

Since $v_1 = c/137$, the result is verified.

Final Answer

$$v_1 = 2.19 \times 10^6 \text{ m/s} = \frac{c}{137} \quad (\text{Hence proved})$$

Question 5

A negatively-charged pion (π^-), whose mass is 273 times the mass of electron, moves in a circular orbit around a proton. It is a pi-mesic hydrogen atom. How many revolutions will the pion make during its lifetime of 10^{-8} s in the ground state of the atom?

Given: $m_\pi = 273 m_e$; lifetime $t = 10^{-8}$ s; $n = 1$; $r_1(\text{electron}) = 5.3 \times 10^{-11}$ m; $v_1 = 2.19 \times 10^6$ m/s

Formula: $r_1(\pi) = \frac{r_1(e)}{273}$ (radius $\propto 1/m$, mass only); v_n is independent of mass;
 $T = \frac{2\pi r_1(\pi)}{v_1}$; $N = \frac{t}{T}$

Solution:

The orbital speed v_1 does not depend on the orbiting particle's mass, so it remains 2.19×10^6 m/s. Only the radius shrinks because it is inversely proportional to mass:

$$r_1(\pi) = \frac{5.3 \times 10^{-11}}{273} = 1.94 \times 10^{-13} \text{ m}$$

Time period of revolution:

$$T = \frac{2\pi \times 1.94 \times 10^{-13}}{2.19 \times 10^6} = 5.56 \times 10^{-19} \text{ s}$$

Number of revolutions in the given lifetime:

$$N = \frac{10^{-8}}{5.56 \times 10^{-19}} = 1.8 \times 10^{10}$$

Final Answer

$$N = 1.8 \times 10^{10} \text{ revolutions}$$

Common Mistake: Students often (wrongly) scale the velocity along with the radius. In the Bohr model v_n is completely independent of the orbiting particle's mass – only r_n and E_n depend on it.

III. Energy of Electron in Different Orbits

Question 6

The energy of electron in the first orbit in hydrogen atom is -13.6 eV. What will be the energy in the second orbit?

★ Likely Exam Question

Given: $E_1 = -13.6$ eV; $n = 2$

Formula: $E_n = -\frac{13.6}{n^2}$ eV

Solution:

$$E_2 = -\frac{13.6}{(2)^2} = -\frac{13.6}{4} = -3.4 \text{ eV}$$

Final Answer

$$E_2 = -3.4 \text{ eV}$$

Question 7

Which level of the doubly-ionised lithium (Li^{++}), with $Z = 3$, has the same energy as the ground-state energy of hydrogen atom ($Z = 1$)? Compare the orbital radii of the two levels.

Given: $Z_{\text{Li}} = 3$; $Z_{\text{H}} = 1$; $E_1(\text{H}) = -13.6$ eV

Formula: $E_n = -\frac{13.6 Z^2}{n^2}$ eV; $r_n = \frac{0.529 n^2}{Z}$ Å

Solution:

Equate the energy of Li^{++} level n to the hydrogen ground-state energy:

$$-\frac{13.6 \times 3^2}{n^2} = -13.6 \Rightarrow \frac{9}{n^2} = 1 \Rightarrow n^2 = 9 \Rightarrow n = 3$$

Ratio of orbital radii:

$$\frac{r_3(\text{Li}^{++})}{r_1(\text{H})} = \frac{n^2/Z_{\text{Li}}}{1^2/Z_{\text{H}}} = \frac{9/3}{1} = 3$$

Final Answer

$$n = 3 \text{ level of Li}^{++}; \quad \frac{r_3(\text{Li}^{++})}{r_1(\text{H})} = 3$$

IV. Energy Emitted/Absorbed in Transitions

Question 8

The electron in a given Bohr orbit has a total energy of -1.5 eV. Calculate its (i) kinetic energy (ii) potential energy and (iii) wavelength of light emitted, when electron makes a transition to the ground state. Ground state energy is -13.6 eV.

Given: $E = -1.5$ eV; $E_1 = -13.6$ eV

Formula: $KE = -E$; $PE = 2E$; $\lambda = \frac{hc}{\Delta E}$

Solution:

(i) Kinetic energy:

$$KE = -E = -(-1.5) = 1.5 \text{ eV}$$

(ii) Potential energy:

$$PE = 2E = 2 \times (-1.5) = -3.0 \text{ eV}$$

(iii) Energy released on transition to ground state:

$$\Delta E = (-1.5) - (-13.6) = 12.1 \text{ eV} = 12.1 \times 1.6 \times 10^{-19} \text{ J} = 1.936 \times 10^{-18} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{1.936 \times 10^{-18}} = 1022.7 \times 10^{-10} \text{ m} = 1022.7 \text{ \AA}$$

Final Answer

(i) $KE = 1.5$ eV (ii) $PE = -3.0$ eV (iii) $\lambda = 1022.7 \text{ \AA}$

Common Mistake: Total energy $E = -KE = PE/2$ holds only in the Bohr model; students often confuse the sign of PE.

Question 9

The energy of an electron in the n th orbit of hydrogen atom is given by $E_n = -(13.6/n^2)$ eV. Find energy required to take the electron from the ground state to the (i) first excited state (ii) second excited state.

★ Likely Exam Question

Given: $E_1 = -13.6$ eV; first excited state $n = 2$; second excited state $n = 3$

Formula: $\Delta E = E_n - E_1$

Solution:(i) To $n = 2$:

$$\Delta E = -\frac{13.6}{4} - (-13.6) = -3.4 + 13.6 = 10.2 \text{ eV}$$

(ii) To $n = 3$:

$$\Delta E = -\frac{13.6}{9} - (-13.6) = -1.51 + 13.6 = 12.09 \text{ eV}$$

Final Answer

(i) 10.2 eV (ii) 12.09 eV

Question 10

Calculate the frequency of the photon, which can excite the electron to -3.4 eV from -13.6 eV .

Given: $E_i = -13.6 \text{ eV}$; $E_f = -3.4 \text{ eV}$; Hint: $\Delta E = h\nu$

Formula: $\nu = \frac{\Delta E}{h}$

Solution:

$$\Delta E = -3.4 - (-13.6) = 10.2 \text{ eV} = 10.2 \times 1.6 \times 10^{-19} = 1.632 \times 10^{-18} \text{ J}$$

$$\nu = \frac{1.632 \times 10^{-18}}{6.6 \times 10^{-34}} = 2.47 \times 10^{15} \text{ Hz}$$

Final Answer

$$\nu = 2.47 \times 10^{15} \text{ Hz}$$

Question 11

The energy of an electron in the n th state of hydrogen is given by $E_n = -(13.6/n^2) \text{ eV}$. Calculate the energy and frequency of a photon radiated when the electron de-excites from the first excited state to the ground state. To which spectral series does this photon belong?

Given: first excited state $n = 2$; ground state $n = 1$

Formula: $\Delta E = E_1 - E_2$; $\nu = \Delta E/h$

Solution:

$$E_1 = -13.6 \text{ eV}, \quad E_2 = -3.4 \text{ eV}$$

$$\Delta E = |-13.6 - (-3.4)| = 10.2 \text{ eV} = 1.632 \times 10^{-18} \text{ J}$$

$$\nu = \frac{1.632 \times 10^{-18}}{6.6 \times 10^{-34}} = 2.47 \times 10^{15} \text{ Hz}$$

Since the electron lands on $n = 1$, the photon belongs to the **Lyman series**.

Final Answer

$$\Delta E = 10.2 \text{ eV}, \nu = 2.47 \times 10^{15} \text{ Hz, Lyman series}$$

Question 12

The energy of the lowest level of hydrogen atom is -13.6 eV . Calculate the energy, frequency and wavelength of the photon emitted by transition from $n = 4$ to $n = 2$.

★ Likely Exam Question

Given: $n_i = 4; n_f = 2; E_1 = -13.6 \text{ eV}$

Formula: $E_n = -\frac{13.6}{n^2} \text{ eV}; \quad \nu = \frac{\Delta E}{h}; \quad \lambda = \frac{c}{\nu}$

Solution:

$$E_4 = -\frac{13.6}{16} = -0.85 \text{ eV}, \quad E_2 = -\frac{13.6}{4} = -3.4 \text{ eV}$$

$$\Delta E = |-0.85 - (-3.4)| = 2.55 \text{ eV} = 2.55 \times 1.6 \times 10^{-19} = 4.08 \times 10^{-19} \text{ J}$$

$$\nu = \frac{4.08 \times 10^{-19}}{6.6 \times 10^{-34}} = 6.18 \times 10^{14} \text{ Hz}$$

$$\lambda = \frac{3.0 \times 10^8}{6.18 \times 10^{14}} = 4.853 \times 10^{-7} \text{ m} = 4853 \text{ Å}$$

Final Answer

$$\Delta E = 2.55 \text{ eV}, \nu = 6.18 \times 10^{14} \text{ Hz}, \lambda = 4853 \text{ Å}$$

Question 13

The energy of electron orbiting in the n th orbit of hydrogen atom is given by $E_n = -(13.6/n^2) \text{ eV}$. Calculate: (i) energy liberated when the electron jumps from the second to the first orbit (ii) wavelength of the light emitted in this transition (iii) the ionisation energy of the atom.

★ Likely Exam Question

Given: $n_i = 2, n_f = 1; E_1 = -13.6 \text{ eV}, E_2 = -3.4 \text{ eV}$

Formula: $\Delta E = E_1 - E_2; \quad \lambda = \frac{hc}{\Delta E}; \quad E_{\text{ionisation}} = -E_1$

Solution:

(i)

$$\Delta E = |-13.6 - (-3.4)| = 10.2 \text{ eV}$$

(ii)

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{10.2 \times 1.6 \times 10^{-19}} = \frac{1.98 \times 10^{-25}}{1.632 \times 10^{-18}} = 1213 \times 10^{-10} \text{ m} = 1213 \text{ Å}$$

(iii)

$$E_{\text{ionisation}} = -E_1 = 13.6 \text{ eV}$$

Final Answer

(i) 10.2 eV (ii) 1213 Å (iii) 13.6 eV

Question 14

The energies of two energy-levels of hydrogen atom are -3.4 eV and -1.5 eV . If electron transition occurs from the higher to the lower level, how much energy will be emitted? What will be the wavelength of the emitted radiation?

Given: $E_{\text{higher}} = -1.5 \text{ eV}$; $E_{\text{lower}} = -3.4 \text{ eV}$

Formula: $\Delta E = E_{\text{higher}} - E_{\text{lower}}$; $\lambda = \frac{hc}{\Delta E}$

Solution:

$$\Delta E = -1.5 - (-3.4) = 1.9 \text{ eV} = 1.9 \times 1.6 \times 10^{-19} = 3.04 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{3.04 \times 10^{-19}} = 6.513 \times 10^{-7} \text{ m} = 6513 \text{ Å}$$

Final Answer $\Delta E = 1.9 \text{ eV}$, $\lambda = 6513 \text{ Å}$

Common Mistake: The less negative energy is the higher energy state (-1.5 eV is higher than -3.4 eV) – do not simply subtract the smaller magnitude from the larger one without checking signs.

Question 15

The energy of hydrogen atom in its ground state is -13.6 eV . The atom is given an energy of 10.2 eV . In which state will it reach? What will be the wavelength of the absorbed photon in this process?

Given: $E_1 = -13.6 \text{ eV}$; energy absorbed = 10.2 eV

Formula: $E_n = E_1 + \Delta E$; $\lambda = \frac{hc}{\Delta E}$

Solution:

$$E_n = -13.6 + 10.2 = -3.4 \text{ eV}$$

Since $E_2 = -13.6/4 = -3.4 \text{ eV}$, the atom reaches the state $n = 2$.

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{10.2 \times 1.6 \times 10^{-19}} = 1213 \times 10^{-10} \text{ m} = 1213 \text{ \AA}$$

Final Answer

$n = 2$ (first excited state); $\lambda = 1213 \text{ \AA}$

V. Excitation and Ionisation Potentials

Question 16

The wavelength of the first spectral line of sodium is $5896 \text{ \AA}$. What is the first excitation potential of sodium atom?

Given: $\lambda = 5896 \text{ \AA} = 5896 \times 10^{-10} \text{ m}$

Formula: $\Delta E = \frac{hc}{\lambda}$; excitation potential (V) = ΔE (in eV)

Solution:

$$\Delta E = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{5896 \times 10^{-10}} = 3.358 \times 10^{-19} \text{ J}$$

$$\Delta E = \frac{3.358 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.1 \text{ eV}$$

Final Answer

First excitation potential = 2.1 V

Question 17

The ionisation-potential of hydrogen atom is 13.6 V. Find out: (i) wavelength of H_β line of Balmer series (ii) shortest wavelength of Lyman series.

★ Likely Exam Question

Given: ionisation energy = 13.6 eV; H_β : $n = 4 \rightarrow n = 2$

Formula: $\Delta E = E_{n_f} - E_{n_i}$; $\lambda = \frac{hc}{\Delta E}$

Solution:

(i) For H_β ($n = 4 \rightarrow 2$):

$$E_4 = -0.85 \text{ eV}, \quad E_2 = -3.4 \text{ eV}, \quad \Delta E = 2.55 \text{ eV}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{2.55 \times 1.6 \times 10^{-19}} = 4853 \times 10^{-10} \text{ m} = 4853 \text{ \AA}$$

(ii) Shortest wavelength of Lyman series corresponds to $n = \infty \rightarrow n = 1$, i.e. $\Delta E = 13.6 \text{ eV}$:

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{13.6 \times 1.6 \times 10^{-19}} = 910 \times 10^{-10} \text{ m} = 910 \text{ \AA}$$

Final Answer

(i) $\lambda(H_\beta) = 4853 \text{ \AA}$ (ii) $\lambda_{\text{shortest, Lyman}} = 910 \text{ \AA}$

Question 18

Calculate the wavelength of the H_α line of hydrogen spectrum. The energy levels of hydrogen atom are given by the expression $E_n = -(13.6/n^2) \text{ eV}$.

Given: H_α line: $n = 3 \rightarrow n = 2$

Formula: $\Delta E = E_2 - E_3; \quad \lambda = \frac{hc}{\Delta E}$

Solution:

$$E_3 = -\frac{13.6}{9} = -1.51 \text{ eV}, \quad E_2 = -\frac{13.6}{4} = -3.4 \text{ eV}$$

$$\Delta E = |-3.4 - (-1.51)| \approx 1.89 \text{ eV} = 1.89 \times 1.6 \times 10^{-19} = 3.024 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{3.024 \times 10^{-19}} = 6547.6 \times 10^{-10} \text{ m} = 6547.6 \text{ \AA}$$

Final Answer

$\lambda(H_\alpha) = 6547.6 \text{ \AA}$

Question 19

The first excitation potential of mercury is 4.86 V. A mercury atom, after absorbing a photon, reaches its first excited state. Find the wavelength of the absorbed photon.

Given: excitation potential = 4.86 V $\Rightarrow \Delta E = 4.86 \text{ eV}$

Formula: $\lambda = \frac{hc}{\Delta E}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{4.86 \times 1.6 \times 10^{-19}} = \frac{1.98 \times 10^{-25}}{7.776 \times 10^{-19}} = 2546 \times 10^{-10} \text{ m} = 2546 \text{ \AA}$$

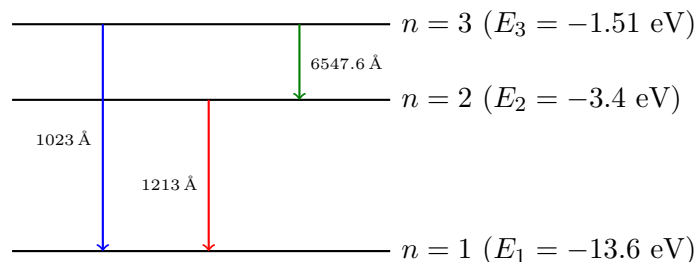
Final Answer

$\lambda = 2546 \text{ \AA}$

Question 20

The first and second excitation potentials of hydrogen atom are respectively 10.2 V and 12.09 V. Draw, on this basis, the energy-level diagram of the atom and show all possible emission transitions in it and mark their corresponding wavelengths.

★ Likely Exam Question



Energy-level diagram of hydrogen showing the three possible emission transitions ($n=2 \rightarrow 1$, $n=3 \rightarrow 1$, $n=3 \rightarrow 2$)

Given: first excitation potential = 10.2 V $\Rightarrow E_2 = -3.4$ eV; second excitation potential = 12.09 V $\Rightarrow E_3 = -1.51$ eV; $E_1 = -13.6$ eV

Formula: $\Delta E = E_{n_f} - E_{n_i}$; $\lambda = \frac{hc}{\Delta E}$

Solution:

Three transitions are possible: $2 \rightarrow 1$, $3 \rightarrow 1$, $3 \rightarrow 2$.

$$\Delta E(2 \rightarrow 1) = 10.2 \text{ eV} \Rightarrow \lambda = \frac{1.98 \times 10^{-25}}{10.2 \times 1.6 \times 10^{-19}} = 1213 \text{ Å}$$

$$\Delta E(3 \rightarrow 1) = 12.09 \text{ eV} \Rightarrow \lambda = \frac{1.98 \times 10^{-25}}{12.09 \times 1.6 \times 10^{-19}} = 1023 \text{ Å}$$

$$\Delta E(3 \rightarrow 2) = 12.09 - 10.2 = 1.89 \text{ eV} \Rightarrow \lambda = \frac{1.98 \times 10^{-25}}{1.89 \times 1.6 \times 10^{-19}} = 6547.6 \text{ Å}$$

Final Answer

$$\lambda(2 \rightarrow 1) = 1213 \text{ Å}, \lambda(3 \rightarrow 1) = 1023 \text{ Å}, \lambda(3 \rightarrow 2) = 6547.6 \text{ Å}$$

Question 21

The ionisation potential of hydrogen atom is 13.6 V. In the lowest energy-level, this atom is ionised by absorbing a photon of 800 Å. What will be the kinetic energy of the released electron?

Given: ionisation energy = 13.6 eV; $\lambda_{\text{photon}} = 800 \text{ Å} = 800 \times 10^{-10} \text{ m}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}$; $KE_{\text{electron}} = E_{\text{photon}} - E_{\text{ionisation}}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{800 \times 10^{-10}} = 2.475 \times 10^{-18} \text{ J} = \frac{2.475 \times 10^{-18}}{1.6 \times 10^{-19}} = 15.47 \text{ eV}$$

$$KE = 15.47 - 13.6 = 1.87 \text{ eV}$$

Final Answer

$$KE = 1.87 \text{ eV}$$

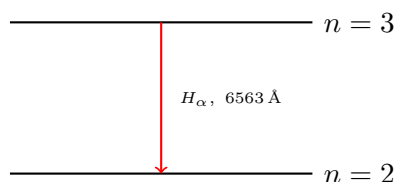
Common Mistake: The photon energy must exceed the ionisation energy; the *excess* energy (not the full photon energy) becomes the electron's kinetic energy.

VI. Series in Hydrogen Spectrum

Question 22

Calculate the wavelength of the first line of the Balmer series of hydrogen. Indicate the transition on an energy-level diagram. (i) What is this line called? (ii) Does it lie in the visible region of the electromagnetic spectrum?

★ Likely Exam Question



First line of the Balmer series: transition $n = 3 \rightarrow n = 2$, the H_α line

Given: $n_i = 3$, $n_f = 2$; $R = 1.097 \times 10^7 \text{ m}^{-1}$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Solution:

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.097 \times 10^7 \times \frac{5}{36} = 1.524 \times 10^6 \text{ m}^{-1}$$

$$\lambda = \frac{1}{1.524 \times 10^6} = 6563 \times 10^{-10} \text{ m} = 6563 \text{ Å}$$

(i) This line is called the H_α **line**. (ii) **Yes**, 6563 Å lies in the visible (red) region of the electromagnetic spectrum.

Final Answer

$\lambda = 6563 \text{ Å}$; (i) H_α line; (ii) Yes, lies in the visible region

Question 23

The wavelength of the first line of Balmer series is 6563 Å. Calculate (i) Rydberg's constant, (ii) ionisation potential of hydrogen atom.

★ Likely Exam Question

Given: $\lambda = 6563 \text{ \AA} = 6563 \times 10^{-10} \text{ m}$ (first Balmer line, $n = 3 \rightarrow n = 2$)

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{9} \right); \quad E_{\text{ionisation}} = Rhc$

Solution:

(i)

$$R = \frac{36}{5\lambda} = \frac{36}{5 \times 6563 \times 10^{-10}} = 1.097 \times 10^7 \text{ m}^{-1}$$

(ii)

$$E_{\text{ionisation}} = Rhc = 1.097 \times 10^7 \times 6.6 \times 10^{-34} \times 3.0 \times 10^8 = 2.172 \times 10^{-18} \text{ J}$$

$$E_{\text{ionisation}} = \frac{2.172 \times 10^{-18}}{1.6 \times 10^{-19}} = 13.6 \text{ eV}$$

Final Answer

(i) $R = 1.097 \times 10^7 \text{ m}^{-1}$ (ii) Ionisation potential = 13.6 V

Question 24

The energy transition in hydrogen atom takes place from energy level $n = 8$ to $n = 1$. Calculate the wavelength of the emitted photon. To which spectral series will this photon belong? To which part of the electromagnetic spectrum will this photon belong? ($R = 1.09 \times 10^7$ per metre.)

★ Likely Exam Question

Given: $n_i = 8, n_f = 1; R = 1.09 \times 10^7 \text{ m}^{-1}$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Solution:

$$\frac{1}{\lambda} = 1.09 \times 10^7 \left(1 - \frac{1}{64} \right) = 1.09 \times 10^7 \times \frac{63}{64} = 1.073 \times 10^7 \text{ m}^{-1}$$

$$\lambda = \frac{1}{1.073 \times 10^7} = 932 \times 10^{-10} \text{ m} = 932 \text{ \AA}$$

Since the electron lands on $n = 1$, this photon belongs to the **Lyman series**, which lies in the **ultraviolet** region of the electromagnetic spectrum.

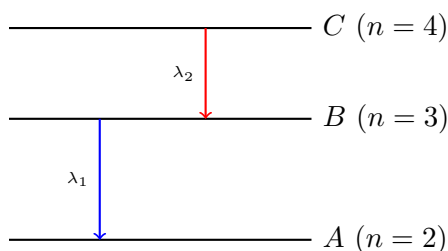
Final Answer

$\lambda = 932 \text{ \AA}$; Lyman series; ultraviolet region

Question 25

As per the given figure, A , B and C are the first, second and third excited states of a hydrogen atom respectively. If the ratio of the two wavelengths (i.e. λ_1/λ_2) is $7/4n$, find the value of n .

★ Likely Exam Question



Energy-level diagram: $A(n=2)$, $B(n=3)$, $C(n=4)$; λ_1 is the transition $B \rightarrow A$, λ_2 is the transition $C \rightarrow B$

Given: $A : n = 2$ (first excited state); $B : n = 3$ (second excited state); $C : n = 4$ (third excited state); λ_1 : transition $B \rightarrow A$; λ_2 : transition $C \rightarrow B$

Formula: $E_n = -\frac{13.6}{n^2} \text{ eV}$; $\lambda \propto \frac{1}{\Delta E}$

Solution:

$$E_A = -\frac{13.6}{4} = -3.4 \text{ eV}, \quad E_B = -\frac{13.6}{9} = -1.511 \text{ eV}, \quad E_C = -\frac{13.6}{16} = -0.85 \text{ eV}$$

$$\Delta E_1(B \rightarrow A) = |-3.4 - (-1.511)| = 1.889 \text{ eV}$$

$$\Delta E_2(C \rightarrow B) = |-1.511 - (-0.85)| = 0.661 \text{ eV}$$

Since $\lambda \propto 1/\Delta E$:

$$\frac{\lambda_1}{\lambda_2} = \frac{\Delta E_2}{\Delta E_1} = \frac{0.661}{1.889} = 0.35 = \frac{7}{20}$$

Comparing with the given ratio $\frac{7}{4n}$:

$$\frac{7}{4n} = \frac{7}{20} \Rightarrow 4n = 20 \Rightarrow n = 5$$

Final Answer

$$n = 5$$

Question 26

Calculate the wavelengths of the third and the fifth spectral lines of Paschen series in hydrogen spectrum when $E = -(2.18 \times 10^{-18})/n^2$ joule.

Given: Paschen series: transitions terminate at $n_f = 3$; third line: $n_i = 6$; fifth line:

$$n_i = 8; E_n = -\frac{2.18 \times 10^{-18}}{n^2} \text{ J}$$

Formula: $\Delta E = E_{n_f} - E_{n_i}; \quad \lambda = \frac{hc}{\Delta E}$

Solution:

Third line ($n = 6 \rightarrow n = 3$):

$$E_3 = -\frac{2.18 \times 10^{-18}}{9} = -2.422 \times 10^{-19} \text{ J}, \quad E_6 = -\frac{2.18 \times 10^{-18}}{36} = -6.056 \times 10^{-20} \text{ J}$$

$$\Delta E = |-2.422 \times 10^{-19} - (-6.056 \times 10^{-20})| = 1.817 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{1.817 \times 10^{-19}} = 10899 \times 10^{-10} \text{ m} = 10899 \text{ Å}$$

Fifth line ($n = 8 \rightarrow n = 3$):

$$E_8 = -\frac{2.18 \times 10^{-18}}{64} = -3.406 \times 10^{-20} \text{ J}$$

$$\Delta E = |-2.422 \times 10^{-19} - (-3.406 \times 10^{-20})| = 2.082 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^8}{2.082 \times 10^{-19}} = 9512 \times 10^{-10} \text{ m} = 9512 \text{ Å}$$

Final Answer

Third line: $\lambda = 10899 \text{ Å}$; Fifth line: $\lambda = 9512 \text{ Å}$

Common Mistake: In the Paschen series all lines terminate at $n_f = 3$, not $n_f = 1$; the third line is the transition $n = 6 \rightarrow 3$ (first line is $4 \rightarrow 3$, second is $5 \rightarrow 3$, third is $6 \rightarrow 3$), not $n = 3 \rightarrow 3$ or similar miscounting.

Question 27

The wavelength of a yellow line of sodium is 5896 Å . Give its wave number.

Given: $\lambda = 5896 \text{ Å} = 5896 \times 10^{-10} \text{ m} = 5896 \times 10^{-8} \text{ cm}$

Formula: $\bar{\nu} = \frac{1}{\lambda}$

Solution:

$$\bar{\nu} = \frac{1}{5896 \times 10^{-8} \text{ cm}} = 16961 \text{ cm}^{-1}$$

Final Answer

$\bar{\nu} = 16961 \text{ per cm}$

Question 28

The wavelength of the first line of Balmer series is 6563 Å. Find the wavelength of the first line of the Lyman series.

★ Likely Exam Question

Given: $\lambda_{\text{Balmer},1} = 6563 \text{ Å}$ ($n = 3 \rightarrow 2$); Lyman first line: $n = 2 \rightarrow 1$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$

Solution:

For the Balmer first line: $\frac{1}{\lambda_B} = R \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5R}{36}$

For the Lyman first line: $\frac{1}{\lambda_L} = R \left(1 - \frac{1}{4} \right) = \frac{3R}{4}$

Taking the ratio (so that R cancels):

$$\frac{\lambda_L}{\lambda_B} = \frac{5R/36}{3R/4} = \frac{5}{36} \times \frac{4}{3} = \frac{5}{27}$$

$$\lambda_L = 6563 \times \frac{5}{27} = 1215 \text{ Å}$$

Final Answer

$$\lambda(\text{Lyman, first line}) = 1215 \text{ Å}$$

Question 29

The wavelength of the first member of the Balmer series is 6563 Å. Find the wavelengths of the second and the last member of the series.

Given: $\lambda_1 = 6563 \text{ Å}$ (first member, $n = 3 \rightarrow 2$); second member: $n = 4 \rightarrow 2$; last member: $n = \infty \rightarrow 2$

Formula: $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{n_i^2} \right)$

Solution:

First member coefficient: $\frac{1}{\lambda_1} \propto \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5}{36}$

Second member ($n = 4 \rightarrow 2$): $\frac{1}{\lambda_2} \propto \left(\frac{1}{4} - \frac{1}{16} \right) = \frac{3}{16}$

$$\lambda_2 = \lambda_1 \times \frac{5/36}{3/16} = 6563 \times \frac{80}{108} = 4861 \text{ Å}$$

Last member ($n = \infty \rightarrow 2$, series limit): $\frac{1}{\lambda_{\text{last}}} \propto \frac{1}{4}$

$$\lambda_{\text{last}} = \lambda_1 \times \frac{5/36}{1/4} = 6563 \times \frac{20}{36} = 3646 \text{ \AA}$$

Final Answer

Second member: $\lambda_2 = 4861 \text{ \AA}$; Last member: $\lambda_{\text{last}} = 3646 \text{ \AA}$

Question 30

The ionisation energy of hydrogen atom is 13.6 eV. Calculate Rydberg's constant for hydrogen.

Given: $E_{\text{ionisation}} = 13.6 \text{ eV}$; Hint: $Rhc = 13.6 \text{ eV}$

Formula: $R = \frac{E_{\text{ionisation}}}{hc}$

Solution:

$$E_{\text{ionisation}} = 13.6 \times 1.6 \times 10^{-19} = 2.176 \times 10^{-18} \text{ J}$$

$$R = \frac{2.176 \times 10^{-18}}{6.6 \times 10^{-34} \times 3.0 \times 10^8} = \frac{2.176 \times 10^{-18}}{1.98 \times 10^{-25}} = 1.0956 \times 10^7 \text{ m}^{-1}$$

Final Answer

$$R = 1.0956 \times 10^7 \text{ m}^{-1}$$

ISC Class XII Physics

Chapter 13 Nuclei

25 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Mass-Energy Relation
- II. Pair-Production and Annihilation of Matter
- III. Mass Defect and Binding Energy
- IV. Q-Value of Nuclear Reactions
- V. Nuclear Fission
- VI. Nuclear Fusion

Formula Sheet

Standard Constants used in this Chapter:

$$m_p = 1.00728 \text{ u} \quad m_n = 1.00867 \text{ u} \quad m_e = 0.00055 \text{ u}$$

$$h = 6.6 \times 10^{-34} \text{ J-s} \quad c = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$1 \text{ MeV} = 1.60 \times 10^{-13} \text{ J} \quad 1 \text{ u} = 931.5 \text{ MeV (i.e. } 1 \text{ u} \times c^2 = 931.5 \text{ MeV)}$$

I. Mass-Energy Relation

Mass-Energy Equivalence

$$E = mc^2$$

E = energy released/absorbed (J); m = mass converted (kg); c = speed of light

Percentage Change in Mass

$$\% \text{ increase} = \frac{\Delta m}{M} \times 100$$

Δm = mass gained/lost; M = original mass of body

II. Pair-Production and Annihilation of Matter

Minimum Energy for Pair Production

$$E_{min} = 2m_e c^2$$

m_e = rest mass of electron (equal to rest mass of positron)

Photon Energy-Wavelength Relation

$$E = \frac{hc}{\lambda}$$

h = Planck's constant; λ = wavelength of the photon

Particle-Antiparticle Annihilation Energy

$$E = 2mc^2$$

m = rest mass of each particle (e.g. proton and antiproton)

III. Mass Defect and Binding Energy

Mass Defect

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{nucleus}$$

Z = atomic number; A = mass number; $M_{nucleus}$ = actual nuclear mass

Binding Energy

$$BE = \Delta m \text{ (in u)} \times 931.5 \text{ MeV}$$

Δm expressed in atomic mass units (u)

Binding Energy per Nucleon

$$\frac{BE}{A}$$

Measure of nuclear stability; higher value $\Rightarrow$ more stable nucleus

IV. Q-Value of Nuclear Reactions

Q-Value

$$Q = (\Sigma m_{reactants} - \Sigma m_{products}) \times 931.5 \text{ MeV}$$

Positive Q = energy released (exothermic); Negative Q = energy absorbed

V. Nuclear Fission

Fission Energy

$$E = \Delta mc^2$$

Δm = mass defect in the fission process

Energy-kWh Conversion

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Used to express large nuclear energies in electrical units

Fission Rate

$$\frac{dN}{dt} = \frac{P}{E_{fission}}$$

P = power output (W); $E_{fission}$ = energy released per fission (J)

VI. Nuclear Fusion

Fusion Energy

$$E = \Delta mc^2$$

Δm = mass defect between fusing nuclei and product nucleus

Useful Energy with Efficiency

$$E_{useful} = \eta \times E_{total}$$

η = fractional efficiency of the generator/reactor

I. Mass-Energy Relation

Question 1

Earth absorbs 10^{22} joule energy everyday from the sun. What will be the percentage increase in the earth's mass (6.0×10^{24} kg) due to this?

Given: $E = 10^{22}$ J, $M = 6.0 \times 10^{24}$ kg, $c = 3 \times 10^8$ m s⁻¹

Formula: $E = \Delta mc^2 \Rightarrow \% \text{ increase} = \frac{\Delta m}{M} \times 100$

Solution:

$$\Delta m = \frac{E}{c^2} = \frac{10^{22}}{(3 \times 10^8)^2} = \frac{10^{22}}{9 \times 10^{16}} = 1.111 \times 10^5 \text{ kg}$$

$$\% \text{ increase} = \frac{\Delta m}{M} \times 100 = \frac{1.111 \times 10^5}{6.0 \times 10^{24}} \times 100 = 1.85 \times 10^{-18} \%$$

Final Answer

Percentage increase in Earth's mass $\approx 1.85 \times 10^{-18} \%$

Common Mistake: Don't forget to multiply by 100 for a percentage, and keep careful track of the very large/small powers of ten – a single exponent slip changes the answer completely.

Question 2

Work out the energy equivalence of the rest-mass of electron (9.1×10^{-31} kg).

Given: $m_e = 9.1 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m s⁻¹

Formula: $E = mc^2$

Solution:

$$E = 9.1 \times 10^{-31} \times (3 \times 10^8)^2 = 9.1 \times 10^{-31} \times 9 \times 10^{16} = 8.19 \times 10^{-14} \text{ J}$$

Final Answer

$E \approx 8.2 \times 10^{-14}$ J

Question 3

Find the rest-mass energy of electron in MeV.

★ Likely Exam Question

Given: $E = 8.2 \times 10^{-14}$ J (from Q2), $1 \text{ MeV} = 1.6 \times 10^{-13}$ J

Formula: $E(\text{MeV}) = \frac{E(\text{J})}{1.6 \times 10^{-13}}$

Solution:

$$E = \frac{8.2 \times 10^{-14}}{1.6 \times 10^{-13}} = 0.5125 \text{ MeV}$$

Final Answer

Rest-mass energy of electron ≈ 0.5125 MeV

Common Mistake: This value (≈ 0.511 MeV) is a standard constant used repeatedly in pair-production questions – memorise it.

Question 4

What is the equivalent energy in electron-volt of 1 gram mass?

Given: $m = 1 \text{ g} = 10^{-3} \text{ kg}$, $c = 3 \times 10^8 \text{ m s}^{-1}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

Formula: $E = mc^2$

Solution:

$$E = 10^{-3} \times (3 \times 10^8)^2 = 10^{-3} \times 9 \times 10^{16} = 9 \times 10^{13} \text{ J}$$

$$E(\text{eV}) = \frac{9 \times 10^{13}}{1.6 \times 10^{-19}} = 5.625 \times 10^{32} \text{ eV}$$

Final Answer

$$E \approx 5.625 \times 10^{32} \text{ eV}$$

Common Mistake: Use $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ here (not $1.6 \times 10^{-13} \text{ J}$ for MeV) – mixing these two conversion factors is a very common slip.

II. Pair-Production and Annihilation of Matter

Question 5

What should be the minimum energy of a γ -ray photon for the production of an electron-positron pair? What can be the maximum wavelength? The rest-mass of electron (also of positron) is 9.1×10^{-31} kg.

Given: $m_e = 9.1 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m s⁻¹, $h = 6.6 \times 10^{-34}$ J-s

Formula: $E_{min} = 2m_e c^2$ $\lambda_{max} = \frac{hc}{E_{min}}$

Solution:

$$E_{min} = 2 \times 9.1 \times 10^{-31} \times (3 \times 10^8)^2 = 2 \times 8.19 \times 10^{-14} = 1.638 \times 10^{-13} \text{ J}$$

$$E_{min}(\text{MeV}) = \frac{1.638 \times 10^{-13}}{1.6 \times 10^{-13}} = 1.02 \text{ MeV}$$

$$\lambda_{max} = \frac{hc}{E_{min}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.638 \times 10^{-13}} = 1.209 \times 10^{-12} \text{ m} = 0.012 \text{ \AA}$$

Final Answer

$$E_{min} \approx 1.02 \text{ MeV}, \quad \lambda_{max} \approx 0.012 \text{ \AA}$$

Common Mistake: Pair production creates *two* particles (electron + positron), so the threshold energy is $2m_e c^2$, not $m_e c^2$ – a very common exam trap.

Question 6

Calculate in joule the energy produced by the interaction of a proton and an antiproton. The mass of proton is 1.6725×10^{-27} kg.

Given: $m_p = 1.6725 \times 10^{-27}$ kg, $c = 3 \times 10^8$ m s⁻¹

Formula: $E = 2mc^2$

Solution:

$$E = 2 \times 1.6725 \times 10^{-27} \times (3 \times 10^8)^2 = 2 \times 1.6725 \times 10^{-27} \times 9 \times 10^{16}$$

$$E = 3.0 \times 10^{-10} \text{ J}$$

Final Answer

$$E \approx 3.0 \times 10^{-10} \text{ J}$$

Common Mistake: Both the proton *and* the antiproton contribute mass to the annihilation – omitting the factor of 2 halves the correct answer.

Question 7

A proton and an antiproton annihilate into two photons of same frequency. What is the wavelength of the photons produced? The mass of proton is $1.67 \times 10^{-27} \text{ kg}$.

Given: $m_p = 1.67 \times 10^{-27} \text{ kg}$, $c = 3 \times 10^8 \text{ m s}^{-1}$, $h = 6.6 \times 10^{-34} \text{ J-s}$

Formula: $E_{total} = 2mc^2$, $E_{photon} = \frac{E_{total}}{2}$, $\lambda = \frac{hc}{E_{photon}}$

Solution:

$$E_{total} = 2 \times 1.67 \times 10^{-27} \times 9 \times 10^{16} = 3.006 \times 10^{-10} \text{ J}$$

Since the two photons carry the total energy equally,

$$E_{photon} = \frac{3.006 \times 10^{-10}}{2} = 1.503 \times 10^{-10} \text{ J}$$

$$\lambda = \frac{hc}{E_{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.503 \times 10^{-10}} = 1.3 \times 10^{-15} \text{ m}$$

Final Answer

$$\lambda \approx 1.3 \times 10^{-15} \text{ m}$$

Common Mistake: Divide the total annihilation energy by 2 before finding λ – each photon carries only half the total energy released.

III. Mass Defect and Binding Energy

Question 8

The mass-defect in a nuclear fusion reaction is 0.3 per cent. What amount of energy will be liberated in 1 kg fusion reaction?

Given: $M = 1$ kg, mass defect = 0.3% of M , $c = 3 \times 10^8$ m s⁻¹

Formula: $E = \Delta mc^2$

Solution:

$$\Delta m = 0.3\% \times 1 \text{ kg} = 0.003 \text{ kg}$$

$$E = 0.003 \times (3 \times 10^8)^2 = 0.003 \times 9 \times 10^{16} = 2.7 \times 10^{14} \text{ J}$$

Final Answer

$$E = 2.7 \times 10^{14} \text{ J}$$

Question 9

The binding energy per nucleon of ${}_8\text{O}^{16}$ is 7.97 MeV and that of ${}_8\text{O}^{17}$ is 7.75 MeV. Find the energy needed to remove a neutron from ${}_8\text{O}^{17}$.

★ Likely Exam Question

Given: $(BE/A)_{\text{O}^{16}} = 7.97$ MeV, $(BE/A)_{\text{O}^{17}} = 7.75$ MeV

Formula: Energy needed = $BE(\text{O}^{17}) - BE(\text{O}^{16})$

Solution:

$$BE(\text{O}^{16}) = 16 \times 7.97 = 127.52 \text{ MeV}$$

$$BE(\text{O}^{17}) = 17 \times 7.75 = 131.75 \text{ MeV}$$

$$\text{Energy needed} = 131.75 - 127.52 = 4.23 \text{ MeV}$$

Final Answer

Energy needed to remove a neutron = 4.23 MeV

Common Mistake: Always convert BE/nucleon to *total* binding energy (multiply by mass number A) before subtracting – subtracting the per-nucleon values directly gives a wrong answer.

Question 10

Bombardment of lithium with protons give rise to the following reaction:



The atomic masses of lithium, hydrogen and helium are 7.016 u, 1.008 u and 4.004 u respectively. Find the energy carried by each α -particle. Given: 1 u = 931 MeV.

Given: $m(\text{Li}^7) = 7.016$ u, $m(\text{H}^1) = 1.008$ u, $m(\text{He}^4) = 4.004$ u, 1 u = 931 MeV

Formula: $\Delta m = [m(\text{Li}^7) + m(\text{H}^1)] - 2m(\text{He}^4)$, $E_{\text{total}} = \Delta m \times 931$ MeV

Solution:

$$\Delta m = (7.016 + 1.008) - 2(4.004) = 8.024 - 8.008 = 0.016 \text{ u}$$

$$E_{\text{total}} = 0.016 \times 931 = 14.896 \text{ MeV}$$

Since two identical α -particles are produced, each carries half the total energy:

$$E_{\alpha} = \frac{14.896}{2} = 7.448 \text{ MeV}$$

Final Answer

Energy carried by each α -particle = 7.448 MeV

Common Mistake: Use the 1 u = 931 MeV value exactly as given in the question (not 931.5 MeV) so the final figure matches the expected answer.

Question 11

The mass defect in a particular reaction is 0.4 g. The amount of energy liberated is $n \times 10^7$ kWh. What is the value of n here? (speed of light = 3×10^8 m/s)

Given: $\Delta m = 0.4$ g = 0.4×10^{-3} kg, $c = 3 \times 10^8$ m s⁻¹

Formula: $E = \Delta mc^2$, 1 kWh = 3.6×10^6 J

Solution:

$$E = (0.4 \times 10^{-3}) \times (3 \times 10^8)^2 = 0.4 \times 10^{-3} \times 9 \times 10^{16} = 3.6 \times 10^{13} \text{ J}$$

$$E(\text{kWh}) = \frac{3.6 \times 10^{13}}{3.6 \times 10^6} = 1 \times 10^7 \text{ kWh}$$

Comparing with $n \times 10^7$ kWh,

$$n = 1$$

Final Answer

$n = 1$

Question 12

Determine the binding energy of the nucleus of ${}_8\text{O}^{16}$. Given: mass of proton = 1.007825 u, mass of neutron = 1.008665 u and atomic mass of ${}_8\text{O}^{16} = 15.994915$ u.

★ Likely Exam Question

Given: $Z = 8$, $A = 16$, $m_p = 1.007825$ u, $m_n = 1.008665$ u, $M = 15.994915$ u

Formula: $\Delta m = Zm_p + (A - Z)m_n - M$, $BE = \Delta m \times 931.5$ MeV

Solution:

$$\Delta m = 8(1.007825) + 8(1.008665) - 15.994915$$

$$\Delta m = 8.062600 + 8.069320 - 15.994915 = 16.131920 - 15.994915 = 0.137005 \text{ u}$$

$$BE = 0.137005 \times 931.5 = 127.616 \text{ MeV}$$

Final Answer

Binding energy of ${}_8\text{O}^{16} = 127.616$ MeV

Common Mistake: Here proton and neutron masses (not the hydrogen atom mass) are given directly – use them as they are; don't further subtract electron mass separately.

Question 13

If we form a lithium nucleus (${}_3\text{Li}^7$) by 4 neutrons and 3 protons, then how much energy will be released? The mass of lithium nucleus is 7.01436 u.

Given: 4 neutrons, 3 protons, $m_n = 1.00867$ u, $m_p = 1.00728$ u, $M(\text{Li}^7) = 7.01436$ u

Formula: $\Delta m = 4m_n + 3m_p - M$, $E = \Delta m \times 931.5$ MeV

Solution:

$$\Delta m = 4(1.00867) + 3(1.00728) - 7.01436$$

$$\Delta m = 4.03468 + 3.02184 - 7.01436 = 7.05652 - 7.01436 = 0.04216 \text{ u}$$

$$E = 0.04216 \times 931.5 = 39.27 \text{ MeV}$$

Final Answer

Energy released = 39.27 MeV

Question 14

How much energy is needed to remove a neutron from the nucleus of ${}_{13}\text{Al}^{27}$? Given: mass of ${}_{13}\text{Al}^{27}$ atom = 26.981541 u, mass of ${}_{13}\text{Al}^{26}$ atom = 25.986895 u, mass of neutron = 1.008665 u.

Given: $M(\text{Al}^{27}) = 26.981541$ u, $M(\text{Al}^{26}) = 25.986895$ u, $m_n = 1.008665$ u

Formula: $E = [M(\text{Al}^{26}) + m_n - M(\text{Al}^{27})] \times 931.5$ MeV

Solution:

$$M(\text{Al}^{26}) + m_n = 25.986895 + 1.008665 = 26.995560 \text{ u}$$

$$\Delta m = 26.995560 - 26.981541 = 0.014019 \text{ u}$$

$$E = 0.014019 \times 931.5 = 13.06 \text{ MeV}$$

Final Answer

Energy needed to remove a neutron = 13.06 MeV

Common Mistake: Set up the mass difference as (products – reactant is reversed here) – the energy *absorbed* equals [mass of fragments] – [mass of original nucleus], the opposite sign convention from a normal exothermic reaction.

Question 15

100 g of ${}_{3}\text{Li}^7$ is converted into ${}_{2}\text{He}^4$ by proton bombardment. Calculate the energy released. Given: mass of ${}_{3}\text{Li}^7$ atom = 7.016 u, mass of ${}_{2}\text{He}^4$ atom = 4.004 u, mass of ${}_{1}\text{H}^1$ atom = 1.008 u. The Avogadro number is $6.02 \times 10^{23} \text{ mol}^{-1}$.

Given: $m = 100$ g of Li^7 , $M(\text{Li}^7) = 7.016$ u, $M(\text{He}^4) = 4.004$ u, $M(\text{H}^1) = 1.008$ u, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

Formula: $\Delta m = [M(\text{Li}^7) + M(\text{H}^1)] - 2M(\text{He}^4)$, $E_{\text{total}} = N \times \Delta m \times 931.5$ MeV

Solution:

Mass defect per reaction:

$$\Delta m = (7.016 + 1.008) - 2(4.004) = 8.024 - 8.008 = 0.016 \text{ u}$$

Energy per reaction:

$$E_1 = 0.016 \times 931.5 = 14.904 \text{ MeV} = 14.904 \times 1.6 \times 10^{-13} \text{ J} = 2.385 \times 10^{-12} \text{ J}$$

Number of Li^7 atoms in 100 g (using atomic mass ≈ 7 g/mol):

$$N = \frac{100}{7} \times 6.02 \times 10^{23} = 8.6 \times 10^{24} \text{ atoms}$$

Total energy released:

$$E_{total} = N \times E_1 = 8.6 \times 10^{24} \times 2.385 \times 10^{-12} \approx 2 \times 10^{13} \text{ J}$$

Final Answer

Total energy released $\approx 2 \times 10^{13} \text{ J}$

Common Mistake: Find the number of moles of Li^7 (mass/molar mass) first, then multiply by N_A – do not multiply 100 g directly by N_A .

IV. Q-Value of Nuclear Reactions

Question 16

Calculate the disintegration energy Q when a heavy nucleus of mass number $A = 240$ with binding energy per nucleon 7.6 MeV is split into two equal fragments of mass number $A' = 120$, each with $\text{BE/nucleon } 8.5 \text{ MeV}$.

★ Likely Exam Question

Given: $A = 240$, $(\text{BE}/A)_{\text{initial}} = 7.6 \text{ MeV}$, $A' = 120$ (two fragments), $(\text{BE}/A)_{\text{final}} = 8.5 \text{ MeV}$

Formula: $Q = BE_{\text{final}} - BE_{\text{initial}}$

Solution:

$$BE_{\text{initial}} = 240 \times 7.6 = 1824 \text{ MeV}$$

$$BE_{\text{final}} = 2 \times (120 \times 8.5) = 2 \times 1020 = 2040 \text{ MeV}$$

$$Q = 2040 - 1824 = 216 \text{ MeV}$$

Final Answer

Disintegration energy $Q = 216 \text{ MeV}$

Common Mistake: Remember there are *two* product fragments – forgetting to double the final binding energy is a very common slip in fission Q -value problems.

V. Nuclear Fission

Question 17

0.5 g mass is decayed in the fission of a part of uranium. How much energy will be obtained by it? Express the result in MeV and in kWh.

Given: $\Delta m = 0.5 \text{ g} = 0.5 \times 10^{-3} \text{ kg}$, $c = 3 \times 10^8 \text{ m s}^{-1}$

Formula: $E = \Delta mc^2$, $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

Solution:

$$E = (0.5 \times 10^{-3}) \times 9 \times 10^{16} = 4.5 \times 10^{13} \text{ J}$$

$$E(\text{MeV}) = \frac{4.5 \times 10^{13}}{1.6 \times 10^{-13}} = 2.8 \times 10^{26} \text{ MeV}$$

$$E(\text{kWh}) = \frac{4.5 \times 10^{13}}{3.6 \times 10^6} = 1.25 \times 10^7 \text{ kWh}$$

Final Answer

$$E \approx 2.8 \times 10^{26} \text{ MeV} = 1.25 \times 10^7 \text{ kWh}$$

Question 18

If the loss of mass in some fission process is 0.10%, then how much energy will be obtained by the fission of 1 kg substance? How many kilowatt-hours of electricity can be obtained by it?

Given: $M = 1 \text{ kg}$, mass loss = 0.10% of M

Formula: $E = \Delta mc^2$, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

Solution:

$$\Delta m = 0.10\% \times 1 \text{ kg} = 1 \times 10^{-3} \text{ kg}$$

$$E = (1 \times 10^{-3}) \times 9 \times 10^{16} = 9 \times 10^{13} \text{ J}$$

$$E(\text{kWh}) = \frac{9 \times 10^{13}}{3.6 \times 10^6} = 2.5 \times 10^7 \text{ kWh}$$

Final Answer

$$E = 9 \times 10^{13} \text{ J} = 2.5 \times 10^7 \text{ kWh}$$

Question 19

The energy released per fission of ${}_{92}\text{U}^{235}$ nucleus is 200 MeV. Compute the energy released in joule if 100 g of ${}_{92}\text{U}^{235}$ is fissioned. Avogadro's number is $6.02 \times 10^{23} \text{ mole}^{-1}$.

★ Likely Exam Question

Given: mass of $\text{U}^{235} = 100 \text{ g}$, $E_{\text{fission}} = 200 \text{ MeV}$, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

Formula: $N = \frac{\text{mass}}{\text{molar mass}} \times N_A$, $E_{\text{total}} = N \times E_{\text{fission}}$

Solution:

Number of moles of U^{235} :

$$n = \frac{100}{235} = 0.4255 \text{ mol}$$

Number of atoms:

$$N = 0.4255 \times 6.02 \times 10^{23} = 2.562 \times 10^{23} \text{ atoms}$$

Total energy released:

$$E_{\text{total}} = N \times E_{\text{fission}} = 2.562 \times 10^{23} \times 200 \text{ MeV} = 5.124 \times 10^{25} \text{ MeV}$$

$$E_{\text{total}}(\text{J}) = 5.124 \times 10^{25} \times 1.6 \times 10^{-13} = 8.2 \times 10^{12} \text{ J}$$

Final Answer

Energy released $\approx 8.2 \times 10^{12} \text{ J}$

Question 20

A U^{235} reactor is producing nuclear energy at a rate of 32,000 kW. How many nuclei of U^{235} are fissioned per second? How much U^{235} would be consumed in 1000 hours? Assume an average energy of 200 MeV released per fission. Take Avogadro's number as $6 \times 10^{23} \text{ mol}^{-1}$.

★ Likely Exam Question

Given: $P = 32,000 \text{ kW} = 3.2 \times 10^7 \text{ W}$, $E_{\text{fission}} = 200 \text{ MeV}$, $t = 1000 \text{ h}$, $N_A = 6 \times 10^{23} \text{ mol}^{-1}$

Formula: $\frac{dN}{dt} = \frac{P}{E_{\text{fission}}}$, $\text{mass consumed} = \frac{N_{\text{total}}}{N_A} \times 235 \text{ g}$

Solution:

Energy released per fission:

$$E_{\text{fission}} = 200 \times 1.6 \times 10^{-13} = 3.2 \times 10^{-11} \text{ J}$$

Fissions per second:

$$\frac{dN}{dt} = \frac{3.2 \times 10^7}{3.2 \times 10^{-11}} = 1.0 \times 10^{18} \text{ fissions/s}$$

Total fissions in 1000 hours ($t = 1000 \times 3600 = 3.6 \times 10^6$ s):

$$N_{\text{total}} = 1.0 \times 10^{18} \times 3.6 \times 10^6 = 3.6 \times 10^{24} \text{ nuclei}$$

Moles of U^{235} consumed:

$$n = \frac{3.6 \times 10^{24}}{6 \times 10^{23}} = 6 \text{ mol}$$

Mass consumed:

$$m = 6 \times 235 = 1410 \text{ g} = 1.41 \text{ kg}$$

Final Answer

Fission rate = 1.0×10^{18} nuclei/sec; mass of U^{235} consumed in 1000 h = 1.41 kg

Common Mistake: Convert 1000 hours into seconds ($\times 3600$) before finding total fissions – mixing time units is the most common error in reactor-rate problems.

Question 21

In nuclear fusion, 0.993 g helium is formed from 1 g hydrogen. If the efficiency of the generator be 5%, then calculate the energy obtained in kilowatt-hour (kWh).

★ Likely Exam Question

Given: mass of hydrogen = 1 g, mass of helium formed = 0.993 g, efficiency $\eta = 5\% = 0.05$

Formula: $\Delta m = m_H - m_{He}$, $E = \Delta mc^2$, $E_{\text{useful}} = \eta \times E$

Solution:

$$\Delta m = 1 - 0.993 = 0.007 \text{ g} = 7 \times 10^{-6} \text{ kg}$$

$$E = (7 \times 10^{-6}) \times 9 \times 10^{16} = 6.3 \times 10^{11} \text{ J}$$

$$E_{\text{useful}} = 0.05 \times 6.3 \times 10^{11} = 3.15 \times 10^{10} \text{ J}$$

$$E_{\text{useful}}(\text{kWh}) = \frac{3.15 \times 10^{10}}{3.6 \times 10^6} = 8.75 \times 10^3 \text{ kWh}$$

Final Answer

Energy obtained $\approx 8.75 \times 10^3$ kWh

Common Mistake: Apply the 5% efficiency factor to the total energy released – do not apply it to the mass defect or skip it altogether.

Question 22

If 200 MeV energy is released in the fission of a single nucleus of ${}_{92}\text{U}^{235}$, how many fissions must occur in one second to produce a power of 1 kW? Also compute the energy released in the complete fission of 0.5 kg of U^{235} . Take Avogadro's number $= 6.02 \times 10^{26} \text{ kmol}^{-1}$.

Given: $E_{\text{fission}} = 200 \text{ MeV}$, $P = 1 \text{ kW} = 1000 \text{ W}$, $\text{mass} = 0.5 \text{ kg}$, $N_A = 6.02 \times 10^{26} \text{ kmol}^{-1}$

Formula: $\frac{dN}{dt} = \frac{P}{E_{\text{fission}}}$, $E_{\text{total}} = \frac{\text{mass}}{\text{molar mass}} \times N_A \times E_{\text{fission}}$

Solution:

Energy per fission:

$$E_{\text{fission}} = 200 \times 1.6 \times 10^{-13} = 3.2 \times 10^{-11} \text{ J}$$

Fissions per second for 1 kW:

$$\frac{dN}{dt} = \frac{1000}{3.2 \times 10^{-11}} = 3.125 \times 10^{13} \text{ fissions/s}$$

For 0.5 kg of U^{235} (molar mass 235 kg/kmol):

$$n = \frac{0.5}{235} = 2.128 \times 10^{-3} \text{ kmol}$$

$$N = 2.128 \times 10^{-3} \times 6.02 \times 10^{26} = 1.281 \times 10^{24} \text{ atoms}$$

$$E_{\text{total}} = 1.281 \times 10^{24} \times 200 \text{ MeV} = 2.56 \times 10^{26} \text{ MeV}$$

Final Answer

Fission rate $= 3.125 \times 10^{13} / \text{sec}$; energy from complete fission of 0.5 kg $\text{U}^{235} = 2.56 \times 10^{26} \text{ MeV}$

VI. Nuclear Fusion

Question 23

In a star, three α -particles (${}_2\text{He}^4$ nuclei) fuse together to form a ${}_6\text{C}^{12}$ nucleus. Compute the energy evolved. The masses of ${}_2\text{He}^4$ atom and ${}_6\text{C}^{12}$ atom are 4.002604 u and 12.000000 u respectively.

Given: $3 \times {}_2\text{He}^4 \rightarrow {}_6\text{C}^{12}$, $M(\text{He}^4) = 4.002604$ u, $M(\text{C}^{12}) = 12.000000$ u

Formula: $\Delta m = 3M(\text{He}^4) - M(\text{C}^{12})$, $E = \Delta m \times 931.5$ MeV

Solution:

$$\Delta m = 3(4.002604) - 12.000000 = 12.007812 - 12.000000 = 0.007812 \text{ u}$$

$$E = 0.007812 \times 931.5 = 7.277 \text{ MeV}$$

Final Answer

Energy evolved = 7.277 MeV

Question 24

Four hydrogen nuclei fuse together to form a helium nucleus with emission of two positrons. Calculate the energy released in the reaction. Given: mass of ${}_1\text{H}^1$ atom = 1.007825 u, mass of ${}_2\text{He}^4$ atom = 4.002603 u, mass of positron = 0.000549 u.

★ Likely Exam Question

Given: $4{}_1\text{H}^1 \rightarrow {}_2\text{He}^4 + 2e^+$, $M(\text{H}^1) = 1.007825$ u, $M(\text{He}^4) = 4.002603$ u, $m_{e^+} = 0.000549$ u

Formula: $\Delta m = 4M(\text{H}^1) - [M(\text{He}^4) + 2m_{e^+}]$, $E = \Delta m \times 931.5$ MeV

Solution:

$$4M(\text{H}^1) = 4 \times 1.007825 = 4.031300 \text{ u}$$

$$M(\text{He}^4) + 2m_{e^+} = 4.002603 + 2(0.000549) = 4.002603 + 0.001098 = 4.003701 \text{ u}$$

$$\Delta m = 4.031300 - 4.003701 = 0.027599 \text{ u}$$

$$E = 0.027599 \times 931.5 = 25.71 \text{ MeV}$$

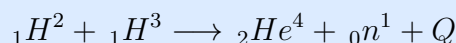
Final Answer

Energy released = 25.71 MeV

Common Mistake: The two emitted positrons carry mass too – their combined mass must be added to the helium mass on the product side before subtracting; leaving them out overstates the mass defect.

Question 25

The following equation represents a fusion reaction:



where Q is the energy released. Given: mass of ${}_1H^2$ atom = 2.014102 u, mass of ${}_1H^3$ atom = 3.016050 u, mass of ${}_2He^4$ atom = 4.002603 u and mass of neutron ${}_0n^1 = 1.008665$ u. Calculate the value of Q .

★ Likely Exam Question

Given: $M(H^2) = 2.014102$ u, $M(H^3) = 3.016050$ u, $M(He^4) = 4.002603$ u, $m_n = 1.008665$ u

Formula: $\Delta m = [M(H^2) + M(H^3)] - [M(He^4) + m_n]$, $Q = \Delta m \times 931.5$ MeV

Solution:

$$M(H^2) + M(H^3) = 2.014102 + 3.016050 = 5.030152 \text{ u}$$

$$M(He^4) + m_n = 4.002603 + 1.008665 = 5.011268 \text{ u}$$

$$\Delta m = 5.030152 - 5.011268 = 0.018884 \text{ u}$$

$$Q = 0.018884 \times 931.5 = 17.59 \text{ MeV}$$

Final Answer

$Q = 17.59$ MeV

Common Mistake: This is the classic D-T fusion reaction that powers fusion reactors – always subtract product mass from reactant mass ($\Sigma m_{\text{reactants}} - \Sigma m_{\text{products}}$) to keep Q positive for an exothermic reaction.

ISC Class XII Physics

Chapter 14 Semiconductor Electronics

3 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. p-n Junction Diode – Forward Bias, Barrier Potential & Dynamic Resistance

Formula Sheet

I. p-n Junction Diode – Forward Bias

Formula: $V_{\text{battery}} = I_{\text{max}}R + V_D$

V_{battery} = battery EMF, I_{max} = maximum safe forward current, R = series resistance, V_D = constant forward voltage drop across the diode.

Formula: $R_{\text{dynamic}} = \frac{\Delta V}{\Delta I}$

R_{dynamic} = dynamic (AC) resistance of the diode, ΔV = change in voltage across diode, ΔI = corresponding change in diode current.

Formula: $I = \frac{V - V_0}{R}$

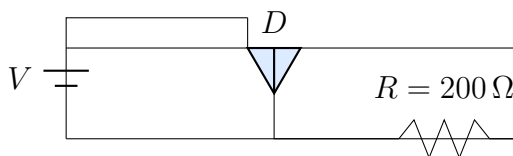
I = current in the circuit, V = applied battery voltage, V_0 = barrier (junction) potential of the diode, R = series resistance.

I. p-n Junction Diode – Forward Bias

Question 1

A p - n junction diode can withstand currents upto a maximum of 10 mA. A resistor $R = 200\ \Omega$ is connected in series with it. The diode, when forward-biased, has a potential drop of 0.5 V across it which is assumed to be independent of current. Find the maximum voltage of the battery used to forward bias the diode.

★ Likely Exam Question



Diode D in series with resistor $R = 200\ \Omega$ and battery V , forward-biased

Given: $I_{\max} = 10\ \text{mA} = 10 \times 10^{-3}\ \text{A}$, $R = 200\ \Omega$, $V_D = 0.5\ \text{V}$

Formula: $V_{\text{battery}} = I_{\max}R + V_D$

Solution:

By Kirchhoff's voltage law, the battery voltage equals the sum of the drop across the resistor and the (constant) drop across the forward-biased diode:

$$V_{\text{battery}} = I_{\max}R + V_D$$

Substituting the values:

$$V_{\text{battery}} = (10 \times 10^{-3}\ \text{A})(200\ \Omega) + 0.5\ \text{V}$$

$$V_{\text{battery}} = 2.0\ \text{V} + 0.5\ \text{V}$$

$$V_{\text{battery}} = 2.5\ \text{V}$$

Final Answer

$$V_{\text{battery}(\max)} = 2.5\ \text{V}$$

Common Mistake: Students often forget to add the diode's forward voltage drop V_D and only calculate $I_{\max}R$, giving an incomplete answer of 2.0 V instead of 2.5 V. Remember: the diode drop is NOT zero even though it is "small" – it must always be included in series-circuit voltage equations.

Question 2

When the voltage drop across a p - n junction diode is increased from 0.65 V to 0.70 V, the change in the diode current is 5 mA. What is the dynamic resistance of the diode?

Given: $\Delta V = 0.70 \text{ V} - 0.65 \text{ V} = 0.05 \text{ V}$, $\Delta I = 5 \text{ mA} = 5 \times 10^{-3} \text{ A}$

Formula: $R_{\text{dynamic}} = \frac{\Delta V}{\Delta I}$

Solution:

The dynamic (AC) resistance of a diode is the ratio of a small change in voltage to the corresponding small change in current:

$$R_{\text{dynamic}} = \frac{\Delta V}{\Delta I} = \frac{0.05 \text{ V}}{5 \times 10^{-3} \text{ A}}$$

$$R_{\text{dynamic}} = \frac{0.05}{0.005} \Omega$$

$$R_{\text{dynamic}} = 10 \Omega$$

Final Answer

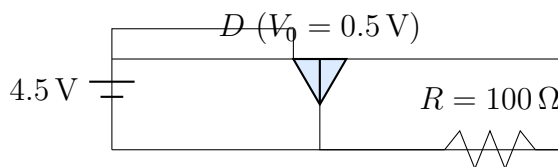
$$R_{\text{dynamic}} = 10 \Omega$$

Common Mistake: Do not confuse dynamic resistance ($\Delta V/\Delta I$, applicable to small changes about an operating point) with static/DC resistance (V/I , the ratio of total voltage to total current). The two are numerically different because the diode's I - V characteristic is non-linear.

Question 3

A p - n junction diode is connected to a series resistor of 100Ω and a 4.5 V DC battery. The barrier potential at the junction is 0.5 V. Find the current in the circuit.

★ Likely Exam Question



Diode D , resistor $R = 100 \Omega$, and 4.5 V battery in series (forward bias)

Given: $V = 4.5 \text{ V}$, $R = 100 \Omega$, $V_0 = 0.5 \text{ V}$ (barrier potential)

Formula: $I = \frac{V - V_0}{R}$

Solution:

The battery voltage is shared between the barrier potential of the forward-biased junction and the voltage drop across the series resistor. Applying Kirchhoff's voltage law:

$$V = IR + V_0 \quad \Rightarrow \quad I = \frac{V - V_0}{R}$$

Substituting the values:

$$\begin{aligned} I &= \frac{4.5 \text{ V} - 0.5 \text{ V}}{100 \, \Omega} = \frac{4.0 \text{ V}}{100 \, \Omega} \\ I &= 0.04 \text{ A} \\ I &= 40 \text{ mA} \end{aligned}$$

Final Answer

$$I = 40 \text{ mA}$$

Common Mistake: A very common exam error is to use the full battery voltage (4.5 V) directly in $I = V/R$, ignoring the barrier potential drop across the junction. The barrier potential must always be subtracted first, since it represents the energy needed to overcome the depletion-region potential barrier before current can flow.